

upvoted 14 times

OmegaLambda7XL9 2 years, 6 months ago

A lil correction,S3 is Object storage not Block Storage

upvoted 4 times

RBSK 3 years, 6 months ago

None meets 100s of TB / file. Bit confusing / misleading

upvoted 7 times

Stpn2me **Most Recent** 2 months, 3 weeks ago

Selected Answer: C

I think of EBS for boot partitions, not for file storage. Any answer with EBS would have been a no.

upvoted 1 times

Clouddon 11 months, 1 week ago

Selected Answer: C

Use EC2 with Amazon EFS to meet the application's need for a shared, scalable, POSIX-compliant, highly available file system with minimal operational management.

upvoted 1 times

SirDNS 1 year, 2 months ago

Selected Answer: C

C: The only option with file system

upvoted 1 times

AwsAbhiKumar 1 year, 4 months ago

Selected Answer: C

HOW EFS is suitable here as it has a maximum file size limit of 52 TiB, so it cannot store a single file that exceeds this size. But in question it has mentioned that output files may vary in size from tens of gigabytes to hundreds of terabytes. Isn't it contradictory?

upvoted 2 times

Dharmarajan 1 year, 4 months ago

Selected Answer: C

The question is not clear. None of the options suit the 100s of TB requirements.

EFS max file size is clearly specified as 49.5TB.

But lets assume for a minute that the app is capable of multipart files, then EFS would be a great fit. (requirement: Must be a file system)

upvoted 2 times

satyaammm 1 year, 5 months ago

Selected Answer: C

EFS is the only file system structure here so it it most suitable.

upvoted 1 times

trinh_le 1 year, 6 months ago

Selected Answer: C

Keyword

Must store in standard file system -> ignore A

Minimum operational -> ignore B

Scalable + Terabytes-> pick C

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - S3 unlimited storage in both scope and size of individual objects

upvoted 1 times

bishtr3 1 year, 10 months ago

C : EFS as It is built to scale on demand to petabytes without disrupting applications, growing and shrinking automatically as you add and remove files.

Multiple compute instances, including Amazon EC2, Amazon ECS, and AWS Lambda, can access an Amazon EFS file system at the same time, providing a common data source for workloads.

upvoted 2 times

HectorCosta 2 years, 1 month ago

Selected Answer: C

Key words: Standard File System and Scales Automatically.

S3 is object Store, hence it fails with the "Standard File System" requirement, so we can discard A.

EBS does not scale automatically, failing with the "Scales Automatically" requirement, so we can discard B and D

upvoted 3 times

sidharthwader 2 years, 3 months ago

C is the only option which supports standard file system when we talk about high availability. EBS scope is within a availability zone but EFS has scope of a region.

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

Standard file system that is highly available: EFS

Autoscaling highly available system: EC2 or ECS or EKS can work

A: Not suitable due to S3 which is BLOB not file system

B: EKS is ok but EBS is not HA

D: EBS is not HA

So by elimination, C is best option.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: C

"File system structure" = EFS, which also meets all the other requirements.

upvoted 4 times

Mikado211 2 years, 6 months ago

Selected Answer: C

Technically the A could work, ECS is often recommended by AWS in case of minimum operational overhead, and S3 is durable and highly scalable BUT it is not a "traditional" file system structure. In an S3 bucket, there is no real file structure, only files and prefixes that simulate a structure.

B is wrong because of EKS which require more management

EFS is recommended for minimum operational overhead instead of EBS.

So C (EC2 + EFS) is recommended here over D (EC2 + EBS).

upvoted 4 times

wantu 2 years, 6 months ago

Selected Answer: C

Palabras clave: autoescalado y ficheros

upvoted 1 times

A company needs to store its accounting records in Amazon S3. The records must be immediately accessible for 1 year and then must be archived for an additional 9 years. No one at the company, including administrative users and root users, can be able to delete the records during the entire 10-year period. The records must be stored with maximum resiliency.

Which solution will meet these requirements?

- A. Store the records in S3 Glacier for the entire 10-year period. Use an access control policy to deny deletion of the records for a period of 10 years.
- B. Store the records by using S3 Intelligent-Tiering. Use an IAM policy to deny deletion of the records. After 10 years, change the IAM policy to allow deletion.
- C. Use an S3 Lifecycle policy to transition the records from S3 Standard to S3 Glacier Deep Archive after 1 year. Use S3 Object Lock in compliance mode for a period of 10 years. **Most Voted**
- D. Use an S3 Lifecycle policy to transition the records from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 1 year. Use S3 Object Lock in governance mode for a period of 10 years.

Correct Answer: C

Community vote distribution

C (100%)

Comments

awsgeek75 **Highly Voted** 2 years, 4 months ago

Selected Answer: C

Only CD provides Object Lock options which is required for stopping admin/root users from deleting.

D is governance mode which is like government, pay enough money and you can do anything. This is not what we want so compliance is the option. C is right choice.

For future, remember

S3 Lock Governance = corrupt government official

S3 Lock Compliance = honest solution architect!

upvoted 25 times

Guru4Cloud **Highly Voted** 2 years, 10 months ago

Selected Answer: C

The key reasons are:

The S3 Lifecycle policy transitions the data to Glacier Deep Archive after 1 year for long-term archival.

S3 Object Lock in compliance mode prevents any user from deleting or overwriting objects for the specified retention period.

Glacier Deep Archive provides very high durability and the lowest storage cost for long-term archival.

Compliance mode ensures no one can override or change the retention settings even if policies change.

This meets all the requirements - immediate access for 1 year, archived for 9 years, unable to delete for 10 years, maximum resiliency

upvoted 10 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: C

Use:

S3 Standard → S3 Glacier Deep Archive (after 1 year) via lifecycle policy

S3 Object Lock (compliance mode, 10 years) to enforce WORM protection

This provides automated archiving, immutability, and maximum data durability — fully aligned with the company's needs.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: C

S3 lifecycle policies are required to transition from immediate availability to archived form. Also the S3 object lock helps prevent deletion of the bucket.

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - S3 Glacier after year 1 in compliance mode with object lock (=immutable lock)

upvoted 2 times

Ruffyit 2 years, 7 months ago

No one at the company, including administrative users and root users, can be able to delete the records during the entire 10-year period = Compliance Mode

upvoted 5 times

axelrodb 2 years, 9 months ago

Selected Answer: C

To meet the requirements of immediately accessible records for 1 year and then archived for an additional 9 years with maximum resiliency, we can use S3 Lifecycle policy to transition records from S3 Standard to S3 Glacier Deep Archive after 1 year. And to ensure that the records cannot be deleted by anyone, including administrative and root users, we can use S3 Object Lock in compliance mode for a period of 10 years. Therefore, the correct answer is option C.

Reference: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.htmls>

upvoted 5 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: C

No one at the company, including administrative users and root users, can be able to delete the records during the entire 10-year period = Compliance Mode

upvoted 2 times

miki111 2 years, 10 months ago

Option C is the correct answer

upvoted 2 times

MutiverseAgent 2 years, 11 months ago

Why not A? Move all files to S3 Glacier instant retrieval (Cheaper than S3) and then move files older than a year to S3 Deep archive.

upvoted 1 times

dhax12 2 years, 8 months ago

Put entire 10 years to Glacier means it's not accessible for the 1 year window. Hence wrong answer.

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: C

To prevent deletion of records during the entire 10-year period, you can utilize S3 Object Lock feature. By enabling it in compliance mode, you can set a retention period on the objects, preventing any user, including administrative and root users, from deleting records.

A: S3 Glacier is suitable for long-term archival, it may not provide immediate accessibility for the first year as required.

B: Intelligent-Tiering may not offer the most cost-effective archival storage option for extended 9-year period. Changing the IAM policy after 10 year to allow deletion also introduces manual steps and potential human error.

D: While S3 One Zone-IA can provide cost savings, it doesn't offer the same level of resiliency as S3 Glacier Deep Archive for long-term archival.

upvoted 4 times

11pantheman11 3 years, 1 month ago

Selected Answer: C

In compliance mode, a protected object version can't be overwritten or deleted by any user, including the root user in your AWS account.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock-overview.html>

upvoted 4 times

athiha 3 years, 3 months ago

Selected Answer: C

Retention Period: A period is specified by Days & Years.

With Retention Compliance Mode, you can't change/adjust (even by the account root user) the retention mode during the retention period while all objects within the bucket are Locked.

With Retention Governance mode, a less restrictive mode, you can grant special permission to a group of users to adjust the Lock settings by using S3:BypassGovernanceRetention.

Legal Hold: It's On/Off setting on an object version. There is no retention period. If you enable Legal Hold on specific object version, you will not be able to delete or override that specific object version. It needs S:PutObjectLegalHold as a permission.

upvoted 5 times

WherecanIstart 3 years, 3 months ago

Selected Answer: C

S3 Glacier Deep Archive all day....

upvoted 1 times

SilentMilli 3 years, 5 months ago

Selected Answer: C

Use an S3 Lifecycle policy to transition the records from S3 Standard to S3 Glacier Deep Archive after 1 year. Use S3 Object Lock in compliance mode for a period of 10 years.

upvoted 2 times

k1kavi1 3 years, 5 months ago

Selected Answer: C

Use S3 Object Lock in compliance mode

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock-overview.html>

upvoted 4 times

pazabal 3 years, 5 months ago

Selected Answer: C

C, A lifecycle set to transition from standard to Glacier deep archive and use lock for the delete requirement

A, B and D don't meet the requirements

upvoted 2 times

A company runs multiple Windows workloads on AWS. The company's employees use Windows file shares that are hosted on two Amazon EC2 instances. The file shares synchronize data between themselves and maintain duplicate copies. The company wants a highly available and durable storage solution that preserves how users currently access the files.

What should a solutions architect do to meet these requirements?

- A. Migrate all the data to Amazon S3. Set up IAM authentication for users to access files.
- B. Set up an Amazon S3 File Gateway. Mount the S3 File Gateway on the existing EC2 instances.
- C. Extend the file share environment to Amazon FSx for Windows File Server with a Multi-AZ configuration. Migrate all the data to FSx for Windows File Server. **Most Voted**
- D. Extend the file share environment to Amazon Elastic File System (Amazon EFS) with a Multi-AZ configuration. Migrate all the data to Amazon EFS.

Correct Answer: C

Community vote distribution

C (98%)

24

Comments

k1kavi1 **Highly Voted** 3 years, 5 months ago

Selected Answer: C

EFS is not supported on Windows instances

<https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/AmazonEFS.html>

Amazon FSx for Windows File Server provides fully managed Microsoft Windows file servers, backed by a fully native Windows file system.

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/what-is.html>

upvoted 21 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 6 months ago

Selected Answer: C

Windows file shares = Amazon FSx for Windows File Server

Hence, the correct answer is C

upvoted 8 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Taking back this answer. As explained in the latest update.

CORRECT

D: Extend the file share environment to Amazon Elastic File System (Amazon EFS) with a Multi-AZ configuration. Migrate all the data to Amazon EFS

upvoted 1 times

pentium75 2 years, 5 months ago

No, users should continue using SMB, which EFS doesn't support

upvoted 7 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: C

Amazon FSx for Windows File Server (Multi-AZ) is purpose-built to support Windows file sharing, with native SMB, high availability, and no user experience changes, making Option C the correct choice.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: C

FSx is used for windows file systems whereas EFS is used for Linux file systems.

upvoted 2 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - use FSx for Windows File Server - why create extra work by migrating to EFS? Add Multi-AZ configuration for availability
upvoted 2 times

MatAlves 1 year, 10 months ago

Amazon EFS (Linux) and Amazon FSx (Windows) provide highly durable and available file systems that can span multiple Availability Zones. Both solutions are designed to deliver high performance, however, when choosing to use network file systems consider the access patterns.

<https://docs.aws.amazon.com/wellarchitected/latest/sap-lens/best-practice-14-3.html>
upvoted 2 times

SMALLE 2 years, 4 months ago

Selected Answer: C

FSx=Windows
<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/what-is.html>
upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

Windows workload rules out S3 and EFS as they cannot be mounted directly on Windows. S3 File Gateway is mainly for on-prem to AWS which is not a requirement here as company is already in AWS.

C meets all the requirements.
upvoted 4 times

jitlathi 2 years, 5 months ago

Selected Answer: C

With Amazon FSx for Windows File Server, you can enjoy a native Windows file server experience with a fully managed, scalable, and highly dependable file storage solution
upvoted 3 times

Ruffyit 2 years, 7 months ago

<https://aws.amazon.com/fsx/windows/faqs/>
Thousands of compute instances and devices can access a file system concurrently.
upvoted 2 times

AWSStudyBuddy 2 years, 7 months ago

Selected Answer: C

With Amazon FSx for Windows File Server, you can enjoy a native Windows file server experience with a fully managed, scalable, and highly dependable file storage solution. Rich administrative features including end-user file recovery, user quotas, and Microsoft Active Directory integration are all provided by this Windows Server-based system.
upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

The key reasons are:

FSx for Windows provides fully managed Windows-native SMB file shares that are accessible from Windows clients.
It allows seamlessly migrating the existing Windows file shares to FSx shares without disrupting users.
The Multi-AZ configuration provides high availability and durability for file storage.
Users can continue to access files the same way over SMB without any changes.
It is optimized for Windows workloads and provides features like user quotas, ACLs, AD integration.
Data is stored on SSDs with automatic backups for resilience.
upvoted 1 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: C

The company wants a highly available and durable storage solution that preserves how users currently access the files = Amazon FSx for Windows File Server
upvoted 2 times

miki111 2 years, 10 months ago

Option C is the correct answer
upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

Migrating all the data to FSx for Windows File Server allows you to preserve existing user access method and maintain compatibility with Windows file shares. Users can continue accessing files using the same method as before, without any disruptions.

A: S3 is a highly durable object storage service, it is not designed to directly host Windows file shares. Implementing IAM authentication for file access would require significant changes to existing user access method.

B: S3 File Gateway can provide access to Amazon S3 objects through standard file protocols, it may not be ideal solution for preserving existing user access method and maintaining Windows file shares.

D: Although Amazon EFS provides highly available and durable file storage, it may not directly support the existing Windows file shares and their access method.

upvoted 5 times

11pantheman11 3 years, 1 month ago

Selected Answer: C

<https://aws.amazon.com/fsx/windows/faqs/>

Thousands of compute instances and devices can access a file system concurrently.

EFS does not support Windows

upvoted 3 times

cheese929 3 years, 1 month ago

Selected Answer: C

C is correct. Amazon FSx for Windows File Server.

upvoted 3 times

A solutions architect is developing a VPC architecture that includes multiple subnets. The architecture will host applications that use Amazon EC2 instances and Amazon RDS DB instances. The architecture consists of six subnets in two Availability Zones. Each Availability Zone includes a public subnet, a private subnet, and a dedicated subnet for databases. Only EC2 instances that run in the private subnets can have access to the RDS databases.

Which solution will meet these requirements?

- A. Create a new route table that excludes the route to the public subnets' CIDR blocks. Associate the route table with the database subnets.
- B. Create a security group that denies inbound traffic from the security group that is assigned to instances in the public subnets. Attach the security group to the DB instances.
- C. Create a security group that allows inbound traffic from the security group that is assigned to instances in the private subnets. Attach the security group to the DB instances. **Most Voted**
- D. Create a new peering connection between the public subnets and the private subnets. Create a different peering connection between the private subnets and the database subnets.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Sinaneos **Highly Voted** 3 years, 8 months ago

Selected Answer: C

- A: doesn't fully configure the traffic flow
- B: security groups don't have deny rules
- D: peering is mostly between VPCs, doesn't really help here

answer is C, most mainstream way

upvoted 55 times

Gary_Phillips_2007 **Highly Voted** 3 years, 3 months ago

Just took the exam today and EVERY ONE of the questions came from this dump. Memorize it all. Good luck.

upvoted 40 times

orhan64 2 years, 10 months ago

Hey bro, did you buy premium access?

upvoted 5 times

PaulGa **Most Recent** 1 year, 9 months ago

Selected Answer: C

Ans C - only one that makes sense: "...a security group that allows inbound traffic from the security group that is assigned to instances in the private subnets" - operative word: "allows"

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

- A: route table that connect... no idea what this option is trying to do but won't work for RDS
- B: SG are deny by default
- D: Peering connection between subnets? No idea what this is but happy to learn if such a thing exists.

C: SG to allow input to private subnet means everything else will be blocked. Attaching this SG to DB instance means it will block everything except the private subnet instances which is where the required EC2 instances are.

upvoted 5 times

AWSSStudyBuddy 2 years, 7 months ago

Selected Answer: C

RDS databases can only be accessed by EC2 instances located in private subnets: From the security group given to instances in the private subnets, the DB instances' security group will permit incoming traffic. Because of this, the RDS databases will only be accessible by EC2 instances located on the private subnets.

Because of its safe architecture, Every other source of incoming traffic will be blocked by the security group that is linked to the database instances. The RDS databases will be better shielded from unwanted access thanks to this.

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

The key reasons are:

Using security groups to control access between resources is a standard practice in VPCs.

The security group attached to the RDS DB instances can allow inbound traffic from the security group for the EC2 instances in the private subnets. This allows only those EC2 instances in the private subnets to connect to the databases, meeting the requirements.

Route tables, peering connections, and denying public subnet access would not achieve the needed selectivity of allowing only the private subnet EC2 instances.

Security groups provide stateful filtering at the instance level for precise access control.

upvoted 3 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: C

Security groups only have allow rules

upvoted 3 times

praveenvky83 2 years, 10 months ago

Selected Answer: C

optoin C

upvoted 1 times

miki111 2 years, 10 months ago

Option C is the correct answer

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

Creating security group that allows inbound traffic from security group assigned to instances in private subnets ensures that only EC2 running in private subnets can access the RDS databases. By associating security group with DB, you restrict access to only instances that belong to designated security group.

A: This approach may help control routing within VPC, it does not address the specific access requirement between EC2 instances and RDS databases.

B: Using a deny rule in a security group can lead to complexities and potential misconfigurations. It is generally recommended to use allow rules to explicitly define access permissions.

D: Peering connections enable communication between different VPCs or VPCs in different regions, and they are not necessary for restricting access between subnets within the same VPC.

upvoted 5 times

Bmarodi 3 years ago

Selected Answer: C

Option C meets the requirements.

upvoted 1 times

Abrar2022 3 years ago

By default, a security group is set up with rules that deny all inbound traffic and permit all outbound traffic.

upvoted 2 times

water314 3 years, 1 month ago

Selected Answer: C

CCCCCCCCC

upvoted 1 times

SilentMilli 3 years, 5 months ago

Selected Answer: C

Create a security group that allows inbound traffic from the security group that is assigned to instances in the private subnets. Attach the security group to the DB instances. This will allow the EC2 instances in the private subnets to have access to the RDS databases while denying access to the

EC2 instances in the public subnets.

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: C

The solution that meets the requirements described in the question is option C: Create a security group that allows inbound traffic from the security group that is assigned to instances in the private subnets. Attach the security group to the DB instances.

In this solution, the security group applied to the DB instances allows inbound traffic from the security group assigned to instances in the private subnets. This ensures that only EC2 instances running in the private subnets can have access to the RDS databases.

upvoted 3 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, creating a new route table that excludes the route to the public subnets' CIDR blocks and associating it with the database subnets, would not meet the requirements because it would block all traffic to the database subnets, not just traffic from the public subnets.

Option B, creating a security group that denies inbound traffic from the security group assigned to instances in the public subnets and attaching it to the DB instances, would not meet the requirements because it would allow all traffic from the private subnets to reach the DB instances, not just traffic from the security group assigned to instances in the private subnets.

Option D, creating a new peering connection between the public subnets and the private subnets and a different peering connection between the private subnets and the database subnets, would not meet the requirements because it would allow all traffic from the private subnets to reach the DB instances, not just traffic from the security group assigned to instances in the private subnets.

upvoted 1 times

Nandan747 3 years, 5 months ago

Selected Answer: C

The real trick is between B and C. A and D are ruled out for obvious reasons.

B is wrong as you cannot have deny type rules in Security groups.

So- C is the right answer.

upvoted 4 times

ashish_t 3 years, 6 months ago

Selected Answer: C

The key is "Only EC2 instances that run in the private subnets can have access to the RDS databases"

The answer is C.

upvoted 2 times

A company has registered its domain name with Amazon Route 53. The company uses Amazon API Gateway in the ca-central-1 Region as a public interface for its backend microservice APIs. Third-party services consume the APIs securely. The company wants to design its API Gateway URL with the company's domain name and corresponding certificate so that the third-party services can use HTTPS.

Which solution will meet these requirements?

- A. Create stage variables in API Gateway with Name="Endpoint-URL" and Value="Company Domain Name" to overwrite the default URL. Import the public certificate associated with the company's domain name into AWS Certificate Manager (ACM).
- B. Create Route 53 DNS records with the company's domain name. Point the alias record to the Regional API Gateway stage endpoint. Import the public certificate associated with the company's domain name into AWS Certificate Manager (ACM) in the us-east-1 Region.
- C. Create a Regional API Gateway endpoint. Associate the API Gateway endpoint with the company's domain name. Import the public certificate associated with the company's domain name into AWS Certificate Manager (ACM) in the same Region. Attach the certificate to the API Gateway endpoint. Configure Route 53 to route traffic to the API Gateway endpoint. **Most Voted**
- D. Create a Regional API Gateway endpoint. Associate the API Gateway endpoint with the company's domain name. Import the public certificate associated with the company's domain name into AWS Certificate Manager (ACM) in the us-east-1 Region. Attach the certificate to the API Gateway APIs. Create Route 53 DNS records with the company's domain name. Point an A record to the company's domain name.

Correct Answer: C

Community vote distribution

C (95%)

5%

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: C

The correct solution to meet these requirements is option C.

To design the API Gateway URL with the company's domain name and corresponding certificate, the company needs to do the following:

1. Create a Regional API Gateway endpoint: This will allow the company to create an endpoint that is specific to a region.
2. Associate the API Gateway endpoint with the company's domain name: This will allow the company to use its own domain name for the API Gateway URL.
3. Import the public certificate associated with the company's domain name into AWS Certificate Manager (ACM) in the same Region: This will allow the company to use HTTPS for secure communication with its APIs.
4. Attach the certificate to the API Gateway endpoint: This will allow the company to use the certificate for securing the API Gateway URL.
5. Configure Route 53 to route traffic to the API Gateway endpoint: This will allow the company to use Route 53 to route traffic to the API Gateway URL using the company's domain name.

upvoted 69 times

t0nx 2 years, 6 months ago

Why the "reveal solution" most of the time gives the wrong answer ?

upvoted 13 times

yaralexie 1 year, 4 months ago

😅 happens to me too

upvoted 1 times

wharftargo 2 years, 4 months ago

i read this before that they can't give 100% of the right answers legally or something
upvoted 9 times

aadityaravi8 2 years, 11 months ago

google bard reply..
upvoted 4 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option C includes all the necessary steps to meet the requirements, hence it is the correct solution.

Options A and D do not include the necessary steps to associate the API Gateway endpoint with the company's domain name and attach the certificate to the endpoint.

Option B includes the necessary steps to associate the API Gateway endpoint with the company's domain name and attach the certificate, but it imports the certificate into the us-east-1 Region instead of the ca-central-1 Region where the API Gateway is located.

upvoted 12 times

masetromain **Highly Voted** 3 years, 8 months ago

Selected Answer: C

I think the answer is C. we don't need to attach a certificate in us-east-1, if is not for cloudfront. In our case the target is ca-central-1.
upvoted 33 times

Valero_ 3 years, 7 months ago

I think that is C too, the target would be the same Region.

<https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-regional-api-custom-domain-create.html>
upvoted 8 times

MutiverseAgent 2 years, 11 months ago

Agree, C is correct by using the API Gateway option "Custom domain names"

<https://docs.aws.amazon.com/apigateway/latest/developerguide/how-to-custom-domains.html>
upvoted 2 times

FlyingHawk **Most Recent** 1 year, 4 months ago

Selected Answer: C

I selected B initially as I mixed API Gateway with CloudFront, for CloudFront, it is a global service, the cert needs to be in us-east-1 region as CF's control plane is in us-east-1 region, but for regional API Gateway, ELB, they should be in the same region as the service.
<https://docs.aws.amazon.com/acm/latest/userguide/import-certificate.html>
<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/cnames-and-https-requirements.html>
upvoted 3 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - as per Buruguduystunstugudunstuy (1 year, 8 months ago) for the reasons therein...
Not sure what Ans D is addressing..
upvoted 2 times

jatric 1 year, 11 months ago

Selected Answer: C

Option C has all the steps to meet the requirement and attach certificate in the same region
upvoted 2 times

ChymKuBoy 1 year, 12 months ago

Selected Answer: C

C for sure
upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

BD are wrong because they are in wrong regions.
A. Does not help with R53 routing to API Gateway and not sure what it's trying to do here
C is correct
upvoted 2 times

bujuman 2 years, 5 months ago

Selected Answer: C

Important
For an API Gateway Regional custom domain name, you must request or import the certificate in the same Region as your API.

upvoted 4 times

debasishdtta 2 years, 5 months ago

Selected Answer: D

All certificates in ACM are regional resources, including the certificates that you import. To use the same certificate with Elastic Load Balancing load balancers in different AWS Regions, you must import the certificate into each Region where you want to use it. To use a certificate with Amazon CloudFront, you must import it into the US East (N. Virginia) Region.
<https://docs.aws.amazon.com/acm/latest/userguide/import-certificate.html>

upvoted 3 times

EtherealBagel 2 years, 6 months ago

Only if the API Gateway is global then the corresponding AWS ACM Certificate must be placed in us-east-1

upvoted 1 times

luongtrann 2 years, 7 months ago

Selected Answer: C

Correct answer

upvoted 1 times

Abitek007 2 years, 8 months ago

Selected Answer: D

A records support Elasticity and load balancing and by default resilience is Key in any configuration in AWS

upvoted 2 times

Abitek007 2 years, 8 months ago

now I am confused, I would have chosen C, but with a Closer look D might be right, because of the A records and again the region used and not stated can be for resilience. I think? can someone clarify

upvoted 2 times

OctavioBatera 2 years, 2 months ago

I think C is the correct answer, because the DNS record in this case must be an alias (cname). DNS A record is for IP address. Here some documentation that can be useful:

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-to-api-gateway.html>

upvoted 3 times

paniya93 2 years, 8 months ago

Selected Answer: C

Explain why this saying a different region which not mentioned in the Q.

upvoted 1 times

Hassao 2 years, 9 months ago

c is right

The other options have various issues:

Option A: Using stage variables and importing certificates into ACM is not sufficient for achieving the requirement of associating a custom domain and certificate with the API Gateway endpoint.

Option B: While it mentions importing the certificate into ACM, it doesn't address the need for a Regional API Gateway or the appropriate region for the certificate.

Option D: Using certificates from the us-east-1 region for a Regional API Gateway might cause issues. Additionally, it doesn't provide clear details on how to associate the domain name and certificate with the API Gateway endpoint.

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

C is the correct solution.

To use a custom domain name with HTTPS for API Gateway:

The API Gateway endpoint needs to be Regional, not private or edge-optimized.

The ACM certificate must be requested in the same region as the API Gateway endpoint.

The custom domain name is then mapped to the Regional API endpoint under API Gateway domain names.

Route 53 is configured to route traffic to the API Gateway regional domain.

The ACM certificate is attached to the API Gateway domain name to enable HTTP

upvoted 1 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: C

Import the public certificate associated with the company's domain name into AWS Certificate Manager (ACM) in the same Region.

upvoted 1 times

A company is running a popular social media website. The website gives users the ability to upload images to share with other users. The company wants to make sure that the images do not contain inappropriate content. The company needs a solution that minimizes development effort.

What should a solutions architect do to meet these requirements?

- A. Use Amazon Comprehend to detect inappropriate content. Use human review for low-confidence predictions.
- B. Use Amazon Rekognition to detect inappropriate content. Use human review for low-confidence predictions. **Most Voted**
- C. Use Amazon SageMaker to detect inappropriate content. Use ground truth to label low-confidence predictions.
- D. Use AWS Fargate to deploy a custom machine learning model to detect inappropriate content. Use ground truth to label low-confidence predictions.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: B

The best solution to meet these requirements would be option B: Use Amazon Rekognition to detect inappropriate content, and use human review for low-confidence predictions.

Amazon Rekognition is a cloud-based image and video analysis service that can detect inappropriate content in images using its pre-trained label detection model. It can identify a wide range of inappropriate content, including explicit or suggestive adult content, violent content, and offensive language. The service provides high accuracy and low latency, making it a good choice for this use case.

upvoted 25 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, using Amazon Comprehend, is not a good fit for this use case because Amazon Comprehend is a natural language processing service that is designed to analyze text, not images.

Option C, using Amazon SageMaker to detect inappropriate content, would require significant development effort to build and train a custom machine learning model. It would also require a large dataset of labeled images to train the model, which may be time-consuming and expensive to obtain.

Option D, using AWS Fargate to deploy a custom machine learning model, would also require significant development effort and a large dataset of labeled images. It may not be the most efficient or cost-effective solution for this use case.

In summary, the best solution is to use Amazon Rekognition to detect inappropriate content in images, and use human review for low-confidence predictions to ensure that all inappropriate content is detected.

upvoted 17 times

masetromain **Highly Voted** 3 years, 8 months ago

Selected Answer: B

Good Answer is B :

<https://docs.aws.amazon.com/rekognition/latest/dg/moderation.html?pg=ln&sec=ft>

upvoted 14 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: B

Amazon Rekognition is a ML service by AWS for image and video analysis so it is the most suitable option here.

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: B

Ans B - I'll go with ans B reluctantly because Rekognition seems to be primarily aimed at video/image content as opposed to language/text... and only because you can add the latter "You can add features [to Rekognition] that detect objects, text, unsafe content, analyze images/videos, and compare faces to your application using Rekognition's APIs." (<https://docs.aws.amazon.com/rekognition/latest/dg/what-is.html>)

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

<https://aws.amazon.com/rekognition/>

Automate and lower the cost of your image recognition and video analysis with machine learning

upvoted 2 times

awsgeek75 2 years, 4 months ago

<https://aws.amazon.com/rekognition/content-moderation/>

Amazon Rekognition Content Moderation automates and streamlines your image and video moderation workflows using machine learning (ML), without requiring ML experience.

upvoted 2 times

slimen 2 years, 7 months ago

Selected Answer: B

comprehend is for NLP

sagemaker is for training and deploying ML and AI models

deploying custom models using fargate requires time and development effort which is not recommended by the question

upvoted 4 times

Ruffyit 2 years, 7 months ago

<https://docs.aws.amazon.com/rekognition/latest/dg/moderation.html?pg=ln&sec=ft>

upvoted 2 times

AWSSStudyBuddy 2 years, 7 months ago

Selected Answer: B

You can easily incorporate image and video analysis to your applications with the help of Amazon Rekognition. Numerous functions are available to it, including as facial analysis, image classification, and object and scene identification.

DetectModerationLabels is an operation that may be used with Amazon Rekognition to identify incorrect content in photos. By using this procedure, photos with violent, drug-related, tobacco-related, alcohol-related, hate-filled, or provocative material can be identified.

upvoted 3 times

Syruis 2 years, 9 months ago

Selected Answer: B

B is the best solution as far

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: B

Amazon Rekognition is a fully managed service that provides image and video analysis capabilities. It can be used to detect inappropriate content in images, such as nudity, violence, and hate speech.

Amazon Rekognition is a good choice for this solution because it is a managed service, which means that the company does not have to worry about managing the infrastructure or the machine learning model. Rekognition is also highly accurate, and it can be used to detect a wide range of inappropriate content

upvoted 2 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: B

Amazon Rekognition to the rescue...whoosh!

upvoted 1 times

cookieMr 2 years, 11 months ago

Using Amazon Rekognition for content moderation is a cost-effective and efficient solution that reduces the need for developing and training custom machine learning models, making it the best option in terms of minimizing development effort.

A. Amazon Comprehend is a natural language processing service provided by AWS, primarily focused on text analysis rather than image analysis.

C. Amazon SageMaker is a comprehensive machine learning service that allows you to build, train, and deploy custom machine learning models. It requires significant development effort to build and train a custom model. In addition, utilizing ground truth to label low-confidence predictions would further add to the development complexity and maintenance overhead.

D. Similar to C, using AWS Fargate to deploy a custom machine learning model requires significant development effort.

upvoted 3 times

krajar 3 years, 2 months ago

Selected Answer: B

Amazon Rekognition is a cloud-based image and video analysis service that can detect inappropriate content in images using its pre-trained label detection model. It can identify a wide range of inappropriate content, including explicit or suggestive adult content, violent content, and offensive language.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

Shasha1 3 years, 6 months ago

B

AWS Rekognition to detect inappropriate content and use human review for low-confidence predictions. This option minimizes development effort because Amazon Rekognition is a pre-built machine learning service that can detect inappropriate content. Using human review for low-confidence predictions allows for more accurate detection of inappropriate content.

upvoted 2 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 1 times

ArielSchivo 3 years, 7 months ago

Selected Answer: B

Option B.

<https://docs.aws.amazon.com/rekognition/latest/dg/a2i-rekognition.html>

upvoted 2 times

A company wants to run its critical applications in containers to meet requirements for scalability and availability. The company prefers to focus on maintenance of the critical applications. The company does not want to be responsible for provisioning and managing the underlying infrastructure that runs the containerized workload.

What should a solutions architect do to meet these requirements?

- A. Use Amazon EC2 instances, and install Docker on the instances.
- B. Use Amazon Elastic Container Service (Amazon ECS) on Amazon EC2 worker nodes.
- C. Use Amazon Elastic Container Service (Amazon ECS) on AWS Fargate. **Most Voted**
- D. Use Amazon EC2 instances from an Amazon Elastic Container Service (Amazon ECS)-optimized Amazon Machine Image (AMI).

Correct Answer: C

Community vote distribution

C (100%)

Comments

masetromain **Highly Voted** 3 years, 8 months ago

Selected Answer: C

Good answer is C:

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without having to manage servers. AWS Fargate is compatible with Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS).

<https://aws.amazon.com/fr/fargate/>

upvoted 28 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: C

Using ECS on Fargate allows you to run containers without the need to manage the underlying infrastructure. Fargate abstracts away the underlying EC2 and provides serverless compute for containers.

A. This option would require manual provisioning and management of EC2, as well as installing and configuring Docker on those instances. It would introduce additional overhead and responsibilities for maintaining the underlying infrastructure.

B. While this option leverages ECS to manage containers, it still requires provisioning and managing EC2 to serve as worker nodes. It adds complexity and maintenance overhead compared to the serverless nature of Fargate.

D. This option still involves managing and provisioning EC2, even though an ECS-optimized AMI simplifies the process of setting up EC2 for running ECS. It does not provide the level of serverless abstraction and ease of management offered by Fargate.

upvoted 8 times

PaulGa **Most Recent** 1 year, 9 months ago

Selected Answer: C

Ans C - use Fargate to do all the management / deployment (which the company doesn't want to do)

upvoted 2 times

Hkayne 2 years, 1 month ago

Selected Answer: C

ECS FARGATE

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

Managed containers = Fargate

upvoted 4 times

Ruffyyit 2 years, 7 months ago

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without having to manage servers. AWS Fargate is compatible with Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS).

upvoted 2 times

AWSStudyBuddy 2 years, 7 months ago

Selected Answer: C

In order to execute containerized apps without having to manage servers, AWS Fargate is a serverless compute engine for Amazon ECS. Amazon Elastic Compute Cloud (Amazon EC2) instance clusters no longer require provisioning, configuring, or scaling thanks to AWS Fargate. So that you can concentrate on developing and maintaining your applications, AWS Fargate handles the monotonous, repetitive labor of managing servers.

upvoted 2 times

Teruteru 2 years, 9 months ago

Option C is the correct answer.

upvoted 1 times

Syruis 2 years, 9 months ago

Selected Answer: C

C for Fargate

upvoted 1 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: C

The company does not want to be responsible for provisioning and managing the underlying infrastructure that runs the containerized workload = Serverless compute for containers = AWS Fargate

upvoted 2 times

miki111 2 years, 10 months ago

Option C is the correct answer

upvoted 1 times

cheese929 3 years, 1 month ago

Selected Answer: C

AWS Fargate is a technology that you can use with Amazon ECS to run containers without having to manage servers or clusters of Amazon EC2 instances.

<https://docs.aws.amazon.com/AmazonECS/latest/userguide/what-is-fargate.html>

upvoted 2 times

SilentMilli 3 years, 5 months ago

Selected Answer: C

ECS + Fargate

upvoted 3 times

gustavtd 3 years, 5 months ago

Selected Answer: C

AWS Fargate will hide all the complexity for you

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: C

C. Use Amazon Elastic Container Service (Amazon ECS) on AWS Fargate.

AWS Fargate is a fully managed container execution environment that runs containers without the need to provision and manage underlying infrastructure. This makes it a good choice for companies that want to focus on maintaining their critical applications and do not want to be responsible for provisioning and managing the underlying infrastructure.

Option A involves installing Docker on Amazon EC2 instances, which would still require the company to manage the underlying infrastructure. Option B involves using Amazon ECS on Amazon EC2 worker nodes, which would also require the company to manage the underlying infrastructure. Option D involves using Amazon EC2 instances from an Amazon ECS-optimized Amazon Machine Image (AMI), which would also require the company to manage the underlying infrastructure.

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

benaws 3 years, 6 months ago

Selected Answer: C

Obviously anything with EC2 in the answer is wrong...
upvoted 1 times

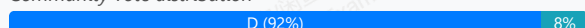
A company hosts more than 300 global websites and applications. The company requires a platform to analyze more than 30 TB of clickstream data each day.

What should a solutions architect do to transmit and process the clickstream data?

- A. Design an AWS Data Pipeline to archive the data to an Amazon S3 bucket and run an Amazon EMR cluster with the data to generate analytics.
- B. Create an Auto Scaling group of Amazon EC2 instances to process the data and send it to an Amazon S3 data lake for Amazon Redshift to use for analysis.
- C. Cache the data to Amazon CloudFront. Store the data in an Amazon S3 bucket. When an object is added to the S3 bucket, run an AWS Lambda function to process the data for analysis.
- D. Collect the data from Amazon Kinesis Data Streams. Use Amazon Kinesis Data Firehose to transmit the data to an Amazon S3 data lake. Load the data in Amazon Redshift for analysis. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: D

Option D is the most appropriate solution for transmitting and processing the clickstream data in this scenario.

Amazon Kinesis Data Streams is a highly scalable and durable service that enables real-time processing of streaming data at a high volume and high rate. You can use Kinesis Data Streams to collect and process the clickstream data in real-time.

Amazon Kinesis Data Firehose is a fully managed service that loads streaming data into data stores and analytics tools. You can use Kinesis Data Firehose to transmit the data from Kinesis Data Streams to an Amazon S3 data lake.

Once the data is in the data lake, you can use Amazon Redshift to load the data and perform analysis on it. Amazon Redshift is a fully managed, petabyte-scale data warehouse service that allows you to quickly and efficiently analyze data using SQL and your existing business intelligence tools.
upvoted 43 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, which involves using AWS Data Pipeline to archive the data to an Amazon S3 bucket and running an Amazon EMR cluster with the data to generate analytics, is not the most appropriate solution because it does not involve real-time processing of the data.

Option B, which involves creating an Auto Scaling group of Amazon EC2 instances to process the data and sending it to an Amazon S3 data lake for Amazon Redshift to use for analysis, is not the most appropriate solution because it does not involve a fully managed service for transmitting the data from the processing layer to the data lake.

Option C, which involves caching the data to Amazon CloudFront, storing the data in an Amazon S3 bucket, and running an AWS Lambda function to process the data for analysis when an object is added to the S3 bucket, is not the most appropriate solution because it does not involve a scalable and durable service for collecting and processing the data in real-time.
upvoted 10 times

MutiverseAgent 2 years, 11 months ago

The question does not say that real-time is needed here
upvoted 3 times

pentium75 2 years, 5 months ago

Question asks how to "transmit and process the clickstream data", NOT how to analyze it. Thus D.
upvoted 2 times

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: D

Option D.

<https://aws.amazon.com/es/blogs/big-data/real-time-analytics-with-amazon-redshift-streaming-ingestion/>
upvoted 17 times

RBSK 3 years, 6 months ago

Unsure if this is right URL for this scenario. Option D is referring to S3 and then Redshift. Whereas URL discuss about eliminating S3 :- We're excited to launch Amazon Redshift streaming ingestion for Amazon Kinesis Data Streams, which enables you to ingest data directly from the Kinesis data stream without having to stage the data in Amazon Simple Storage Service (Amazon S3). Streaming ingestion allows you to achieve low latency in the order of seconds while ingesting hundreds of megabytes of data into your Amazon Redshift cluster.

upvoted 5 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: D

Kinesis data firehouse is the most suitable for streaming data and Redshift is the most suitable for large data sets.
upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: D

Ans D - using Kinesis Streams / Firehouse (data in/out) is fast and reliable. Using Redshift allows all sorts of permutations of data analyses and interfacing to user apps
upvoted 2 times

effiecancode 1 year, 11 months ago

D is the best option
upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

A: Not sure how recent this question is but Data Pipeline is not really a product AWS is recommending anymore
<https://docs.aws.amazon.com/datapipeline/latest/DeveloperGuide/what-is-datapipeline.html>

B: 30TB of clickstream data could be done with EC2 but it would be challenging

C: CloudFront is for CDN and caching and mostly outgoing data, not incoming.

D: Kinesis, S3 data lake and Redshift will work perfectly for this case

upvoted 5 times

clumsyninja4life 2 years, 5 months ago

Selected Answer: A

The answer should be A. Clickstream does not mean real time, it just means they capture user interactions on the web page. Kinesis data streaming is not required. Furthermore, redshift is a data warehousing solution, it cant run complex analysis as well as EMR. My vote goes for A
upvoted 1 times

pentium75 2 years, 5 months ago

Question asks how to "transmit and process the clickstream data", NOT how to analyze it. Also question does NOT ask how to archive the data (as is mentioned in A). Thus D.

upvoted 1 times

Reckless_Jas 2 years, 9 months ago

when you see clickstream data, think about Kinesis Data Stream
upvoted 6 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

The key reasons are:

Kinesis Data Streams can continuously capture and ingest high volumes of clickstream data in real-time. This handles the large 30TB daily data intake
Kinesis Firehose can automatically load the streaming data into S3. This creates a data lake for further analysis.

Firehose can transform and analyze the data in flight before loading to S3 using Lambda. This enables real-time processing.

The data in S3 can be easily loaded into Amazon Redshift for interactive analysis at scale.

Kinesis auto scales to handle the high data volumes. Minimal effort is needed for infrastructure management.

upvoted 2 times

miki111 2 years, 10 months ago

Option D is the correct answer
upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: D

A. This option utilizes S3 for data storage and EMR for analytics, Data Pipeline is not ideal service for real-time streaming data ingestion and processing. It is better suited for batch processing scenarios.

B. This option involves managing and scaling EC2, which adds operational overhead. It is also not real-time streaming solution. Additionally, use of Redshift for analyzing clickstream data might not be most efficient or cost-effective approach.

C. CloudFront is CDN service and is not designed for real-time data processing or analytics. While using Lambda to process data can be an option, it may not be most efficient solution for processing large volumes of clickstream data.

Therefore, collecting the data from Kinesis Data Streams, using Kinesis Data Firehose to transmit it to S3 data lake, and loading it into Redshift for analysis is the recommended approach. This combination provides scalable, real-time streaming solution with storage and analytics capabilities that can handle high volume of clickstream data.

upvoted 2 times

Rahulbit34 3 years, 1 month ago

Clickstream is the key - Answer is D

upvoted 1 times

PaoloRoma 3 years, 2 months ago

Selected Answer: A

I am going to be unpopular here and I'll go for A). Even if here are other services that offer a better experience, data Pipeline can do the job here. "you can use AWS Data Pipeline to archive your web server's logs to Amazon Simple Storage Service (Amazon S3) each day and then run a weekly Amazon EMR (Amazon EMR) cluster over those logs to generate traffic reports"

<https://docs.aws.amazon.com/datapipeline/latest/DeveloperGuide/what-is-datapipeline.html> In the question there is no specific timing requirement for analytics. Also the EMR cluster job can be scheduled be executed daily.

Option D) is a valid answer too, however with Amazon Redshift Streaming Ingestion "you can connect to Amazon Kinesis Data Streams data streams and pull data directly to Amazon Redshift without staging data in S3" <https://aws.amazon.com/redshift/redshift-streaming-ingestion>. So in this scenario Kinesis Data Firehose and S3 are redundant.

upvoted 6 times

MutiverseAgent 2 years, 11 months ago

I think I agree with you, I does not make sense in option D) using Amazon Kinesis Data Firehose to transmit the data to an Amazon S3 data lake and then to Redshift, as you can send directly the data from Firehose to Redshift.

upvoted 2 times

juanrasus2 2 years, 7 months ago

Also the Kinesis family is related to real time or near real time services. This is not a requirement at all. We have to process data daily, but not need to do it in real time

upvoted 2 times

pentium75 2 years, 5 months ago

Question asks how to "transmit and process the clickstream data", NOT how to analyze it. This picture shows exactly scenario D:

Producer - Kinesis - Intermediate S3 bucket - Redshift

<https://d2908q01vomqb2.cloudfront.net/b6692ea5df920cad691c20319a6fffd7a4a766b8/2020/07/30/StreamTransformAnalyzeKinesisLambdaRedshift1.png>

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

studis 3 years, 5 months ago

It is C.

The image in here <https://aws.amazon.com/kinesis/data-firehose/> shows how kinesis can send data collected to firehose who can send it to Redshift. It is also possible to use an intermediary S3 bucket between firehose and redshift. See image in here

<https://aws.amazon.com/blogs/big-data/stream-transform-and-analyze-xml-data-in-real-time-with-amazon-kinesis-aws-lambda-and-amazon-redshift/>

upvoted 1 times

pentium75 2 years, 5 months ago

Makes sense, but this is D, not C

upvoted 1 times

sebasta 3 years, 6 months ago

Why not A?

You can collect data with AWS Data Pipeline and then analyze it with EMR. Whats wrong with this option?

upvoted 4 times

bearcandy 3 years, 6 months ago

It's not A, the wording is tricky! It says "to archive the data to S3" - there is no mention of archiving in the question, so it has to be D :)

upvoted 3 times

pentium75 2 years, 5 months ago

And, the the question is not asking about analyzing the data at all, just about "transmitting and processing".

upvoted 1 times

Wpcorgan 3 years, 6 months ago

D is correct

upvoted 1 times

A company has a website hosted on AWS. The website is behind an Application Load Balancer (ALB) that is configured to handle HTTP and HTTPS separately. The company wants to forward all requests to the website so that the requests will use HTTPS. What should a solutions architect do to meet this requirement?

- A. Update the ALB's network ACL to accept only HTTPS traffic.
- B. Create a rule that replaces the HTTP in the URL with HTTPS.
- C. Create a listener rule on the ALB to redirect HTTP traffic to HTTPS. **Most Voted**
- D. Replace the ALB with a Network Load Balancer configured to use Server Name Indication (SNI).

Correct Answer: C

Community vote distribution

C (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: C

C. Create a listener rule on the ALB to redirect HTTP traffic to HTTPS.

To meet the requirement of forwarding all requests to the website so that the requests will use HTTPS, a solutions architect can create a listener rule on the ALB that redirects HTTP traffic to HTTPS. This can be done by creating a rule with a condition that matches all HTTP traffic and a rule action that redirects the traffic to the HTTPS listener. The HTTPS listener should already be configured to accept HTTPS traffic and forward it to the target group.

upvoted 31 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A. Updating the ALB's network ACL to accept only HTTPS traffic is not a valid solution because the network ACL is used to control inbound and outbound traffic at the subnet level, not at the listener level.

Option B. Creating a rule that replaces the HTTP in the URL with HTTPS is not a valid solution because this would not redirect the traffic to the HTTPS listener.

Option D. Replacing the ALB with a Network Load Balancer configured to use Server Name Indication (SNI) is not a valid solution because it would not address the requirement to redirect HTTP traffic to HTTPS.

upvoted 19 times

masetromain **Highly Voted** 3 years, 8 months ago

Selected Answer: C

Answer C :

https://docs.aws.amazon.com/fr_fr/elasticloadbalancing/latest/application/create-https-listener.html

<https://aws.amazon.com/fr/premiumsupport/knowledge-center/elb-redirect-http-to-https-using-alb/>

upvoted 15 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: C

To redirect all HTTP traffic to HTTPS efficiently and securely, use ALB listener rules

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: C

Redirecting the network on to HTTPS is the requirement here.

upvoted 1 times

cookieMr 1 year, 8 months ago

Selected Answer: C

A. Network ACLs operate at subnet level and control inbound and outbound traffic. Updating the network ACL alone will not enforce the redirection of HTTP to HTTPS.

B. This approach would require modifying application code or server configuration to perform URL rewrite. It is not an optimal solution as it adds complexity and potential maintenance overhead. Moreover, it does not leverage the ALB's capabilities for handling HTTP-to-HTTPS redirection.

D. While NLB can handle SSL/TLS termination using SNI for routing requests to different services, replacing the ALB solely to enforce HTTP-to-HTTPS redirection would be an unnecessary and more complex solution.

Therefore, the recommended approach is to create a listener rule on the ALB to redirect HTTP traffic to HTTPS. By configuring a listener rule, you can define a redirect action that automatically directs HTTP requests to their corresponding HTTPS versions.

upvoted 5 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - don't re-invent; just re-direct

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

<https://repost.aws/knowledge-center/elb-redirect-http-to-https-using-alb>

Steps 6-8 tells exactly how to do this:

"6. Select a load balancer, and then choose HTTP Listener.

7. Under Rules, choose View/edit rules.

8. Choose Edit Rule to modify the existing default rule to redirect all HTTP requests to HTTPS. Or, insert a rule between the existing rules (if appropriate for your use case)."

upvoted 3 times

Ruffyit 2 years, 7 months ago

C. Create a listener rule on the ALB to redirect HTTP traffic to HTTPS.

upvoted 3 times

AWSStudyBuddy 2 years, 7 months ago

Selected Answer: C

This solution meets all of the requirements:

Forward all requests to the website so that the requests will use HTTPS: The ALB can be configured to redirect all HTTP traffic to HTTPS.

The other options are not as good for this scenario:

A. Updating the ALB's network ACL to accept only HTTPS traffic will prevent users from accessing the website using HTTP.

B. Creating a rule that replaces the HTTP in the URL with HTTPS will not prevent users from accessing the website using HTTP.

D. Replacing the ALB with a Network Load Balancer configured to use Server Name Indication (SNI) is not necessary because the ALB can be configured to redirect all HTTP traffic to HTTPS.

upvoted 3 times

Tom123456ac 2 years, 8 months ago

I hate this question description "The company wants to forward all requests to the website so that the requests will use HTTPS."

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

The best solution is to create a listener rule on the Application Load Balancer (ALB) to redirect HTTP traffic to HTTPS (option C).

Here is why:

ALB listener rules allow you to redirect traffic from one listener port (e.g. 80 for HTTP) to another (e.g. 443 for HTTPS). This achieves the goal to forward all requests over HTTPS.

Network ACLs control traffic at the subnet level and cannot distinguish between HTTP and HTTPS requests to implement a redirect (option A incorrect).

Replacing HTTP with HTTPS in the URL happens at the client side. It does not redirect at the ALB (option B incorrect).

Network Load Balancers work at the TCP level and do not understand HTTP or HTTPS protocols. So they cannot redirect in this manner (option D incorrect).

upvoted 8 times

miki111 2 years, 10 months ago

Option C is the correct answer

upvoted 1 times

Abrar2022 3 years ago

A solutions architect should create listen rules to direct http traffic to https.

upvoted 1 times

cheese929 3 years, 1 month ago

Selected Answer: C

C is correct. Traffic redirection will solve it.

upvoted 2 times

elarningtakai 3 years, 2 months ago

Selected Answer: C

This rule can be created in the following way:

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.
2. In the navigation pane, choose Load Balancers.
3. Select the ALB and choose Listeners.
4. Choose View/edit rules and then choose Add rule.
5. In the Add Rule dialog box, choose HTTPS.
6. In the Default action dialog box, choose Redirect to HTTPS.
7. Choose Save rules.

This listener rule will redirect all HTTP requests to HTTPS, ensuring that all traffic is encrypted.

upvoted 5 times

mell1222 3 years, 3 months ago

Selected Answer: C

Configure an HTTPS listener on the ALB: This step involves setting up an HTTPS listener on the ALB and configuring the security policy to use a secure SSL/TLS protocol and cipher suite.

Create a redirect rule on the ALB: The redirect rule should be configured to redirect all incoming HTTP requests to HTTPS. This can be done by creating a redirect rule that redirects HTTP requests on port 80 to HTTPS requests on port 443.

Update the DNS record: The DNS record for the website should be updated to point to the ALB's DNS name, so that all traffic is routed through the ALB.

Verify the configuration: Once the configuration is complete, the website should be tested to ensure that all requests are being redirected to HTTPS. This can be done by accessing the website using HTTP and verifying that the request is redirected to HTTPS.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

A company is developing a two-tier web application on AWS. The company's developers have deployed the application on an Amazon EC2 instance that connects directly to a backend Amazon RDS database. The company must not hardcode database credentials in the application. The company must also implement a solution to automatically rotate the database credentials on a regular basis.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Store the database credentials in the instance metadata. Use Amazon EventBridge (Amazon CloudWatch Events) rules to run a scheduled AWS Lambda function that updates the RDS credentials and instance metadata at the same time.
- B. Store the database credentials in a configuration file in an encrypted Amazon S3 bucket. Use Amazon EventBridge (Amazon CloudWatch Events) rules to run a scheduled AWS Lambda function that updates the RDS credentials and the credentials in the configuration file at the same time. Use S3 Versioning to ensure the ability to fall back to previous values.
- C. Store the database credentials as a secret in AWS Secrets Manager. Turn on automatic rotation for the secret. Attach the required permission to the EC2 role to grant access to the secret. **Most Voted**
- D. Store the database credentials as encrypted parameters in AWS Systems Manager Parameter Store. Turn on automatic rotation for the encrypted parameters. Attach the required permission to the EC2 role to grant access to the encrypted parameters.

Correct Answer: C

Community vote distribution

C (100%)

Comments

KVK16 **Highly Voted** 3 years, 7 months ago

Selected Answer: C

Secrets manager supports Autorotation unlike Parameter store.
upvoted 25 times

JesseeS 3 years, 7 months ago

Parameter store does not support autorotation.
upvoted 10 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: C

The correct solution is C. Store the database credentials as a secret in AWS Secrets Manager. Turn on automatic rotation for the secret. Attach the required permission to the EC2 role to grant access to the secret.

AWS Secrets Manager is a service that enables you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. By storing the database credentials as a secret in Secrets Manager, you can ensure that they are not hardcoded in the application and that they are automatically rotated on a regular basis. To grant the EC2 instance access to the secret, you can attach the required permission to the EC2 role. This will allow the application to retrieve the secret from Secrets Manager as needed.

upvoted 16 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, storing the database credentials in the instance metadata and using a Lambda function to update them, would not meet the requirement of not hardcoding the credentials in the application.

Option B, storing the database credentials in an encrypted S3 bucket and using a Lambda function to update them, would also not meet this requirement, as the application would still need to access the credentials from the configuration file.

Option D, storing the database credentials as encrypted parameters in AWS Systems Manager Parameter Store, would also not meet this requirement, as the application would still need to access the encrypted parameters in order to use them.

upvoted 10 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: C

Use Secrets Manager for:

Secure storage of secrets

Built-in automatic rotation

Easy IAM integration with EC2

upvoted 1 times

DatNX 1 year, 3 months ago

Selected Answer: C

- automatically rotate credentials on regular basis -> AWS Secret Manager

upvoted 2 times

Kyleskii 1 year, 4 months ago

Selected Answer: C

Keyword: database credentials.

AWS secrets managers will handle it.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: C

Rotating the credentials is a feature provided by AWS Secrets Manager

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - Secrets Manager, provides rotation - and also a lot more API calls

upvoted 1 times

soufiyane 2 years, 2 months ago

Selected Answer: C

parameter store does not have auto rotation

upvoted 1 times

Atul6969 2 years, 3 months ago

Selected Answer: C

test kjlshfjkh jfskjfnkj kj bskjfb kj kjs bfkjs b kjf

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

Secrets Manager is purpose built for this scenario

AB are wrong and insecure way of doing this

D Parameter store with encrypted string can be used for this but is not ideal choice and AFAIK it does not support automatic rotation without extra programming

upvoted 1 times

1Alpha1 2 years, 5 months ago

Selected Answer: C

C - "Auto Rotation"

upvoted 1 times

Ruffyit 2 years, 7 months ago

AWS Secrets Manager is a service that enables you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. By storing the database credentials as a secret in Secrets Manager, you can ensure that they are not hardcoded in the application and that they are automatically rotated on a regular basis. To grant the EC2 instance access to the secret, you can attach the required permission to the EC2 role.

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

Storing the credentials in AWS Secrets Manager and enabling automatic rotation meets the requirements with the least operational overhead. The EC2 instance role just needs permission to access the secret, and Secrets Manager handles rotating the credentials automatically on a schedule.

upvoted 1 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: C

Key Autorotation = AWS Secrets Manager

upvoted 2 times

miki111 2 years, 10 months ago

Option C is the right answer.

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

Storing the credentials in Secrets Manager provides dedicated and secure management. With automatic rotation enabled, Secrets Manager handles the credential updates automatically. Attaching the necessary permissions to the EC2 role allows the application to securely access the secret.

This approach minimizes operational overhead and provides a secure and managed solution for credential management.

upvoted 2 times

Bmarodi 3 years ago

Selected Answer: C

The solution that meets the requirements with the least operational overhead, is option C.

upvoted 1 times

A company is deploying a new public web application to AWS. The application will run behind an Application Load Balancer (ALB). The application needs to be encrypted at the edge with an SSL/TLS certificate that is issued by an external certificate authority (CA). The certificate must be rotated each year before the certificate expires.

What should a solutions architect do to meet these requirements?

- A. Use AWS Certificate Manager (ACM) to issue an SSL/TLS certificate. Apply the certificate to the ALB. Use the managed renewal feature to automatically rotate the certificate.
- B. Use AWS Certificate Manager (ACM) to issue an SSL/TLS certificate. Import the key material from the certificate. Apply the certificate to the ALB. Use the managed renewal feature to automatically rotate the certificate.
- C. Use AWS Certificate Manager (ACM) Private Certificate Authority to issue an SSL/TLS certificate from the root CA. Apply the certificate to the ALB. Use the managed renewal feature to automatically rotate the certificate.
- D. Use AWS Certificate Manager (ACM) to import an SSL/TLS certificate. Apply the certificate to the ALB. Use Amazon EventBridge (Amazon CloudWatch Events) to send a notification when the certificate is nearing expiration. Rotate the certificate manually. **Most Voted**

Correct Answer: D

Community vote distribution

D (94%)

6%

Comments

Sinaneos **Highly Voted** 3 years, 7 months ago

Selected Answer: D

It's a third-party certificate, hence AWS cannot manage renewal automatically. The closest thing you can do is to send a notification to renew the 3rd party certificate.

upvoted 64 times

mabotega **Highly Voted** 3 years, 7 months ago

Selected Answer: D

It is D, because ACM does not manage the renewal process for imported certificates. You are responsible for monitoring the expiration date of your imported certificates and for renewing them before they expire.

Check this question on the link below:

Q: What types of certificates can I create and manage with ACM?

https://www.amazonaws.cn/en/certificate-manager/faqs/#Managed_renewal_and_deployment

upvoted 22 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: D

Since the certificate is from an external CA:

You import it into ACM.

Apply it to the ALB.

Set up EventBridge notifications to alert before expiration.

You must manually rotate it each year.

upvoted 2 times

Dharmarajan 1 year, 4 months ago

Selected Answer: D

D is the most appropriate. C seems like it might work, but the word "Private" makes it ineligible answer.

Although I would say is to have ACM renew the public certificate automatically per the renewal method supported:

https://www.amazonaws.cn/en/certificate-manager/faqs/#Managed_renewal_and_deployment

Thanks @mabotega for the link to the doc.

upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: D

Since the certificate is external it needs to be imported and thus D is the most suitable option.

upvoted 1 times

MrAnhQuan 1 year, 5 months ago

Selected Answer: D

It's D, because ACM doesn't manage the renewal process for imported certificates.

upvoted 2 times

Tjazz04 1 year, 6 months ago

Selected Answer: D

Ans. D

Keyword: issued by an external certificate authority

ACM cannot rotate certificate automatically if from external CA.

upvoted 3 times

PaulGa 1 year, 9 months ago

Selected Answer: D

Ans D - hint: we're importing the certificate "...Use AWS Certificate Manager (ACM) to import an SSL/TLS certificate"

upvoted 3 times

Anji195 2 years, 4 months ago

Yes it's D. Here is a clear explanation.

Imported certificates – If you want to use a third-party certificate with Amazon CloudFront, Elastic Load Balancing, or Amazon API Gateway, you may import it into ACM using the AWS Management Console, AWS CLI, or ACM APIs. ACM can not renew imported certificates, but it can help you manage the renewal process. You are responsible for monitoring the expiration date of your imported certificates and for renewing them before they expire. You can use ACM CloudWatch metrics to monitor the expiration dates of an imported certificates and import a new third-party certificate to replace an expiring one.

https://www.amazonaws.cn/en/certificate-manager/faqs/#Managed_renewal_and_deployment

upvoted 5 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

"certificate that is issued by an external certificate authority (CA)"

AB will create a new certificate in AWS

C will also create a new certificate but this is not what PCA are for *(<https://docs.aws.amazon.com/privateca/latest/userguide/PcaWelcome.html>)

D: Import the certificate is correct answer

upvoted 1 times

1Alpha1 2 years, 5 months ago

Selected Answer: D

D - "External CA" --> 'Update Manually'

upvoted 1 times

xdkonorek2 2 years, 7 months ago

Selected Answer: A

internal CA are typically trusted only within the organization unless you manually distribute and trust the root certificate elsewhere

external CA:

Certificates from a well-known external CA are trusted by most browsers and systems by default

<https://docs.aws.amazon.com/acm/latest/userguide/acm-certificate.html>

"Public certificates that you request through ACM are obtained from Amazon Trust Services, an Amazon managed public certificate authority (CA). ... Any browser, application, or OS that includes the Amazon or Starfield roots will trust public certificates obtained from ACM."

The answer is A, different story if they said external certificate

upvoted 3 times

Ruffyit 2 years, 7 months ago

: What types of certificates can I create and manage with ACM?

https://www.amazonaws.cn/en/certificate-manager/faqs/#Managed_renewal_and_deployment

upvoted 1 times

est3la21 2 years, 9 months ago

answer is D

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

The key points are:

Obtain certificate from external CA, not ACM

Import the external certificate into ACM

Apply imported certificate to the ALB

Set up EventBridge rule to trigger notification on certificate expiration

Manually renew and rotate the external certificate each year.

upvoted 2 times

miki111 2 years, 10 months ago

Option D is the right answer.

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: D

D: With this approach, you import the third-party certificate into ACM, which allows you to centrally manage and apply it to the ALB. By configuring CloudWatch Events, you can receive notifications when the certificate is close to expiring, prompting you to manually initiate the rotation process.

A & B: These options assume that the SSL/TLS certificate can be issued directly by ACM. However, since the requirement specifies that the certificate should be issued by an external certificate authority (CA), this option is not suitable.

C: ACM Private Certificate Authority is used when you want to create your own private CA and issue certificates from it. It does not support certificates issued by external CAs. Therefore, this option is not suitable for the given requirement.

upvoted 3 times

A company runs its infrastructure on AWS and has a registered base of 700,000 users for its document management application. The company intends to create a product that converts large .pdf files to .jpg image files. The .pdf files average 5 MB in size. The company needs to store the original files and the converted files. A solutions architect must design a scalable solution to accommodate demand that will grow rapidly over time.

Which solution meets these requirements MOST cost-effectively?

A. Save the .pdf files to Amazon S3. Configure an S3 PUT event to invoke an AWS Lambda function to convert the files to .jpg format and store them back in Amazon S3. **Most Voted**

B. Save the .pdf files to Amazon DynamoDB. Use the DynamoDB Streams feature to invoke an AWS Lambda function to convert the files to .jpg format and store them back in DynamoDB.

C. Upload the .pdf files to an AWS Elastic Beanstalk application that includes Amazon EC2 instances, Amazon Elastic Block Store (Amazon EBS) storage, and an Auto Scaling group. Use a program in the EC2 instances to convert the files to .jpg format. Save the .pdf files and the .jpg files in the EBS store.

D. Upload the .pdf files to an AWS Elastic Beanstalk application that includes Amazon EC2 instances, Amazon Elastic File System (Amazon EFS) storage, and an Auto Scaling group. Use a program in the EC2 instances to convert the file to .jpg format. Save the .pdf files and the .jpg files in the EBS store.

Correct Answer: A

Community vote distribution

A (99%)

Comments

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: A

Option A. Elastic BeanStalk is expensive, and DocumentDB has a 400KB max to upload files. So Lambda and S3 should be the one.
upvoted 52 times

raffaello44 3 years, 7 months ago

is lambda scalable as an EC2 ?

upvoted 5 times

EtherealBagel 2 years, 6 months ago

lambda has near infinite scale

upvoted 6 times

rob74 3 years, 7 months ago

In addition to this Lambda is paid only when used....

upvoted 7 times

mrbottomwood 3 years, 5 months ago

I'm thinking when you wrote DocumentDB you meant it as DynamoDB...yes?

upvoted 6 times

benjl 3 years, 5 months ago

Yes, DynamoDB has 400KB limit for the item.

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/ServiceQuotas.html>

upvoted 8 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: A

B. Using DynamoDB for storing and processing large .pdf files would not be cost-effective due to storage and throughput costs associated with DynamoDB.

C. Using Elastic Beanstalk with EC2 and EBS storage can work, but it may not be most cost-effective solution. It involves managing the underlying infrastructure and scaling manually.

D. Similar to C, using Elastic Beanstalk with EC2 and EFS storage can work, but it may not be most cost-effective solution. EFS is a shared file storage service and may not provide optimal performance for conversion process, especially as demand and file sizes increase.

A. leverages Lambda and the scalable and cost-effective storage of S3. With Lambda, you only pay for actual compute time used during the file conversion, and S3 provides durable and scalable storage for both .pdf files and .jpg files. The S3 PUT event triggers Lambda to perform conversion, eliminating need to manage infrastructure and scaling, making it most cost-effective solution for this scenario.

upvoted 10 times

Clouddon Most Recent 11 months, 1 week ago

Selected Answer: A

S3 + Lambda provides event-driven, scalable, and serverless file processing.
Best choice for large-scale file upload and transformation workflows.

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: A

Ans A - a simple get and PUT back to S3 bucket. At "...average 5 MB" the returned .jpeg files should be smaller than standard S3

upvoted 1 times

shasi07 1 year, 11 months ago

BeanStack is expensive solution, and dynamoDB have a limitation of 400KB max to upload files. So Lambda an S3 should be the one.

upvoted 1 times

Uzbekistan 2 years, 2 months ago

Given the company's requirement for access to both AWS and on-premises file storage with minimum latency and no significant changes to existing file access patterns, the most suitable option is:

A. Deploy and configure Amazon FSx for Windows File Server on AWS. Move the on-premises file data to FSx for Windows File Server. Reconfigure the workloads to use FSx for Windows File Server on AWS.

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

S3 is the only scalable option for such a large user base in cost effective way.
BCD can work but will be extremely costly

upvoted 1 times

Ruffyit 2 years, 7 months ago

B. Using DynamoDB for storing and processing large .pdf files would not be cost-effective due to storage and throughput costs associated with DynamoDB.

C. Using Elastic Beanstalk with EC2 and EBS storage can work, but it may not be most cost-effective solution. It involves managing the underlying infrastructure and scaling manual

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

Option A is the most cost-effective solution that meets the requirements. Here is why:

Storing the PDFs in Amazon S3 is inexpensive and scalable storage.

Using S3 events to trigger Lambda functions to do the file conversion is a serverless approach that scales automatically. No need to manage EC2 instances.

Lambda usage is charged only for compute time used, which is cost-efficient for spiky workloads like this.

Storing the converted JPGs back in S3 keeps the storage scalable and cost-effective.

upvoted 4 times

RDX19 2 years, 10 months ago

Selected Answer: A

Option A is right answer since Dynamo DB has size limitations.

upvoted 2 times

miki111 2 years, 10 months ago

Option A is the right answer.

upvoted 1 times

Bmarodi 3 years ago

Selected Answer: A

The solution meets these requirements most cost-effectively is option A.
upvoted 1 times

Bmarodi 3 years ago

Selected Answer: A

I think the best solution is A.
Ref. <https://s3.amazonaws.com/doc/s3-developer-guide/RESTObjectPUT.html>
upvoted 1 times

Abrar2022 3 years ago

Since this requires a cost-effect solution then you can use Lambda to convert pdf files to jpeg and store them on S3. Lambda is serverless, so only pay when you use it and automatically scales to cope with demand.
upvoted 1 times

srirajav 3 years, 1 month ago

if Option A is correct, however storing the data back to the same S3, wont it cause infinite looping, it's not best practice right storing a object that is processed by Lambda function to the same S3 bucket, it has chances to cause infinite Loop and then if the option B cant we increase the limits of Dynamo DB requesting AWS?
upvoted 3 times

bedwal2020 3 years, 1 month ago

In question, it is never mentioned that the jpg files will also be stored in same s3 bucket. We can have different s3 buckets right ?
upvoted 2 times

cheese929 3 years, 1 month ago

Selected Answer: A

Answer A is the most cost effective solution that meets the requirement
upvoted 1 times

channn 3 years, 2 months ago

Selected Answer: A

Key words: MOST cost-effectively, so S3 + Lambda
upvoted 1 times

A company has more than 5 TB of file data on Windows file servers that run on premises. Users and applications interact with the data each day.

The company is moving its Windows workloads to AWS. As the company continues this process, the company requires access to AWS and on-premises file storage with minimum latency. The company needs a solution that minimizes operational overhead and requires no significant changes to the existing file access patterns. The company uses an AWS Site-to-Site VPN connection for connectivity to AWS.

What should a solutions architect do to meet these requirements?

- A. Deploy and configure Amazon FSx for Windows File Server on AWS. Move the on-premises file data to FSx for Windows File Server. Reconfigure the workloads to use FSx for Windows File Server on AWS.
- B. Deploy and configure an Amazon S3 File Gateway on premises. Move the on-premises file data to the S3 File Gateway. Reconfigure the on-premises workloads and the cloud workloads to use the S3 File Gateway.
- C. Deploy and configure an Amazon S3 File Gateway on premises. Move the on-premises file data to Amazon S3. Reconfigure the workloads to use either Amazon S3 directly or the S3 File Gateway, depending on each workload's location.
- D. Deploy and configure Amazon FSx for Windows File Server on AWS. Deploy and configure an Amazon FSx File Gateway on premises. Move the on-premises file data to the FSx File Gateway. Configure the cloud workloads to use FSx for Windows File Server on AWS. Configure the on-premises workloads to use the FSx File Gateway. **Most Voted**

Correct Answer: D

Community vote distribution

D (76%)

A (22%)

2

Comments

sba21 **Highly Voted** 3 years, 8 months ago

Selected Answer: D

<https://www.examtopycs.com/discussions/amazon/view/83281-exam-aws-certified-solutions-architect-associate-saa-c02/>
upvoted 26 times

MutiverseAgent 2 years, 11 months ago

Agree answer is D)

Requirements are:

- "Users and applications interact with the data each day"
- "the company requires access to AWS and on-premises file storage with minimum latency"

Explanation: Answer A) will work with the same on-prem <> aws latency as in answer D) as both use the VPN Connection. Having said this, by using an Amazon FSx File Gateway on premise as the D) scenario mentioned, all users will have a great benefit on using the cache that the FSx File Gateway has on their daily workloads. And that is part of the requirements: "users", "each day", "latency"

upvoted 5 times

MrAWS **Highly Voted** 3 years, 4 months ago

D IS WRONG - Its used for caching. you cannot 'Move the on-premises file data to the FSx File Gateway.' which is stated in answer D. It pretty sure AWS employee's are spamming this site with the wrong answers intentionally.

upvoted 20 times

DarthVaper 2 years, 8 months ago

What's the problem with it being a cache? They did say "the company requires access to AWS and on-premises file storage with minimum latency." Not discarding what you said but what's wrong here?

upvoted 4 times

chagantik90 2 years, 6 months ago

you don't move data to gateway, its cached when people use those files from Fsx server

upvoted 1 times

dsshahu01 2 years, 3 months ago

The problem with cache is - It needs to be refreshed, which is an overkill for a migration project

The cache refresh requires another solution since the users/applications interact with it daily which means it does modify often.

upvoted 2 times

Rabbit117 2 years ago

I think D is correct. The FSx File gateway will look like any other Windows file share to the on-prem clients so when they write, or "Copy" data to the share it will be cached locally and also written to the FSx service in the cloud. <https://docs.aws.amazon.com/filegateway/latest/filegateway-fsx-concepts.html>

upvoted 3 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: D

FSx for Windows + FSx File Gateway provides the best hybrid file system for Windows environments with low-latency on-prem access, fully managed cloud storage, and no major app changes.

upvoted 3 times

senrix 1 year ago

Selected Answer: D

D will address the "latency" requirement

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: D

On reconsidering, A might be incorrect. First part of the answer looks ok, but later it says "Configure apps to use FSx in AWS". So that doesn't make sense - what about the on-prem data?

So perhaps it's better to simply move all data to FSx on AWS. I think none of the choices are perfect. So D comes close.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

D is incorrect choice. The documentation describes the FSx File Gateway as:

"FSx for Windows File Server makes all the shares on the server available as shares on your Amazon FSx File Gateway." - so it is simply a fronting for the FSx Windows File Server, not an independent storage FSx system.

So I would choose A as a more appropriate response.

upvoted 1 times

dariar 1 year, 4 months ago

Selected Answer: A

You can't migrate data to FSx File Gateway

Migrate your files to FSx for Windows File Server. For more information, see Migrating existing file storage to FSx for Windows File Server.

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: A

Ans A - it can only be A or D. With D introducing unnecessary operations for an eventual AWS migration

upvoted 1 times

iyiola_daniel 1 year, 9 months ago

Selected Answer: A

The question says the company is moving its workload from windows to AWS. Plus D involves operational overhead.

upvoted 3 times

MatAlves 1 year, 10 months ago

Selected Answer: D

Bad question.

A - Doesn't meet the "low latency" requirement due to VPN connection.

D - It's not possible to "move file data to FSx File Gateway". Instead, it provides improved latency with local caching.

B and C - wrong, since the question requires "no significant changes to the existing file access patterns"

I'd go with D, but not 100% happy with it.

upvoted 5 times

hieunt.hus 1 year, 10 months ago

Selected Answer: A

A. Deploy and configure Amazon FSx for Windows File Server on AWS. Move the on-premises file data to FSx for Windows File Server. Reconfigure the workloads to use FSx for Windows File Server on AWS.

upvoted 2 times

jatric 1 year, 11 months ago

Selected Answer: D

keyword as company continue to migrate workload from om-prem to cloud

upvoted 2 times

Abhi2024 2 years ago

I would go with option A as question mentions "solution that minimizes operational overhead" . Adding FSx File Gateway would add more complexity. They already have Site to Site VPN. This way min change required.

upvoted 2 times

ManikRoy 2 years, 1 month ago

Selected Answer: D

Will go with option D as option A is missing the Fsx File gateway component deployed on-premise which will provide the low latency access.

upvoted 3 times

3680113 2 years ago

Option D is Correct, D provides low latency options for both of premium and the cloud loads.

upvoted 2 times

3680113 2 years ago

on-prem

upvoted 1 times

stormbreaker7 2 years, 1 month ago

Selected Answer: A

min. operational overhead.

upvoted 1 times

Uzbekistan 2 years, 2 months ago

Selected Answer: A

Option D suggests deploying both Amazon FSx for Windows File Server on AWS and an Amazon FSx File Gateway on premises. While this option may provide a solution for accessing on-premises file data in AWS, it introduces additional complexity and potential overhead that may not be necessary given the company's existing AWS Site-to-Site VPN connectivity.

Here are some reasons why Option D may not be the most suitable choice:

Complexity, Cost, Redundancy and Operational Overhead.

upvoted 1 times

Rabbit117 2 years ago

A VPN or DX connection is required for FSx File gateway. Option D provides low mantiance as FSx for Windows is a managed service and the file gateway will provide the low latency access required by the on-prem services.

upvoted 1 times

3680113 2 years ago

Both Amazon FSx file gateway on-prem and Amazon FSx for windows file server goes hand in hand. So there is no way you can use the file gateway on-Prem and not use the Fsx for windows file server. Your conclusion of adding more overhead does not hold. Hence D is the perfect answer.

Configure the cloud workloads to use FSx for Windows File Server on AWS. Configure the on-premises workloads to use the FSx File Gateway.

upvoted 1 times

LIORAGE 2 years, 2 months ago

Selected Answer: D

D: FSX File gateway is nessary for communication between on-premise and aws FSX for window

upvoted 2 times

A hospital recently deployed a RESTful API with Amazon API Gateway and AWS Lambda. The hospital uses API Gateway and Lambda to upload reports that are in PDF format and JPEG format. The hospital needs to modify the Lambda code to identify protected health information (PHI) in the reports.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use existing Python libraries to extract the text from the reports and to identify the PHI from the extracted text.
- B. Use Amazon Textract to extract the text from the reports. Use Amazon SageMaker to identify the PHI from the extracted text.
- C. Use Amazon Textract to extract the text from the reports. Use Amazon Comprehend Medical to identify the PHI from the extracted text. **Most Voted**
- D. Use Amazon Rekognition to extract the text from the reports. Use Amazon Comprehend Medical to identify the PHI from the extracted text.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: C

The correct solution is C: Use Amazon Textract to extract the text from the reports. Use Amazon Comprehend Medical to identify the PHI from the extracted text.

Option C: Using Amazon Textract to extract the text from the reports, and Amazon Comprehend Medical to identify the PHI from the extracted text, would be the most efficient solution as it would involve the least operational overhead. Textract is specifically designed for extracting text from documents, and Comprehend Medical is a fully managed service that can accurately identify PHI in medical text. This solution would require minimal maintenance and would not incur any additional costs beyond the usage fees for Textract and Comprehend Medical.

upvoted 21 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A: Using existing Python libraries to extract the text and identify the PHI from the text would require the hospital to maintain and update the libraries as needed. This would involve operational overhead in terms of keeping the libraries up to date and debugging any issues that may arise.

Option B: Using Amazon SageMaker to identify the PHI from the extracted text would involve additional operational overhead in terms of setting up and maintaining a SageMaker model, as well as potentially incurring additional costs for using SageMaker.

Option D: Using Amazon Rekognition to extract the text from the reports would not be an effective solution, as Rekognition is primarily designed for image recognition and would not be able to accurately extract text from PDF or JPEG files.

upvoted 8 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: C

Use Textract to extract text from medical reports (PDFs, JPEGs) and Comprehend Medical to detect PHI — with minimal code and maximum accuracy

upvoted 2 times

satyaamm 1 year, 5 months ago

Selected Answer: C

Textract is more suitable than Rekognition as it is build to scan text documents and Comprehend helps identify PHI.

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - Textract to 'read' data; Comprehend to assess whether its PHI

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

Textract = Extract text from PDF/images
Comprehend Medical = PHI

ABD are wrong products for this requirement so won't achieve the results

upvoted 3 times

djgodzilla 2 years, 5 months ago

Selected Answer: C

Both Rekognition and Textract possess the ability to detect text within images, yet they are optimized for differing applications.

Rekognition specializes in identifying text located spatially within an image, for instance, words displayed on street signs, t-shirts, or license plates. Its typical use cases encompass visual search, content filtering, deriving insights from content, among others. However, it's not the ideal choice for images containing more than 100 words, as this exceeds its limitation.

On the other hand, Textract is tailored more towards processing documents and PDFs, offering a comprehensive suite for Optical Character Recognition (OCR). It proves useful in scenarios involving financial reports, medical records, receipts, ID documents, and more.

upvoted 4 times

Ruffyit 2 years, 7 months ago

The correct solution is C: Use Amazon Textract to extract the text from the reports. Use Amazon Comprehend Medical to identify the PHI from the extracted text.

Option C: Using Amazon Textract to extract the text from the reports, and Amazon Comprehend Medical to identify the PHI from the extracted text, would be the most efficient solution as it would involve the least operational overhead. Textract is specifically designed for extracting text from documents, and Comprehend Medical is a fully managed service that can accurately identify PHI in medical text. This solution would require minimal maintenance and would not incur any additional costs beyond the usage fees for Textract and Comprehend Medical.

upvoted 2 times

AWSStudyBuddy 2 years, 7 months ago

Selected Answer: C

• Amazon Textract: This program is made to extract text and data from scanned documents, such as pictures and PDFs. It helps to retain the formatting of the report by automatically extracting text while preserving the document's layout.

Identifying and extracting medical information, including protected health information (PHI), from unstructured text is the specialty of Amazon Comprehend Medical. Medical entities that are frequently included in reporting on healthcare, such as ailments, drugs, and more, can be recognized by it.

upvoted 2 times

Chiquitabandita 2 years, 9 months ago

with the choices here, I would go with C, but if offered, I would use amazon textract for the text and use Macie to do the scanning of text files, not comprehend.

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

Here's why:

Amazon Textract has built-in support to extract text from PDFs and images, eliminating the need to build this yourself with Python libraries. Amazon Comprehend Medical has pre-trained machine learning models to identify PHI entities out-of-the-box, avoiding the need to train your own SageMaker model.

Using these fully managed AWS services minimizes operational overhead of maintaining machine learning models yourself.

upvoted 1 times

miki111 2 years, 10 months ago

Option C is the right answer.

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: C

C leverages capabilities of Textract, which is a service that automatically extracts text and data from documents, including PDF and JPEG. By using Textract, hospital can extract text content from reports without need for additional custom code or libraries.

Once text is extracted, hospital can then use Comprehend Medical, a natural language processing service specifically designed for medical text, to analyze and identify PHI. It can recognize medical entities such as medical conditions, treatments, and patient information.

A. suggests using existing Python libraries, which would require hospital to develop and maintain custom code for text extraction and PHI identification.

B and D involve using Textract along with SageMaker or Rekognition, respectively, for PHI identification. While these options could work, they introduce additional complexity by incorporating machine learning models and training.

upvoted 2 times

channn 3 years, 2 months ago

Key word: hospital!

upvoted 1 times

alexiscloud 3 years, 2 months ago

Answer C:

upvoted 1 times

Chirantan 3 years, 5 months ago

Selected Answer: C

Amazon Textract is a machine learning (ML) service that automatically extracts text, handwriting, and data from scanned documents.

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

SONA_M_ 3 years, 5 months ago

WHY OPTION D IS WRONG

upvoted 1 times

s_fun 3 years, 5 months ago

D is wrong only because Amazon Rekognition doesn't read text, only explicit image contents.

upvoted 3 times

mj61 3 years, 4 months ago

B/C you use TextTract to extract text not Rekognition.

upvoted 1 times

A company has an application that generates a large number of files, each approximately 5 MB in size. The files are stored in Amazon S3. Company policy requires the files to be stored for 4 years before they can be deleted. Immediate accessibility is always required as the files contain critical business data that is not easy to reproduce. The files are frequently accessed in the first 30 days of the object creation but are rarely accessed after the first 30 days.

Which storage solution is MOST cost-effective?

- A. Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 Glacier 30 days from object creation. Delete the files 4 years after object creation.
- B. Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) 30 days from object creation. Delete the files 4 years after object creation.
- C. Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days from object creation. Delete the files 4 years after object creation. **Most Voted**
- D. Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days from object creation. Move the files to S3 Glacier 4 years after object creation.

Correct Answer: C

Community vote distribution

C (65%)

B (19%)

A (17%)

Comments

Six_Fingered_Jose **Highly Voted** 3 years, 7 months ago

Selected Answer: C

i think C should be the answer here,
> Immediate accessibility is always required as the files contain critical business data that is not easy to reproduce

If they do not explicitly mention that they are using Glacier Instant Retrieval, we should assume that Glacier -> takes more time to retrieve and may not meet the requirements

upvoted 92 times

JayBee65 3 years, 6 months ago

You can make that assumption, but I think it would be wrong to make it. It does not state they are not using Glacier Instant Retrieval, and it's use would be the logical choice in this question, so I'm going for A

upvoted 8 times

sandordini 2 years, 2 months ago

"With S3 Glacier Instant Retrieval, you can save up to 68% on storage costs compared to using the S3 Standard-Infrequent Access (S3 Standard-IA) storage class, when your data is accessed once per quarter." Repeat: once per quarter! Plus if you keep all the data forever, even if on Glacier, it will cost more over time, than storing on a higher-cost storage for a period of time (here: 4 yrs). A does not delete.

upvoted 3 times

sandordini 2 years, 2 months ago

Sorry! A also deletes... My bad... Still. once a quarter.

upvoted 1 times

syh_rapha 3 years, 6 months ago

I think his assumption is correct because if you go to AWS documentation (<https://aws.amazon.com/s3/storage-classes/glacier/>) they clearly mention: "S3 Glacier Flexible Retrieval (formerly S3 Glacier)". So since this question doesn't specify the S3 Glacier class, then it would default to flexible retrieval (which ofc is not equal to Instant Retrieval).

upvoted 11 times

slackbot 2 years, 9 months ago

why everybody assumed files must be deleted after 4 years. they said files "can" be deleted, and not "must" be deleted. ideally store the files in glacier after 4 years

upvoted 3 times

wearrexdzw3123 2 years, 7 months ago

Because it requires the lowest cost

upvoted 5 times

Kumaran1508 3 years ago

Yeah, Correct answer is C

Because even if you assume the glacier class as Instant Retrieval. As per the Instant Retrieval class the immediate availability is only once per quarter. But in question it is clearly mentioned that the files should be immediately available anytime.

upvoted 7 times

ninjawrz **Highly Voted** 3 years, 7 months ago

Selected Answer: A

Most COST EFFECTIVE

A: S3 Glacier Instant Retrieval is a new storage class that delivers the fastest access to archive storage, with the same low latency and high-throughput performance as the S3 Standard and S3 Standard-IA storage classes. You can save up to 68 percent on storage costs as compared with using the S3 Standard-IA storage class when you use the S3 Glacier Instant Retrieval storage class and pay a low price to retrieve data.

upvoted 31 times

wearrexdzw3123 2 years, 7 months ago

Glaciers usually take some time to retrieve

upvoted 1 times

babayomi 1 year, 9 months ago

COST EFFECTIVE IS THE DIAL WORD HERE. Answer is A

upvoted 2 times

123jh10 3 years, 7 months ago

I think A is the answer, too. As S3 Glacier has 3 different classes and one of them is able to retrieve objects in milliseconds (Instant Retrieval). As no class has been specified, we can assume the one that meets requirements is selected.

upvoted 6 times

GasGasJDM 3 years, 7 months ago

I agree with this, since no specific information about Glacier modality, we can assume that Instant Retrieval is a possibility and as far as I know, this is cheaper than IA.

upvoted 1 times

RBSK 3 years, 6 months ago

And we have a requirement (Most cost effective solution) hence A is the winner

upvoted 2 times

wh1t4k3r 3 years, 6 months ago

In the other hand, you need to choose a tier when going for glacier, so my previous comment is not stating well. The question is tricky, I change my mind: agree with you on this one

upvoted 2 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: C

Use S3 Standard for 30 days, S3 Standard-IA after that, and retain objects for 4 years, then delete.

upvoted 1 times

bora4motion 1 year, 1 month ago

Selected Answer: C

"Immediate accessibility is always required " and then "Cost effective"- that leaves us with C.

upvoted 1 times

bora4motion 1 year, 1 month ago

Selected Answer: C

I'm going with C - the files might be required instantly even if rarely accessed.

upvoted 1 times

KeshavP 1 year, 3 months ago

Selected Answer: C

S3 Glacier Instant Retrieval is more expensive than S3 Standard-IA for both storage and retrieval requests.

S3 Standard-IA is 68% cheaper for storage than S3 Standard, while S3 Glacier Instant Retrieval is only 46% cheaper

upvoted 1 times

Vandaman 1 year, 3 months ago

Selected Answer: A

Answer is A. Files are rarely accessed after 30 days and they require the MOST cost effective solution. Then the answer is Glacier.

upvoted 1 times

kyd0nix 1 year, 4 months ago

Selected Answer: C

If Glacier class is not specified, I could assume it is referring to S3 Glacier Flexible Retrieval (formerly S3 Glacier). Then, the retrieval of data is not immediate as requested. The best option would be glacier Instant Retrieval, but it's not specified... so the best answer that fits is C.

upvoted 1 times

babyalttabs 1 year, 4 months ago

Selected Answer: C

C is the answer. While both "S3 Standard-Infrequent Access" and "S3 One Zone-Infrequent Access" are storage classes on AWS S3 designed for data accessed infrequently, the key difference is that "S3 One Zone-IA" stores data in a single Availability Zone, making it cheaper but with a higher risk of data loss if that zone fails, whereas "S3 Standard-IA" spreads data across multiple Availability Zones for increased redundancy and availability; making it a better choice for critical data that needs to be readily accessible even if rarely accessed.

upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: C

S3 IA is the most suitable here.

upvoted 1 times

PSH123 1 year, 6 months ago

Selected Answer: B

onezone IA is cheap more than standard IA

upvoted 4 times

0de7d1b 1 year, 6 months ago

Option C is the most cost-effective solution because it provides immediate access for the first 30 days and optimizes for cost by transitioning the file to S3 Standard-IA after that period. It also complies with the company's retention policy.

upvoted 1 times

PaPi_z 1 year, 8 months ago

Selected Answer: C

Immediate accessibility

upvoted 3 times

anminfang 1 year, 9 months ago

It should be C. The critical point is Immediate accessibility, only S3-IA can provide this.

upvoted 2 times

MatAlves 1 year, 10 months ago

Selected Answer: C

"Immediate accessibility is always required". It can't be A, since Glacier Instant Retrieval has a limit of times the data can be accessed.

"S3 Glacier Retrieval is the lowest cost archive storage with milliseconds retrieval for rarely accessed data. It is ideal for data that is accessed once or twice per quarter, and that requires immediate access."

upvoted 2 times

MatAlves 1 year, 10 months ago

If you do so more than ~4 times per year, you would pay more for Glacier Instant Retrieval than for S3 Standard.

upvoted 1 times

jatric 1 year, 11 months ago

Selected Answer: C

immediately accessible for 4 years but rarely use - S3 standard IA.

Correct answer is C

upvoted 2 times

MoSaleh8890 1 year, 4 months ago

Why not A, with Glacier Instant Retrieval?

upvoted 1 times

Duckydoo 2 years ago

Selected Answer: C

Keep in mind that "Glacier Flexible Retrieval" was previously named just "Glacier" (before they added Instant Retrieval). Regardless, they should have made it clear which Glacier class they are referring to since which one you select/assume can impact the correctness.

upvoted 3 times

A company hosts an application on multiple Amazon EC2 instances. The application processes messages from an Amazon SQS queue, writes to an Amazon RDS table, and deletes the message from the queue. Occasional duplicate records are found in the RDS table. The SQS queue does not contain any duplicate messages.

What should a solutions architect do to ensure messages are being processed once only?

- A. Use the CreateQueue API call to create a new queue.
- B. Use the AddPermission API call to add appropriate permissions.
- C. Use the ReceiveMessage API call to set an appropriate wait time.
- D. Use the ChangeMessageVisibility API call to increase the visibility timeout. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

KVK16 **Highly Voted** 3 years, 7 months ago

Selected Answer: D

In case of SQS - multi-consumers if one consumer has already picked the message and is processing, in meantime other consumer can pick it up and process the message there by two copies are added at the end. To avoid this the message is made invisible from the time its picked and deleted after processing. This visibility timeout is increased according to max time taken to process the message

upvoted 53 times

JayBee65 3 years, 6 months ago

To add to this "The VisibilityTimeout in SQS is a time frame that the message can be hidden so that no others can consume it except the first consumer who calls the ReceiveMessageAPI." The API ChangeMessageVisibility changes this value.

upvoted 21 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: D

To ensure that messages are being processed only once, a solutions architect should use the ChangeMessageVisibility API call to increase the visibility timeout which is Option D.

The visibility timeout determines the amount of time that a message received from an SQS queue is hidden from other consumers while the message is being processed. If the processing of a message takes longer than the visibility timeout, the message will become visible to other consumers and may be processed again. By increasing the visibility timeout, the solutions architect can ensure that the message is not made visible to other consumers until the processing is complete and the message can be safely deleted from the queue.

Option A (Use the CreateQueue API call to create a new queue) would not address the issue of duplicate message processing.

Option B (Use the AddPermission API call to add appropriate permissions) is not relevant to this issue.

Option C (Use the ReceiveMessage API call to set an appropriate wait time) is also not relevant to this issue.

upvoted 18 times

karbob 3 years, 5 months ago

not relevant to this issue. ??? what is added value

upvoted 3 times

Buruguduystunstugudunstuy 3 years, 3 months ago

Option B (Use the AddPermission API call to add appropriate permissions) is not relevant to this issue because it deals with setting permissions for accessing an SQS queue, which is not related to preventing duplicate records in the RDS table.

Option C (Use the ReceiveMessage API call to set an appropriate wait time) is not relevant to this issue because it is related to configuring how long the ReceiveMessage API call should wait for new messages to arrive in the SQS queue before returning an empty response. It does not address the issue of duplicate records in the RDS table.

upvoted 8 times

Clouddon Most Recent 11 months, 1 week ago

Selected Answer: D

To prevent duplicate processing of SQS messages, ensure that the visibility timeout is long enough to cover entire processing duration. Use the `ChangeMessageVisibility` API to adjust it dynamically if needed.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: D

I chose D as well, I understand the rationale behind it. But I think it is named incorrectly. Should this not be "InvisibilityTimeout" instead of "VisibilityTimeout"?

If the message becomes visible again after a timeout, because SQS assumes it failed processing - then it remains invisible and after timeout it becomes visible again... So that's where my point of contention is.

upvoted 2 times

Rcosmos 1 year, 5 months ago

Selected Answer: D

Explicação:

Causa provável do problema:

O problema ocorre porque o tempo de visibilidade (visibility timeout) configurado na fila do Amazon SQS é muito curto.

Quando uma instância EC2 lê uma mensagem da fila e começa a processá-la, se o processamento não for concluído antes que o tempo de visibilidade expire, a mensagem volta para a fila e pode ser processada novamente por outra instância. Isso leva a registros duplicados na tabela RDS.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: D

Visibility timeout is used to manage multiple users.

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: D

Ans D - once the message is selected in the queue 'hide' it from other users/apps so it is only processed once, using API 'ChangeMessageVisibility'

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

AB: Irrelevant

C: This is for long polling not for execution <https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-short-and-long-polling.html#sqs-long-polling>

D: Visibility is correct fix issue because over here other SQS clients are seeing the same message back in the queue when the previous processor is taking longer than expected to process the message

upvoted 1 times

Subhrangsu 2 years, 8 months ago

I also opt for D, but asking does increasing MessageVisibilityTimeout good always?

upvoted 2 times

awsgeek75 2 years, 4 months ago

This parameter is the timeout after which the SQS assumes that the processing has failed and makes the item visible to other queue processors. It is normal to increase the timeout if it takes longer to process items.

upvoted 1 times

miki111 2 years, 10 months ago

Option D is the right answer.

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: D

The visibility timeout is the duration during which SQS prevents other consumers from receiving and processing the same message. By increasing the visibility timeout, you allow more time for the processing of a message to complete before it becomes visible to other consumers.

Option A, creating a new queue, does not address the issue of concurrent processing and duplicate records. It would only create a new queue, which is not necessary for solving the problem.

Option B, adding permissions, also does not directly address the issue of duplicate records. Permissions are necessary for accessing the SQS queue but not for preventing concurrent processing.

Option C, setting an appropriate wait time using the ReceiveMessage API call, does not specifically prevent duplicate records. It can help manage the rate at which messages are received from the queue but does not address the issue of concurrent processing.

upvoted 4 times

cheese929 3 years, 1 month ago

Selected Answer: D

D is correct

upvoted 1 times

alexiscloud 3 years, 2 months ago

Answer D:

visibility timeout beings when amazon SQS return a message

upvoted 1 times

test_devops_aws 3 years, 2 months ago

Selected Answer: D

D = ChangeMessageVisibility

upvoted 1 times

dev1978 3 years, 4 months ago

In theory, between reception and changing visibility, you can have multiple consumers. Question is not good as it won't guarantee not executing twice.

upvoted 1 times

techhb 3 years, 5 months ago

Selected Answer: D

Increaseing visibility timeout makes sure message is not visible for time taken to process the message.

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

A solutions architect is designing a new hybrid architecture to extend a company's on-premises infrastructure to AWS. The company requires a highly available connection with consistent low latency to an AWS Region. The company needs to minimize costs and is willing to accept slower traffic if the primary connection fails.

What should the solutions architect do to meet these requirements?

- A. Provision an AWS Direct Connect connection to a Region. Provision a VPN connection as a backup if the primary Direct Connect connection fails. **Most Voted**
- B. Provision a VPN tunnel connection to a Region for private connectivity. Provision a second VPN tunnel for private connectivity and as a backup if the primary VPN connection fails.
- C. Provision an AWS Direct Connect connection to a Region. Provision a second Direct Connect connection to the same Region as a backup if the primary Direct Connect connection fails.
- D. Provision an AWS Direct Connect connection to a Region. Use the Direct Connect failover attribute from the AWS CLI to automatically create a backup connection if the primary Direct Connect connection fails.

Correct Answer: A

Community vote distribution

A (93%)

5%

Comments

KVK16 **Highly Voted** 3 years, 7 months ago

Selected Answer: A

Direct Connect + VPN best of both
upvoted 21 times

mabotega **Highly Voted** 3 years, 7 months ago

Selected Answer: A

Direct Connect goes through 1 Gbps, 10 Gbps or 100 Gbps and the VPN goes up to 1.25 Gbps.

<https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/aws-direct-connect-vpn.html>

upvoted 19 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: A

Use Direct Connect for primary low-latency connection, and a VPN backup for cost-effective failover.
upvoted 2 times

akeed 1 year, 2 months ago

Selected Answer: D

Highly available means the system is up and running most of the time, not 100% of the time. Therefore, minor outages are acceptable.
upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

Since the direct connect takes a lot of time to setup and is much more expensive, the point of consideration is - if those are not in the criteria, we assume Direct connection is alright.
upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: A

And A - as primary use Direct Connect to provide a solid connection. Company is not too worried about backup (other than it works) so use cheaper VPN Site-to-Site.

upvoted 3 times

48cd959 2 years, 2 months ago

Selected Answer: A

Ans A-

Direct Connect because company needs consistent connection.

As a back up, company wants cheaper solution so VPN Site to Site connection should be okay.

upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

HA low latency + minimize cost + acceptable slow traffic if primary fails

B: VPN tunnel will be slow

C: 2 direct connect will be expensive

D: Backup connection for what?

A: Direct connect + VPN as a backup works

upvoted 6 times

Ruffyt 2 years, 7 months ago

A highly available connection with consistent low latency = AWS Direct Connect

Minimize costs and accept slower traffic if the primary connection fails = VPN connection

upvoted 6 times

benacert 2 years, 9 months ago

A is the right choice to save cost

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

Highly available connectivity using Direct Connect for consistent low latency and high throughput.

Cost optimization by using a VPN as a slower, lower cost backup for when Direct Connect fails.

Automatic failover to the VPN when Direct Connect fails.

upvoted 3 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: A

A highly available connection with consistent low latency = AWS Direct Connect

Minimize costs and accept slower traffic if the primary connection fails = VPN connection

upvoted 1 times

hsinchang 2 years, 10 months ago

Selected Answer: A

Slower traffic when primary fails, so the backup plan needs a cheaper solution, and the primary requires high performance, so A.

upvoted 1 times

oguzbeliren 2 years, 10 months ago

Even though, there are a lots of variable affecting the cost of the connection, VPN connection is cheaper than the Direct Connect most of the time since VPN Connection doesn't require any dedicated physical circuit involved.

upvoted 1 times

miki111 2 years, 10 months ago

Option A is the right answer.

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: A

Options B and C propose using multiple VPN connections for private connectivity and as backups. While VPNs can serve as backups, they may not provide the same level of consistent low latency and high availability as Direct Connect connections. Additionally, provisioning multiple VPN tunnels can increase operational complexity and costs.

Option D suggests using the Direct Connect failover attribute from the AWS CLI to automatically create a backup connection if the primary Direct Connect connection fails. While this approach can be automated, it does not provide the same level of immediate failover capabilities as having a separate backup connection in place.

Therefore, option A, provisioning an AWS Direct Connect connection to a Region and provisioning a VPN connection as a backup, is the most suitable solution that meets the company's requirements for connectivity, cost-effectiveness, and high availability.

upvoted 4 times

th3k33n 3 years, 1 month ago

Selected Answer: A

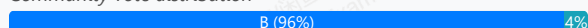
highly available - > direct connect because connection can go up to 10GBPs and VPN 1.5GBPs as backup
upvoted 2 times

A company is running a business-critical web application on Amazon EC2 instances behind an Application Load Balancer. The EC2 instances are in an Auto Scaling group. The application uses an Amazon Aurora PostgreSQL database that is deployed in a single Availability Zone. The company wants the application to be highly available with minimum downtime and minimum loss of data. Which solution will meet these requirements with the LEAST operational effort?

- A. Place the EC2 instances in different AWS Regions. Use Amazon Route 53 health checks to redirect traffic. Use Aurora PostgreSQL Cross-Region Replication.
- B. Configure the Auto Scaling group to use multiple Availability Zones. Configure the database as Multi-AZ. Configure an Amazon RDS Proxy instance for the database. **Most Voted**
- C. Configure the Auto Scaling group to use one Availability Zone. Generate hourly snapshots of the database. Recover the database from the snapshots in the event of a failure.
- D. Configure the Auto Scaling group to use multiple AWS Regions. Write the data from the application to Amazon S3. Use S3 Event Notifications to launch an AWS Lambda function to write the data to the database.

Correct Answer: B

Community vote distribution



Comments

SilentMilli **Highly Voted** 3 years, 5 months ago

Selected Answer: B

By configuring the Auto Scaling group to use multiple Availability Zones, the application will be able to continue running even if one Availability Zone goes down. Configuring the database as Multi-AZ will also ensure that the database remains available in the event of a failure in one Availability Zone. Using an Amazon RDS Proxy instance for the database will allow the application to automatically route traffic to healthy database instances, further increasing the availability of the application. This solution will meet the requirements for high availability with minimal operational effort.

upvoted 37 times

KVK16 **Highly Voted** 3 years, 7 months ago

Selected Answer: B

RDS Proxy for Aurora <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 8 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: B

Multi-AZ EC2 + Multi-AZ Aurora + RDS Proxy = High availability + minimal downtime + low operational overhead.

upvoted 1 times

AshishDhole 1 year, 4 months ago

Selected Answer: B

Highly Available = Multi AZ

DR = Multiple Region

upvoted 3 times

Rcosmos 1 year, 5 months ago

Selected Answer: B

Conclusão:

A configuração B oferece alta disponibilidade tanto para as instâncias do EC2 quanto para o banco de dados Aurora PostgreSQL com o mínimo esforço operacional. Isso atende aos requisitos de tempo de inatividade mínimo e perda mínima de dados.

upvoted 1 times

PaulGa 1 year, 9 months ago

Selected Answer: B

Ans B - good explanation by SilentMilli (1yr 8mth ago). The database Proxy is key.
upvoted 2 times

SaurabhTiwari1 1 year, 9 months ago

Selected Answer: B

Keyword - Highly available
Use multiple availability zones
upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

A: Different region doesn't help
C: Would have made sense if it wasn't restricting to one AZ.
D: Regions + S3 + Lambda = Operational effort extreme
B: Although not entirely sure how RDS Proxy helps because it is for connection pooling but it is the only workable solution using multi AZ
upvoted 3 times

dkw2342 2 years, 3 months ago

One of the benefits of Amazon RDS Proxy is that it can improve application recovery time after database failovers. While RDS Proxy supports both MySQL as well as PostgreSQL engines, in this post, we will use a MySQL test workload to demonstrate how RDS Proxy reduces client recovery time after failover by up to 79% for Amazon Aurora MySQL and by up to 32% for Amazon RDS for MySQL.

-> contributes to minimum downtime req
upvoted 1 times

dkw2342 2 years, 3 months ago

PS: <https://aws.amazon.com/blogs/database/improving-application-availability-with-amazon-rds-proxy/>
upvoted 1 times

MiniYang 2 years, 6 months ago

Selected Answer: A

The company wants to minimize costs and is willing to accept slower traffic if the primary connection fails, it may be tempted to choose a VPN connection as a backup, in which case the answer is A. Cost-Effectiveness: VPN connections are generally more economical than AWS Direct Connect especially for low to moderate bandwidth needs.
Backup connection: A VPN connection can serve as a more cost-effective backup if the primary Direct Connect connection fails, even if it may be slower. Acceptance of slower traffic: The question clearly states that the company is willing to accept slower traffic if the primary connection fails, which implies a tolerance for connection speeds.
upvoted 1 times

pentium75 2 years, 5 months ago

This is about the previous question ;)
upvoted 6 times

asulhi 2 years, 8 months ago

Selected Answer: B

ASG and MultiAZ is the best answer
upvoted 1 times

benacert 2 years, 9 months ago

B is the right answer
upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: B

Option B requires the least operational effort to meet the high availability and minimum downtime/data loss requirements.

The key points are:

Use an Auto Scaling group across multiple AZs for high availability of the EC2 instances.
Configure the Aurora DB as Multi-AZ for high availability, automatic failover, and minimum data loss.
Use RDS Proxy for connection pooling to the DB for performance
upvoted 2 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: B

Highly available, Minimum downtime and Minimum loss of data = Auto Scaling group on Multi-AZ, Database on Multi-AZ, Amazon RDS Proxy.
upvoted 1 times

miki111 2 years, 10 months ago

Option B is the right answer.

upvoted 1 times

hiepdz98 2 years, 11 months ago

Selected Answer: B

B is correct answer

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: B

A. This approach provides geographic redundancy, it introduces additional complexity and operational effort, including managing replication, handling latency, and potentially higher data transfer costs.

C. While snapshots can be used for data backup and recovery, they do not provide real-time failover capabilities and can result in significant data loss if a failure occurs between snapshots.

D. While this approach offers some decoupling and scalability benefits, it adds complexity to the data flow and introduces additional overhead for data processing.

In comparison, option B provides a simpler and more streamlined solution by utilizing multiple AZs, Multi-AZ configuration for the database, and RDS Proxy for improved connection management. It ensures high availability, minimal downtime, and minimum loss of data with the least operational effort.

upvoted 6 times

Abrar2022 3 years ago

@Wajif the reason why it's not A is because the question mentions High availability and nothing to do with region. You can achieve HA without spanning multiple regions. Also B is incorrect because ALB are region specific and span across multiple AZ with that specific region (not cross region).

upvoted 1 times

A company's HTTP application is behind a Network Load Balancer (NLB). The NLB's target group is configured to use an Amazon EC2 Auto Scaling group with multiple EC2 instances that run the web service.

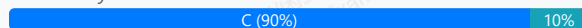
The company notices that the NLB is not detecting HTTP errors for the application. These errors require a manual restart of the EC2 instances that run the web service. The company needs to improve the application's availability without writing custom scripts or code.

What should a solutions architect do to meet these requirements?

- A. Enable HTTP health checks on the NLB, supplying the URL of the company's application.
- B. Add a cron job to the EC2 instances to check the local application's logs once each minute. If HTTP errors are detected, the application will restart.
- C. Replace the NLB with an Application Load Balancer. Enable HTTP health checks by supplying the URL of the company's application. Configure an Auto Scaling action to replace unhealthy instances. **Most Voted**
- D. Create an Amazon Cloud Watch alarm that monitors the UnhealthyHostCount metric for the NLB. Configure an Auto Scaling action to replace unhealthy instances when the alarm is in the ALARM state.

Correct Answer: C

Community vote distribution



Comments

123jhl0 **Highly Voted** 3 years, 7 months ago

Selected Answer: C

I would choose A, as NLB supports HTTP and HTTPS Health Checks, BUT you can't put any URL (as proposed), only the node IP addresses. So, the solution is C.

upvoted 32 times

Ack3rman 3 years, 6 months ago

can you elaborate more pls

upvoted 3 times

BlueVolcano1 3 years, 4 months ago

NLBs support HTTP, HTTPS and TCP health checks:

<https://docs.aws.amazon.com/elasticloadbalancing/latest/network/target-group-health-checks.html> (check HealthCheckProtocol)

But NLBs only accept either selecting EC2 instances or IP addresses directly as targets. You can't provide a URL to your endpoints, only a health check path (if you're using HTTP or HTTPS health checks).

upvoted 11 times

km142646 3 years, 1 month ago

What's the difference between endpoint URL and health check path?

upvoted 2 times

majubmo 3 years ago

A URL includes the hostname. The health check path is only the path portion. For example,

URL = <https://i-0123456789abcdef.us-west-2.compute.internal/index.html>

health check path = /index.html

upvoted 15 times

jerry1 1 year, 2 months ago

I think even http health check is feasible, option A does not provide any suggestion when health check fails

upvoted 1 times

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: C

Option C. NLB works at Layer 4 so it does not support HTTP/HTTPS. The replacement for the ALB is the best choice.
upvoted 20 times

BlueVolcano1 3 years, 4 months ago

That's incorrect. NLB does support HTTP and HTTPS (and TCP) health checks.
<https://docs.aws.amazon.com/elasticloadbalancing/latest/network/target-group-health-checks.html>

There just isn't an answer option that reflects that. My guess is that the question and/or answer options are outdated.
upvoted 8 times

Agatsume 1 year ago

NLB doesn't support HTTP health checks..only TCP
upvoted 1 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: C

Switching to an ALB with HTTP health checks enables application-level monitoring and automated recovery, making it the best solution with minimal operational overhead.
upvoted 2 times

AwsAbhiKumar 1 year, 4 months ago

Selected Answer: C

NLB only supports TCP-level health checks, it cannot detect HTTP errors
upvoted 2 times

satyaammm 1 year, 5 months ago

Selected Answer: C

ALB is more suitable for HTTP requests.
upvoted 3 times

PaulGa 1 year, 9 months ago

Selected Answer: C

Ans C - Yup, something's happening at the Application level so replace NLB with ALB
upvoted 4 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

NLB is for network errors and low level traffic stuff
ALB is for application so C is the only realistic option here
upvoted 9 times

kel2023 2 years, 5 months ago

Selected Answer: A

NLB does support HTTP/HTTPS Health Checks.

I saw other people's comments, it seems like the question was rephrased. The comments were highlighting "application URL", but I don't see words on the question.
upvoted 3 times

ignajtpolandstrong 2 years, 5 months ago

Selected Answer: C

You can use HTTP/HTTPS ONLY when Target is ALB.
By default it is TCP.
<https://docs.aws.amazon.com/elasticloadbalancing/latest/network/target-group-health-checks.html#health-check-settings>
HealthCheckProtocol
The protocol the load balancer uses when performing health checks on targets. The possible protocols are HTTP, HTTPS, and TCP. The default is the TCP protocol. If the target type is ALB, the supported health check protocols are HTTP and HTTPS.
upvoted 4 times

tom_cruise 2 years, 8 months ago

Selected Answer: C

ALB allows you to specify the path which helps to check the error. NLB cannot do that.
upvoted 4 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

The key points are:

Use an Application Load Balancer (ALB) instead of a Network Load Balancer (NLB) since ALBs support HTTP health checks.

Configure HTTP health checks on the ALB to monitor the application health.

Use an Auto Scaling action triggered by the ALB health checks to automatically replace unhealthy instances.

upvoted 4 times

miki111 2 years, 10 months ago

Option C is the right answer.

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

A. NLB, but NLB's health checks are designed for TCP/UDP protocols and lack the advanced features specific to HTTP applications provided by ALB.

B. This approach involves custom scripting and manual intervention, which contradicts the requirement of not writing custom scripts or code.

D. Since the NLB does not detect HTTP errors, relying solely on the UnhealthyHostCount metric may not accurately capture the health of the application instances.

Therefore, C is the recommended choice for improving the application's availability without custom scripting or code. By replacing the NLB with an ALB, enabling HTTP health checks, and configuring Auto Scaling to replace unhealthy instances, the company can ensure that only healthy instances are serving traffic, enhancing the application's availability automatically.

upvoted 10 times

Gizmo2022 1 year, 6 months ago

Thank you for explaining this, I'm currently studying for the exam and your explanation was helpful.

upvoted 1 times

Abrar2022 3 years ago

Replace the NLB (layer 4 udp and tcp) with an Application Load Balancer - ALB (layer 7) supports http and https requests.

upvoted 1 times

datz 3 years, 2 months ago

Selected Answer: C

must be C

Application availability: NLB cannot assure the availability of the application. This is because it bases its decisions solely on network and TCP-layer variables and has no awareness of the application at all. Generally, NLB determines availability based on the ability of a server to respond to ICMP ping or to correctly complete the three-way TCP handshake. ALB goes much deeper and is capable of determining availability based on not only a successful HTTP GET of a particular page but also the verification that the content is as was expected based on the input parameters.

upvoted 1 times

datz 3 years, 2 months ago

Also A doesn't offer what bellow in C offers...

Configure an Auto Scaling action to replace unhealthy instances

upvoted 1 times

Tony1980 3 years, 4 months ago

Answer is C

A solution architect can use Amazon EC2 Auto Scaling health checks to automatically detect and replace unhealthy instances in the EC2 Auto Scaling group. The health checks can be configured to check the HTTP errors returned by the application and terminate the unhealthy instances. This will ensure that the application's availability is improved, without requiring custom scripts or code.

upvoted 1 times

aakashkumar1999 3 years, 4 months ago

I will go with A as Network load balancer supports HTTP and HTTPS health checks, maybe the answer is outdated.

upvoted 2 times

pentium75 2 years, 5 months ago

But you'd need to check the health of the individual nodes, NOT "the URL of the company's application" which points to the Load Balancer.

upvoted 2 times

A company runs a shopping application that uses Amazon DynamoDB to store customer information. In case of data corruption, a solutions architect needs to design a solution that meets a recovery point objective (RPO) of 15 minutes and a recovery time objective (RTO) of 1 hour.

What should the solutions architect recommend to meet these requirements?

- A. Configure DynamoDB global tables. For RPO recovery, point the application to a different AWS Region.
- B. Configure DynamoDB point-in-time recovery. For RPO recovery, restore to the desired point in time. **Most Voted**
- C. Export the DynamoDB data to Amazon S3 Glacier on a daily basis. For RPO recovery, import the data from S3 Glacier to DynamoDB.
- D. Schedule Amazon Elastic Block Store (Amazon EBS) snapshots for the DynamoDB table every 15 minutes. For RPO recovery, restore the DynamoDB table by using the EBS snapshot.

Correct Answer: B

Community vote distribution

B (100%)

Comments

123jh10 **Highly Voted** 3 years, 7 months ago

Selected Answer: B

A - DynamoDB global tables provides multi-Region, and multi-active database, but it not valid "in case of data corruption". In this case, you need a backup. This solutions isn't valid.

****B**** - Point in Time Recovery is designed as a continuous backup juts to recover it fast. It covers perfectly the RPO, and probably the RTO.

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/PointInTimeRecovery.html>

C - A daily export will not cover the RPO of 15min.

D - DynamoDB is serverless... so what are these EBS snapshots taken from???

upvoted 50 times

LionelSid 3 years, 4 months ago

Yes, it is possible to take EBS snapshots of a DynamoDB table. The process for doing this involves the following steps:

Create a new Amazon Elastic Block Store (EBS) volume from the DynamoDB table.

Stop the DynamoDB service on the instance.

Detach the EBS volume from the instance.

Create a snapshot of the EBS volume.

Reattach the EBS volume to the instance.

Start the DynamoDB service on the instance.

You can also use AWS Data pipeline to automate the above process and schedule regular snapshots of your DynamoDB table.

Note that, if your table is large and you want to take a snapshot of it, it could take a long time and consume a lot of bandwidth, so it's recommended to use the Global Tables feature from DynamoDB in order to have a Multi-region and Multi-master DynamoDB table, and you can snapshot each region separately.

upvoted 4 times

piavik 3 years, 2 months ago

What is "DynamoDB service on the instance" ?

upvoted 3 times

pentium75 2 years, 5 months ago

DynamoDB is a native cloud service, there is no "instance" that you could "stop", or detach an "EBS volume" from.

upvoted 2 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: B

The best solution to meet the RPO and RTO requirements would be to use DynamoDB point-in-time recovery (PITR). This feature allows you to restore your DynamoDB table to any point in time within the last 35 days, with a granularity of seconds. To recover data within a 15-minute RPO, you would simply restore the table to the desired point in time within the last 35 days.

To meet the RTO requirement of 1 hour, you can use the DynamoDB console, AWS CLI, or the AWS SDKs to enable PITR on your table. Once enabled, PITR continuously captures point-in-time copies of your table data in an S3 bucket. You can then use these point-in-time copies to restore your table to any point in time within the retention period.

CORRECT

Option B. Configure DynamoDB point-in-time recovery. For RPO recovery, restore to the desired point in time.

upvoted 15 times

Buruguduystunstugudunstuy 3 years, 5 months ago

WRONG

Option A (configuring DynamoDB global tables) would not meet the RPO requirement, as global tables are designed to replicate data to multiple regions for high availability, but they do not provide a way to restore data to a specific point in time.

Option C (exporting data to S3 Glacier) would not meet the RPO or RTO requirements, as S3 Glacier is a cold storage service with a retrieval time of several hours.

Option D (scheduling EBS snapshots) would not meet the RPO requirement, as EBS snapshots are taken on a schedule, rather than continuously. Additionally, restoring a DynamoDB table from an EBS snapshot can take longer than 1 hour, so it would not meet the RTO requirement.

upvoted 7 times

satyaamm Most Recent 1 year, 5 months ago

Selected Answer: B

DynamoDB Point-In-Time-Recovery is the most suitable option here for recovering an object.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - just use the built-in DynamoDB PITR... simples...

upvoted 1 times

ManikRoy 2 years, 1 month ago

Selected Answer: B

With point-in-time recovery, you can restore that table to any point in time during the last 35 days. After you enable point-in-time recovery, you can restore to any point in time from five minutes before the current time until 35 days ago. DynamoDB maintains incremental backups of your table.

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

A: Scalability across regions which is not required

C: Glacier exports and backup restore won't meet 1 hour RPO time

D EBS for DynamoDB table? Sounds impractical

B: DynamoDB point-in-time recovery is for this scenario.

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/PointInTimeRecovery.html>

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: B

The best option to meet the RPO of 15 minutes and RTO of 1 hour is B) Configure DynamoDB point-in-time recovery. For RPO recovery, restore to the desired point in time.

The key points:

DynamoDB point-in-time recovery can restore to any point in time within the last 35 days. This supports an RPO of 15 minutes.

Restoring from a point-in-time backup meets the 1 hour RTO.

Point-in-time recovery is specifically designed to restore DynamoDB tables with second-level granularity.

upvoted 1 times

cookieMr 2 years, 11 months ago

A. Global tables provide multi-region replication for disaster recovery purposes, they may not meet the desired RPO of 15 minutes without additional configuration and potential data loss.

C. Exporting and importing data on a daily basis does not align with the desired RPO of 15 minutes.

D. EBS snapshots can be used for data backup, they are not directly applicable to DynamoDB and cannot provide the desired RPO and RTO without custom implementation.

In comparison, option B utilizing DynamoDB's built-in point-in-time recovery functionality provides the most straightforward and effective solution for meeting the specified RPO of 15 minutes and RTO of 1 hour. By enabling PITR and restoring the table to the desired point in time, the company can recover the customer information with minimal data loss and within the required time frame.

upvoted 3 times

Abrar2022 3 years ago

The answer is in the question. Read the question again!!! Option B. Configure DynamoDB point-in-time recovery. For RPO recovery, restore to the desired point in time.

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

Shasha1 3 years, 6 months ago

B is correct

DynamoDB point-in-time recovery allows the solutions architect to recover the DynamoDB table to a specific point in time, which would meet the RPO of 15 minutes. This feature also provides an RTO of 1 hour, which is the desired recovery time objective for the application. Additionally, configuring DynamoDB point-in-time recovery does not require any additional infrastructure or operational effort, making it the best solution for this scenario.

Option D is not correct because scheduling Amazon EBS snapshots for the DynamoDB table every 15 minutes would not meet the RPO or RTO requirements. While EBS snapshots can be used to recover data from a DynamoDB table, they are not designed to provide real-time data protection or recovery capabilities

upvoted 1 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 1 times

SimonPark 3 years, 7 months ago

Selected Answer: B

B is the answer

upvoted 1 times

BoboChow 3 years, 7 months ago

Selected Answer: B

I think DynamoDB global tables also work here, but Point in Time Recovery is a better choice

upvoted 1 times

Kikiokiki 3 years, 7 months ago

I THINK B.

<https://dynobase.dev/dynamodb-point-in-time-recovery/>

upvoted 1 times

priya2224 3 years, 7 months ago

answer is D

upvoted 1 times

123jhl0 3 years, 7 months ago

DynamoDB is serverless, so no storage snapshots available. <https://aws.amazon.com/dynamodb/>

upvoted 2 times

A company runs a photo processing application that needs to frequently upload and download pictures from Amazon S3 buckets that are located in the same AWS Region. A solutions architect has noticed an increased cost in data transfer fees and needs to implement a solution to reduce these costs.

How can the solutions architect meet this requirement?

- A. Deploy Amazon API Gateway into a public subnet and adjust the route table to route S3 calls through it.
- B. Deploy a NAT gateway into a public subnet and attach an endpoint policy that allows access to the S3 buckets.
- C. Deploy the application into a public subnet and allow it to route through an internet gateway to access the S3 buckets.
- D. Deploy an S3 VPC gateway endpoint into the VPC and attach an endpoint policy that allows access to the S3 buckets.

Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: D

CORRECT

The correct answer is Option D. Deploy an S3 VPC gateway endpoint into the VPC and attach an endpoint policy that allows access to the S3 buckets

By deploying an S3 VPC gateway endpoint, the application can access the S3 buckets over a private network connection within the VPC, eliminating the need for data transfer over the internet. This can help reduce data transfer fees as well as improve the performance of the application. The endpoint policy can be used to specify which S3 buckets the application has access to.

upvoted 47 times

Buruguduystunstugudunstuy 3 years, 5 months ago

WRONG

Option A, deploying Amazon API Gateway into a public subnet and adjusting the route table, would not address the issue of data transfer fees as the application would still be transferring data over the internet.

Option B, deploying a NAT gateway into a public subnet and attaching an endpoint policy, would not address the issue of data transfer fees either as the NAT gateway is used to enable outbound internet access for instances in a private subnet, rather than for connecting to S3.

Option C, deploying the application into a public subnet and allowing it to route through an internet gateway, would not reduce data transfer fees as the application would still be transferring data over the internet.

upvoted 13 times

KVK16 **Highly Voted** 3 years, 7 months ago

Selected Answer: D

To reduce costs get rid of NAT Gateway , VPC endpoint to S3

upvoted 24 times

Gizmo2022 **Most Recent** 1 year, 7 months ago

D is the correct answer

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - remove the internet connection by using a more efficient private VPC direct to S3

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

S3 VPC Gateway is the cheapest solution as it does not use any billable traffic within same region
upvoted 3 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: D

Prevent traffic from traversing the internet = Gateway VPC endpoint for S3.
upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

The best solution to reduce data transfer costs for an application frequently accessing S3 buckets in the same region is option D - Deploy an S3 VPC gateway endpoint into the VPC and attach an endpoint policy that allows access to the S3 buckets.

The key points:

- S3 gateway endpoints allow private connections between VPCs and S3 without going over the public internet.
- This avoids data transfer fees for traffic between the VPC and S3 within the same region.
- An endpoint policy controls access to specific S3 buckets.

upvoted 4 times

cookieMr 2 years, 11 months ago

Selected Answer: D

A. API Gateway can serve as a proxy for S3 requests, it adds unnecessary complexity and additional costs compared to a direct VPC endpoint.

B. Using a NAT gateway for accessing S3 introduces unnecessary data transfer costs as traffic would still flow over the internet.

C. This approach would incur data transfer fees as the traffic would go through the public internet.

In comparison, option D using an S3 VPC gateway endpoint provides a direct and cost-effective solution for accessing S3 buckets within the same Region. By keeping the data transfer within the AWS network infrastructure, it helps reduce data transfer fees and provides secure access to the S3 resources.

upvoted 4 times

Bmarodi 3 years ago

Selected Answer: D

Option D is correct answer.
upvoted 1 times

Erbug 3 years, 4 months ago

To answer this question, I need to know the comparison of the types of gateway of costs, please give me a tip about that issue.
upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D
upvoted 1 times

9014 3 years, 6 months ago

Selected Answer: D

The answer is D:- Actually, the Application (EC2) is running in the same region...instead of going to the internet, data can be copied through the VPC endpoint...so there will be no cost because data is not leaving the AWS infra
upvoted 1 times

JayBee65 3 years, 6 months ago

Can somebody please explain this question? Are we assuming the application is running in AWS and that adding the gateway endpoint avoids the need for the EC2 instance to access the internet and thus avoid costs? Thanks a lot.
upvoted 2 times

SR0611 3 years, 6 months ago

Yes correct
upvoted 1 times

Wpcorgan 3 years, 6 months ago

D is correct
upvoted 1 times

yd_h 3 years, 7 months ago

Selected Answer: D
FYI :

- There is no additional charge for using gateway endpoints.
- Interface endpoints are priced at ~ \$0.01/per AZ/per hour. Cost depends on the Region
- S3 Interface Endpoints resolve to private VPC IP addresses and are routable from outside the VPC (e.g via VPN, Direct Connect, Transit Gateway, etc). S3 Gateway Endpoints use public IP ranges and are only routable from resources within the VPC.

upvoted 5 times

123jh10 3 years, 7 months ago

Selected Answer: D

Close question to the Question #4, with same solution.

upvoted 3 times

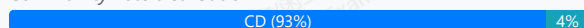
A company recently launched Linux-based application instances on Amazon EC2 in a private subnet and launched a Linux-based bastion host on an Amazon EC2 instance in a public subnet of a VPC. A solutions architect needs to connect from the on-premises network, through the company's internet connection, to the bastion host, and to the application servers. The solutions architect must make sure that the security groups of all the EC2 instances will allow that access.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

- A. Replace the current security group of the bastion host with one that only allows inbound access from the application instances.
- B. Replace the current security group of the bastion host with one that only allows inbound access from the internal IP range for the company.
- C. Replace the current security group of the bastion host with one that only allows inbound access from the external IP range for the company. **Most Voted**
- D. Replace the current security group of the application instances with one that allows inbound SSH access from only the private IP address of the bastion host. **Most Voted**
- E. Replace the current security group of the application instances with one that allows inbound SSH access from only the public IP address of the bastion host.

Correct Answer: CD

Community vote distribution



Comments

Six_Fingered_Jose **Highly Voted** 3 years, 7 months ago

Selected Answer: CD

C because from on-prem network to bastion through internet (using on-prem resource's public IP),
D because bastion and ec2 is in same VPC, meaning bastion can communicate to EC2 via it's private IP address
upvoted 51 times

slimen **Highly Voted** 2 years, 7 months ago

Selected Answer: CD

on-prem -----> bastion host (we use internet, means that we need external IPs of the company)
bastion host -----> private subnet (we use private IP since we are in the same AWS network)
upvoted 8 times

PaulGa **Most Recent** 1 year, 8 months ago

Selected Answer: CD

Ans C,D - as per Six_Fingered_Jose (1 year, 10 mth ago)
upvoted 1 times

lofzee 2 years ago

Selected Answer: CD

CD - easy
upvoted 1 times

Anthony_Rodrigues 2 years ago

Selected Answer: CD

I've noticed that it is very important to focus on the logic for the solution, not just services.

For example, in this question, the goal is to access the application instances only from the bastion while keeping them in the private subnet, which already suggests that the SSH connection must be allowed for the bastion private IP. This is answer D.

On the other hand, the bastion must accept connections only from the company's premises, which already eliminates option A. Option B is wrong because internal IP is used only internally; in this case, the connection will be through the internet, which means that it must be the external IP; therefore, answer C.

upvoted 8 times

awsgeek75 2 years, 4 months ago

Selected Answer: CD

C: Bastion in public subnet should only allow access from public IP of the company

D: app instance in private subnet should only allow access from bastion

ABD are wrong choices

Here is a working example on AWS docs if you want to learn about Bastion setup

<https://aws.amazon.com/solutions/implementations/linux-bastion/>

upvoted 2 times

awsgeek75 2 years, 4 months ago

Good way to remember this one is to think of movie scene where someone is visiting a prisoner in a prison and talks to them from behind a glass using a 2 way phone.

Visitor is in company

Visitor area is public subnet

Prisoner area is private subnet

Phone is bastion

Visitor (company) must only be allowed to use the phone (bastion) from the public area (public subnet) and the phone (bastion) must only be allowed to talk to the prisoner in the prisoner area (private subnet)

upvoted 25 times

1dfed2b 2 years, 2 months ago

really good explanation!!!

upvoted 5 times

Marco_St 2 years, 6 months ago

Selected Answer: BD

the question mentioned from on-prem network to bastion through the company's internet then it should use the internal IP range not external ip ranges. so BD

upvoted 2 times

awsgeek75 2 years, 4 months ago

How would you know the internal IP range of the company? B is wrong.

upvoted 1 times

ATInnovandoJuntos 2 years, 6 months ago

https://en.wikipedia.org/wiki/Network_address_translation

That's the reason is C and not B

upvoted 2 times

viru 2 years, 5 months ago

Company's internet is internet - unless its company's intranet

upvoted 2 times

wearrexdzw3123 2 years, 7 months ago

Why are there always such unclear questions?

upvoted 4 times

tom_cruise 2 years, 8 months ago

Selected Answer: CD

Key: through the company's internet connection

upvoted 3 times

prabhjot 2 years, 8 months ago

Option B - inbound access from the internal IP range for the company. This step ensures that only internal IP addresses from your company's network can access the bastion host, enhancing security and then Option D

upvoted 1 times

Subhrangsu 2 years, 8 months ago

Please check first comments from top of them:

Help2023

WherecanIstart

Burugduystunstugudunstuy

upvoted 1 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: CD

Allows inbound access from the external IP range for the company. Then allow inbound SSH access from only the private IP address of the bastion host.

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: CD

C. This will restrict access to the bastion host from the specific IP range of the on-premises network, ensuring secure connectivity. This step ensures that only authorized users from the on-premises network can access the bastion host.

D. This step enables SSH connectivity from the bastion host to the application instances in the private subnet. By allowing inbound SSH access only from the private IP address of the bastion host, you ensure that SSH access is restricted to the bastion host only.

upvoted 3 times

stanleyjade 3 years, 1 month ago

the internal and external IP range is not clear

upvoted 4 times

PLN6302 2 years, 9 months ago

yes same for me

upvoted 1 times

pentium75 2 years, 5 months ago

The admin is supposed to use "the company's Internet connection", NOT a VPN tunnel or DirectConnect. Thus the connection originates from the company's public/external IP.

upvoted 2 times

km142646 3 years, 1 month ago

The private/public IP address thing is confusing. Ideally, the private instances inbound rule would just allow traffic from the security group of the bastion host.

upvoted 2 times

Spiffaz 3 years, 3 months ago

Why external and not internal?

upvoted 2 times

TariqKipkemei 3 years, 3 months ago

Because the traffic goes through the public internet. In the public internet, public IP(external IP) is used.

upvoted 7 times

Help2023 3 years, 3 months ago

Selected Answer: CE

Application is in private subnet

Bastion Host is in public subnet

D does not make sense because the bastion host is in public subnet and they don't have a private IP but only a public IP address attached to them. The IP wanting to connect is Public as well.

Bastion host in public subnet allows external IP (via internet) of the company to access it. Which then leaves us to give permission to the application private subnet and for that the private subnet with the application accepts the IP coming from Bastion Host by changing its SG. C&E

upvoted 2 times

WheracanIstart 3 years, 3 months ago

Bastion host in public subnet because it has a public IP and a NAT Gateway that can route traffic out of your AWS VPC but it does have the ability to access the private subnet using private IP since it's not leaving AWS to access the private subnet. So C&D are the right answers.

upvoted 3 times

sidharthwader 2 years, 3 months ago

You are right E is also fine but it's not a best thing to do Using private IP is always better than using public IP unless the situation demands the use of Public IP.

upvoted 1 times

A solutions architect is designing a two-tier web application. The application consists of a public-facing web tier hosted on Amazon EC2 in public subnets. The database tier consists of Microsoft SQL Server running on Amazon EC2 in a private subnet. Security is a high priority for the company.

How should security groups be configured in this situation? (Choose two.)

- A. Configure the security group for the web tier to allow inbound traffic on port 443 from 0.0.0.0/0. **Most Voted**
- B. Configure the security group for the web tier to allow outbound traffic on port 443 from 0.0.0.0/0.
- C. Configure the security group for the database tier to allow inbound traffic on port 1433 from the security group for the web tier. **Most Voted**
- D. Configure the security group for the database tier to allow outbound traffic on ports 443 and 1433 to the security group for the web tier.
- E. Configure the security group for the database tier to allow inbound traffic on ports 443 and 1433 from the security group for the web tier.

Correct Answer: AC

Community vote distribution

AC (97%)

3%

Comments

Athena **Highly Voted** 3 years, 7 months ago

Selected Answer: AC

Web Server Rules: Inbound traffic from 443 (HTTPS) Source 0.0.0.0/0 - Allows inbound HTTPS access from any IPv4 address
Database Rules : 1433 (MS SQL) The default port to access a Microsoft SQL Server database, for example, on an Amazon RDS instance

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/security-group-rules-reference.html>

upvoted 30 times

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: AC

EC2 web on public subnets + EC2 SQL on private subnet + security is high priority. So, Option A to allow HTTPS from everywhere. Plus option C to allow SQL connection from the web instance.

upvoted 19 times

Errrrrrwin **Most Recent** 3 months, 3 weeks ago

Selected Answer: CE

What about outbound traffic, do we need to specify?

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: AC

HTTPS (443) from the world

MS SQL server sock connection in 1433 from the private interfaces of the public web tier servers.

Security groups can be added to other security group rules. Hence A,C.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: AC

A & C are the only correct options here.

upvoted 1 times

Himedere 1 year, 5 months ago

Selected Answer: AC

Web server is public: should allow all (0.0.0.0/0) inbound traffic from 443 (HTTPS)
MSSQL Server is private: it should only allow inbound traffic from the web tier from 1433.
443 is HTTPS port, so it's not necessary on MSSQL Server.
upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: AC

Ans A, C - allow public access for all input, but control access to database: source 0.0.0.0/0, control access on port 1433 (MS SQL)
upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: AC

SG are blocked by default and stateful so
A: Allows inbound traffic from web to the HTTPS default port on web servers
B: Outbound is not required if inbound is configured due to stateful nature of SG
C: 1433 is SQL default so allow access from web-tier only
D: Opens up the database to web on 1433 port
E: opens up 443 port unnecessarily on the DB tier so less secure
AC is the most secure config
upvoted 5 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: AC

Allow inbound traffic on port 443 from 0.0.0.0/0 on the web tier. Then allow inbound traffic on port 1433 from the security group for the web tier on the database tier.
upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: AC

The security group for the web tier should allow inbound traffic on port 443 from 0.0.0.0/0. This will allow clients to connect to the web tier using HTTPS. The security group for the web tier should also allow outbound traffic on port 443 to 0.0.0.0/0. This will allow the web tier to connect to the internet to download updates and other resources.

The security group for the database tier should allow inbound traffic on port 1433 from the security group for the web tier. This will allow the web tier to connect to the database tier to access data. The security group for the database tier should not allow outbound traffic on ports 443 and 1433 to the security group for the web tier. This will prevent the database tier from being exposed to the public internet.
upvoted 4 times

cookieMr 2 years, 11 months ago

Selected Answer: AC

A. This configuration allows external users to access the web tier over HTTPS (port 443). However, it's important to note that it is generally recommended to restrict the source IP range to a more specific range rather than allowing access from 0.0.0.0/0 (anywhere). This would limit access to only trusted sources.

C. By allowing inbound traffic on port 1433 (default port for Microsoft SQL Server) from the security group associated with the web tier, you ensure that the database tier can only be accessed by the EC2 instances in the web tier. This provides a level of isolation and restricts direct access to the database tier from external sources.
upvoted 3 times

Abrar2022 3 years ago

DB tier: Port 1433 is the known standard for SQL server and should be used.
web tier on port 443 (HTTPS)
upvoted 4 times

beginnercloud 3 years ago

Selected Answer: AC

AC is correct
upvoted 1 times

WherecanIstart 3 years, 3 months ago

A & C are the correct answer.

Inbound traffic to the web tier on port 443 (HTTPS)
The web tier will then access the Database tier on port 1433 - inbound.
upvoted 1 times

techhb 3 years, 5 months ago

Selected Answer: AC

AC 443-http inbound and 1433-sql server

Security group => focus on inbound traffic since by default outbound traffic is allowed

upvoted 2 times

aba2s 3 years, 5 months ago

Selected Answer: AC

Security group => focus on inbound traffic since by default outbound traffic is allowed

upvoted 2 times

orionizzie 3 years, 5 months ago

why both are inbound rules

upvoted 2 times

kraken21 3 years, 2 months ago

Because security groups are stateful.

upvoted 2 times

A company wants to move a multi-tiered application from on premises to the AWS Cloud to improve the application's performance. The application consists of application tiers that communicate with each other by way of RESTful services. Transactions are dropped when one tier becomes overloaded. A solutions architect must design a solution that resolves these issues and modernizes the application.

Which solution meets these requirements and is the MOST operationally efficient?

A. Use Amazon API Gateway and direct transactions to the AWS Lambda functions as the application layer. Use Amazon Simple Queue Service (Amazon SQS) as the communication layer between application services. **Most Voted**

B. Use Amazon CloudWatch metrics to analyze the application performance history to determine the servers' peak utilization during the performance failures. Increase the size of the application server's Amazon EC2 instances to meet the peak requirements.

C. Use Amazon Simple Notification Service (Amazon SNS) to handle the messaging between application servers running on Amazon EC2 in an Auto Scaling group. Use Amazon CloudWatch to monitor the SNS queue length and scale up and down as required.

D. Use Amazon Simple Queue Service (Amazon SQS) to handle the messaging between application servers running on Amazon EC2 in an Auto Scaling group. Use Amazon CloudWatch to monitor the SQS queue length and scale up when communication failures are detected.

Correct Answer: A

Community vote distribution

A (82%)

Other

Comments

gcmrjbr **Highly Voted** 3 years, 7 months ago

Agree with A>>> Lambda = serverless + autoscale (modernize), SQS= decouple (no more drops)
upvoted 46 times

LuckyAro **Highly Voted** 3 years, 4 months ago

Selected Answer: A

The catch phrase is "scale up when communication failures are detected" Scaling should not be based on communication failures, that'll be crying over spilled milk ! or rather too late. So D is wrong.

upvoted 26 times

mauroicardi 2 years, 3 months ago

Spot on

upvoted 1 times

remand 3 years, 4 months ago

it says "one tier becomes overloaded" , Not communication failure...

upvoted 3 times

LuckyAro 3 years, 4 months ago

D says: "Use Amazon CloudWatch to monitor the SQS queue length and scale up when communication failures are detected".

upvoted 4 times

mc0226 **Most Recent** 1 year, 3 months ago

Selected Answer: A

B. Analyze performance with CloudWatch and increase EC2 instance size:

Vertical scaling (increasing instance size) can address temporary performance issues but is less cost-effective and scalable. It does not modernize the architecture or address dropped transactions due to overloading.

C. Use SNS with EC2 instances and Auto Scaling:

While SNS can facilitate messaging, it is a publish/subscribe model, which is less suited for tier-to-tier decoupling. SQS is a better fit for reliable message queuing and processing.

D. Use SQS with EC2 instances and Auto Scaling:

While this addresses decoupling and scaling, it retains the burden of managing and scaling EC2 instances. A serverless solution like Lambda and API Gateway provides greater operational efficiency and modernization.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

Clearly A, it significantly reduces operational work. All the other options involve some maintenance and cause additional operational work to maintain the EC2 instances and ASG.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: A

Using serverless services like API Gateway and AWS Lambda and SQS helps resolve this problem with the least operational overhead.

upvoted 1 times

19d92c7 1 year, 7 months ago

why its not c?

upvoted 1 times

manabpokhrel7 1 year, 7 months ago

Operational overhead plus SNS isn't durable like SQS

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - keep it simple: API Gateway + Lambda + SQS

D won't work: it relies upon a failure detected - by then it's too late)

upvoted 2 times

344bba0 1 year, 11 months ago

Selected Answer: A

The answer choice D cannot be the answer because CloudWatch is not an appropriate way to monitor transaction failures. I would have been more confused about it if CloudWatch hadn't been mentioned.

upvoted 2 times

SMALLE 2 years, 4 months ago

RESTful services = API Gateways

<https://docs.aws.amazon.com/apigateway/latest/developerguide/welcome.html>

upvoted 4 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

Least operational overhead = API Gateway + Lambda + SQS

BC wrong applications

D: Will work but more operational overhead than A with less resilience to failures

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: A

You want to "modernize the application", that a modern application is serverless, in any case a modern application does NOT use EC2 instances. Also managing EC2 instances (with the OS etc pp) is NOT "operationally efficient". Thus not D.

upvoted 4 times

anikolov 2 years, 5 months ago

Selected Answer: C

For me the solution could be based on using SNS with multiple topics to organize communication between different tiers (Using Subscriber for one and Producer for another topic to proceed with transactions over multi-tiers). CloudWatch to monitor SNS topics queue length and scale up/down based on counts of messages (NumberOfMessagesPublished)

upvoted 1 times

pentium75 2 years, 5 months ago

Wouldn't the scaling take some time? If there is queue length, then one tier IS already overloaded.

upvoted 2 times

MadMarc 1 year, 11 months ago

The question asks for the most operationally efficient solution. The answer A is serverless, while the answer C is VM-based. There is no scenario where managing VMs is operationally more efficient than going serverless.

If Operational Efficiency is what you look for, then you should always consider going serverless.

upvoted 2 times

MrPCarrot 2 years, 6 months ago

A is the perfect answer no need for the ASG

upvoted 2 times

xdkonorek2 2 years, 7 months ago

Selected Answer: D

D is better because in answer A there is a bottleneck on a SQS - service app, D is as operationally efficient as A and solves the above issue

upvoted 2 times

tom_cruise 2 years, 8 months ago

Selected Answer: A

ASG is not as efficient as Lambda!

upvoted 2 times

vijaykamal 2 years, 8 months ago

I feel the answer is D, Lambda would increase the complexity and overhead and it has limitation of running for 15 min.

upvoted 4 times

pentium75 2 years, 5 months ago

How would Lambda increase the operational (!) complexity over handling and scaling servers (and their operating systems and patches etc.) on EC2?

upvoted 2 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: A

MOST operationally efficient = Serverless = AWS Lambda functions, Amazon Simple Queue Service

upvoted 1 times

A company receives 10 TB of instrumentation data each day from several machines located at a single factory. The data consists of JSON files stored on a storage area network (SAN) in an on-premises data center located within the factory. The company wants to send this data to Amazon S3 where it can be accessed by several additional systems that provide critical near-real-time analytics. A secure transfer is important because the data is considered sensitive.

Which solution offers the MOST reliable data transfer?

- A. AWS DataSync over public internet
- B. AWS DataSync over AWS Direct Connect **Most Voted**
- C. AWS Database Migration Service (AWS DMS) over public internet
- D. AWS Database Migration Service (AWS DMS) over AWS Direct Connect

Correct Answer: B

Community vote distribution

B (100%)

Comments

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: B

DMS is for databases and here refers to "JSON files". Public internet is not reliable. So best option is B.

upvoted 39 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: B

CORRECT

The most reliable solution for transferring the data in a secure manner would be option B: AWS DataSync over AWS Direct Connect.

AWS DataSync is a data transfer service that uses network optimization techniques to transfer data efficiently and securely between on-premises storage systems and Amazon S3 or other storage targets. When used over AWS Direct Connect, DataSync can provide a dedicated and secure network connection between your on-premises data center and AWS. This can help to ensure a more reliable and secure data transfer compared to using the public internet.

upvoted 21 times

Buruguduystunstugudunstuy 3 years, 5 months ago

WRONG

Option A, AWS DataSync over the public internet, is not as reliable as using Direct Connect, as it can be subject to potential network issues or congestion.

Option C, AWS Database Migration Service (DMS) over the public internet, is not a suitable solution for transferring large amounts of data, as it is designed for migrating databases rather than transferring large amounts of data from a storage area network (SAN).

Option D, AWS DMS over AWS Direct Connect, is also not a suitable solution, as it is designed for migrating databases and may not be efficient for transferring large amounts of data from a SAN.

upvoted 14 times

doorahmie 3 years, 4 months ago

explanation about D option is good

upvoted 2 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: B

AWS DataSync supports Encryption and Compression and it is much more safer over a Direct Connect.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - keep it simple... private netwk + direct connection

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

near-real-time + large data + secure = DataSync over DirectConnect

A: Less secure due to public internet

C: Slow and not secure

D: Slow even if more secure

DC may not even work as we don't know if there is a DB on other side but even if it was there, it is less preferred way

upvoted 3 times

JTruong 2 years, 5 months ago

Any DMS related-service will not be efficient because DMS can only process JSON files UPTO 2 MB in size

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Tasks.CustomizingTasks.TableMapping.SelectionTransformation.html

so B is CORRECT

upvoted 2 times

Ruffyit 2 years, 7 months ago

AWS DataSync is a data transfer service that uses network optimization techniques to transfer data efficiently and securely between on-premises storage systems and Amazon S3 or other storage targets. When used over AWS Direct Connect, DataSync can provide a dedicated and secure network connection between your on-premises data center and AWS. This can help to ensure a more reliable and secure data transfer compared to using the public internet.

upvoted 4 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: B

Secure and Most reliable transfer = AWS DataSync over AWS Direct Connect

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: B

AWS DataSync is designed for large scale, high speed data transfer between on-prem and S3.

Using AWS Direct Connect provides a dedicated, private connection for reliable, consistent data transfer.

DataSync seamlessly handles data replication, encryption, recovery etc.

upvoted 3 times

MNotABot 2 years, 11 months ago

Not over public hence AC out / DMS is for databases and here refers to "JSON files".

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: B

DataSync is a service specifically designed for data transfer and synchronization between on-premises storage systems and AWS storage services like S3. It provides reliable and efficient data transfer capabilities, ensuring the secure movement of large volumes of data.

By leveraging Direct Connect, which establishes a dedicated network connection between the on-premises data center and AWS, the data transfer is conducted over a private and dedicated network link. This approach offers increased reliability, lower latency, and consistent network performance compared to transferring data over the public internet.

Database Migration Service is primarily focused on database migration and replication, and it may not be the most appropriate tool for general-purpose data transfer like JSON files.

Transferring data over the public internet may introduce potential security risks and performance variability due to factors like network congestion, latency, and potential interruptions.

upvoted 3 times

beginnercloud 3 years ago

Best option and correct is B

upvoted 1 times

Abrar2022 3 years ago

Selected Answer: B

as Ariel suggested and rightly so.....DMS is for databases and here refers to "JSON files". Public internet is not reliable. so B

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B. DMS is not needed as there is no Database migration requirement.

upvoted 1 times

Wajif 3 years, 6 months ago

Selected Answer: B

Public internet options automatically out being best-effort. DMS is for database migration service and here they have to just transfer the data to S3. Hence B.

upvoted 2 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 1 times

A company needs to configure a real-time data ingestion architecture for its application. The company needs an API, a process that transforms data as the data is streamed, and a storage solution for the data.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Deploy an Amazon EC2 instance to host an API that sends data to an Amazon Kinesis data stream. Create an Amazon Kinesis Data Firehose delivery stream that uses the Kinesis data stream as a data source. Use AWS Lambda functions to transform the data. Use the Kinesis Data Firehose delivery stream to send the data to Amazon S3.
- B. Deploy an Amazon EC2 instance to host an API that sends data to AWS Glue. Stop source/destination checking on the EC2 instance. Use AWS Glue to transform the data and to send the data to Amazon S3.
- C. Configure an Amazon API Gateway API to send data to an Amazon Kinesis data stream. Create an Amazon Kinesis Data Firehose delivery stream that uses the Kinesis data stream as a data source. Use AWS Lambda functions to transform the data. Use the Kinesis Data Firehose delivery stream to send the data to Amazon S3. **Most Voted**
- D. Configure an Amazon API Gateway API to send data to AWS Glue. Use AWS Lambda functions to transform the data. Use AWS Glue to send the data to Amazon S3.

Correct Answer: C

Community vote distribution

C (96%)

4%

Comments

123jhl0 **Highly Voted** 3 years, 7 months ago

Selected Answer: C

(A) - You don't need to deploy an EC2 instance to host an API - Operational overhead

(B) - Same as A

(**C**) - Is the answer

(D) - AWS Glue gets data from S3, not from API GW. AWS Glue could do ETL by itself, so don't need lambda. Non sense.

<https://aws.amazon.com/glue/>

upvoted 44 times

Futurebones 3 years, 1 month ago

I don't understand is why we should use Lambda in between to transform data. To me, Kinesis data firehose is enough as it is an extract, transform, and load (ETL) service.

upvoted 5 times

Remy_d 2 years, 8 months ago

It is because they assume that Kinesis Data Firehose built-in transformations are not enough. So you have to use specific lambda transformation.

Please refer to this link : [https://aws.amazon.com/kinesis/data-](https://aws.amazon.com/kinesis/data-firehose/#:~:text=Amazon%20Kinesis%20Data%20Firehose%20is,data%20stores%2C%20and%20analytics%20services.)

[firehose/#:~:text=Amazon%20Kinesis%20Data%20Firehose%20is,data%20stores%2C%20and%20analytics%20services.](https://aws.amazon.com/kinesis/data-firehose/#:~:text=Amazon%20Kinesis%20Data%20Firehose%20is,data%20stores%2C%20and%20analytics%20services.)

upvoted 9 times

TariqKipkemei **Highly Voted** 2 years, 10 months ago

Selected Answer: C

The company needs an API = Amazon API Gateway API

A real-time data ingestion = Amazon Kinesis data stream

A process that transforms data = AWS Lambda functions

Kinesis Data Firehose delivery stream to send the data to Amazon S3

A storage solution for the data = Amazon S3

upvoted 32 times

PaulGa **Most Recent** 1 year, 8 months ago

Selected Answer: C

Ans C - the API is given so just configure Kinesis stream/Firehose to use it

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

C is least operational overhead

A: EC2 is overhead in this scenario

B: Same as A

D: Glue is not real time data streaming

upvoted 4 times

Mikado211 2 years, 6 months ago

Selected Answer: C

It looks overengineered, but as it works, let's go for the C

upvoted 5 times

Ruffyt 2 years, 7 months ago

The company needs an API = Amazon API Gateway API

A real-time data ingestion = Amazon Kinesis data stream

A process that transforms data = AWS Lambda functions

Kinesis Data Firehose delivery stream to send the data to Amazon S3

A storage solution for the data = Amazon S3

upvoted 4 times

peekingpicker 2 years, 8 months ago

Selected Answer: D

"a real-time data ingestion"

isn't firehose not realtime ? Kinesis FireHose is "Near" Real-time . It has 60 seconds gap. I think it should be D

upvoted 2 times

rlamberti 2 years, 7 months ago

The real-time part (data ingestion) will be performed by Kinesis Data Stream and API Gateway. After this, the transformation and storage of the data don't need to be in real-time, since it was already ingested, so Kinesis Firehose + Lambda is perfect. C makes sense to me.

upvoted 5 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

Option C provides the least operational overhead to meet the requirements:

API Gateway provides the API

Kinesis Data Streams ingests the real-time data

Lambda functions transform the data

Firehose delivers the data to S3 storage

The key advantages are:

Serverless architecture requires minimal operational overhead

Fully managed ingestion, processing and storage services

No need to manage EC2 instances

upvoted 2 times

diabloexodia 2 years, 11 months ago

Requirements:

API- API gateway

Real time data ingestion - AWS Kinesis data stream

ETL(Extract Transform Load) - Kinesis Firehose

Storage- S3

upvoted 3 times

tamefi5512 2 years, 11 months ago

Selected Answer: C

C - is the answer

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

C. By leveraging these services together, you can achieve a real-time data ingestion architecture with minimal operational overhead. The data flows from the API Gateway to the Kinesis data stream, undergoes transformations with Lambda, and is then sent to S3 via the Kinesis Data Firehose delivery stream for storage.

A. This adds operational overhead as you need to handle EC2 management, scaling, and maintenance. It is less efficient compared to using a serverless solution like API Gateway.

B. It requires deploying and managing an EC2 to host the API and configuring Glue. This adds operational overhead, including EC2 management and potential scalability limitations.

D. It still requires managing and configuring Glue, which adds operational overhead. Additionally, it may not be the most efficient solution as Glue is primarily used for ETL scenarios, and in this case, real-time data transformation is required.

upvoted 2 times

winzzhhzzhh 3 years ago

Selected Answer: D

I am gonna choose D for this.

Kinesis Data Stream + Data Firehose will add up to the operational overhead, plus it is "Near real-time", not a real time solution.

Lambda functions scale automatically, so no management of scaling/compute resources is needed.

AWS Glue handles the data storage in S3, so no management of that is needed.

upvoted 2 times

UnluckyDucky 3 years, 2 months ago

Gotta love all those chatgpt answers y'all are throwing at us.

Kinesis Firehose is NEAR real-time, not real-time like your bots tell you.

upvoted 2 times

pentium75 2 years, 5 months ago

Stem is about "real-time data INGESTION", not real-time processing.

upvoted 2 times

bullrem 3 years, 4 months ago

Selected Answer: C

option C is the best solution. It uses Amazon Kinesis Data Firehose which is a fully managed service for delivering real-time streaming data to destinations such as Amazon S3. This service requires less operational overhead as compared to option A, B, and D. Additionally, it also uses Amazon API Gateway which is a fully managed service for creating, deploying, and managing APIs. These services help in reducing the operational overhead and automating the data ingestion process.

upvoted 1 times

Burugduystunstugudunstuy 3 years, 5 months ago

Selected Answer: C

Option C is the solution that meets the requirements with the least operational overhead.

In Option C, you can use Amazon API Gateway as a fully managed service to create, publish, maintain, monitor, and secure APIs. This means that you don't have to worry about the operational overhead of deploying and maintaining an EC2 instance to host the API.

Option C also uses Amazon Kinesis Data Firehose, which is a fully managed service for delivering real-time streaming data to destinations such as Amazon S3. With Kinesis Data Firehose, you don't have to worry about the operational overhead of setting up and maintaining a data ingestion infrastructure.

upvoted 1 times

Burugduystunstugudunstuy 3 years, 5 months ago

Finally, Option C uses AWS Lambda, which is a fully managed service for running code in response to events. With AWS Lambda, you don't have to worry about the operational overhead of setting up and maintaining a server to run the data transformation code.

Overall, Option C provides a fully managed solution for real-time data ingestion with minimal operational overhead.

upvoted 2 times

Burugduystunstugudunstuy 3 years, 5 months ago

Option A is incorrect because it involves deploying an EC2 instance to host an API, which adds operational overhead in the form of maintaining and securing the instance.

Option B is incorrect because it involves deploying an EC2 instance to host an API and disabling source/destination checking on the instance. Disabling source/destination checking can make the instance vulnerable to attacks, which adds operational overhead in the form of securing the instance.

upvoted 2 times

Burugduystunstugudunstuy 3 years, 5 months ago

Option D is incorrect because it involves using AWS Glue to send the data to Amazon S3, which adds operational overhead in the form of maintaining and securing the AWS Glue infrastructure.

Overall, Option C is the best choice because it uses fully managed services for the API, data transformation, and data delivery, which minimizes operational overhead.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

A company needs to keep user transaction data in an Amazon DynamoDB table. The company must retain the data for 7 years. What is the MOST operationally efficient solution that meets these requirements?

- A. Use DynamoDB point-in-time recovery to back up the table continuously.
- B. Use AWS Backup to create backup schedules and retention policies for the table. **Most Voted**
- C. Create an on-demand backup of the table by using the DynamoDB console. Store the backup in an Amazon S3 bucket. Set an S3 Lifecycle configuration for the S3 bucket.
- D. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to invoke an AWS Lambda function. Configure the Lambda function to back up the table and to store the backup in an Amazon S3 bucket. Set an S3 Lifecycle configuration for the S3 bucket.

Correct Answer: B

Community vote distribution

B (100%)

Comments

123jhl0 **Highly Voted** 3 years, 7 months ago

Selected Answer: B

Answer is B

"Amazon DynamoDB offers two types of backups: point-in-time recovery (PITR) and on-demand backups. (==> D is not the answer)

PITR is used to recover your table to any point in time in a rolling 35 day window, which is used to help customers mitigate accidental deletes or writes to their tables from bad code, malicious access, or user error. (==> A isn't the answer)

On demand backups are designed for long-term archiving and retention, which is typically used to help customers meet compliance and regulatory requirements.

This is the second of a series of two blog posts about using AWS Backup to set up scheduled on-demand backups for Amazon DynamoDB. Part 1 presents the steps to set up a scheduled backup for DynamoDB tables from the AWS Management Console." (==> Not the DynamoDB console and C isn't the answer either)

<https://aws.amazon.com/blogs/database/part-2-set-up-scheduled-backups-for-amazon-dynamodb-using-aws-backup/>

upvoted 52 times

LuckyAro 3 years, 4 months ago

I think the answer is C because of storage time.

upvoted 2 times

jerry1 1 year, 2 months ago

i think another reason for why C is not correct is the keyword "on-demand" which is not operation efficient (as this should be an ongoing practice)

upvoted 1 times

MutiverseAgent 2 years, 11 months ago

Dynamo backups cannot be scheduled or sent to S3, so answer should be B)

1) <https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/BackupRestore.html>

2) <https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Backup.Tutorial.html>

upvoted 2 times

app12 2 years, 5 months ago

In the very same link you shared it says that you CAN send backups to S3

https://youtu.be/4INEu_hw30Q?t=54

upvoted 2 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: B

The most operationally efficient solution that meets these requirements would be to use option B, which is to use AWS Backup to create backup schedules and retention policies for the table.

AWS Backup is a fully managed backup service that makes it easy to centralize and automate the backup of data across AWS resources. It allows you to create backup policies and schedules to automatically back up your DynamoDB tables on a regular basis. You can also specify retention policies to ensure that your backups are retained for the required period of time. This solution is fully automated and requires minimal maintenance, making it the most operationally efficient option.

upvoted 18 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, using DynamoDB point-in-time recovery, is also a viable option but it requires continuous backup, which may be more resource-intensive and may incur higher costs compared to using AWS Backup.

Option C, creating an on-demand backup of the table and storing it in an S3 bucket, is also a viable option but it requires manual intervention and does not provide the automation and scheduling capabilities of AWS Backup.

Option D, using Amazon EventBridge (CloudWatch Events) and a Lambda function to back up the table and store it in an S3 bucket, is also a viable option but it requires more complex setup and maintenance compared to using AWS Backup.

upvoted 15 times

Dharmarajan Most Recent 1 year, 4 months ago

Selected Answer: B

Anytime the "Operational efficiency" comes into picture, one should consider the options with serverless or managed services. DynamoDB is a serverless service, and AWS Backup is a managed service. Others involve operational work in terms of maintenance of backups etc.

upvoted 2 times

satyaamm 1 year, 5 months ago

Selected Answer: B

Using AWS Backup is most suitable here for retaining data for 7 years.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - there are no special parameters specified for the backup (eg. PITR snapshots) so it doesn't need to be elaborate

upvoted 2 times

OctavioBatera 2 years, 2 months ago

Selected Answer: B

Agreed with option B is the right one. AWS backup retention goes from 1 day to 100 years (or even indefinitely, if you do not enter a retention period), so will meet the requirements.

upvoted 2 times

cheroh_tots 2 years, 3 months ago

Why is the answer not C?

upvoted 2 times

1dfed2b 2 years, 3 months ago

As I see, we are looking for the most operationally efficient solution. So it's B, but the most cost effective is - C (but it isn't a question).

upvoted 2 times

psyll0n 2 years, 4 months ago

https://youtu.be/g4WPLFXLwDE?si=nTWqqDcBe_Y6dtI3

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

Operational efficiency is always a managed service from AWS. AWS Backup is the right one in this case so B is right answer

upvoted 2 times

awsgeek75 2 years, 5 months ago

Answer is simply B as it is MOST operationally efficient. Other options are "distractors" to confuse everyone.

upvoted 2 times

viru 2 years, 5 months ago

Selected Answer: B

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/backuprestore_HowItWorksAWS.html

upvoted 1 times

Mikado211 2 years, 6 months ago

Selected Answer: B

Well a 7 years TTL on the dynamoDB records could be the simplest to answer the question, so B for the "retention policies". And since the B also propose AWS backup with a retention time at 7 years, why not.

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: B

The key advantages of using AWS Backup are:

Fully managed backup service requiring minimal operational overhead
Built-in scheduling, retention policies, and backup monitoring
Supports point-in-time restore for DynamoDB
Automated and scalable solution

upvoted 1 times

tamefi5512 2 years, 11 months ago

Selected Answer: B

B - is the answer because its easy to setup via AWS Backup & It indicates the keyword "MOST Operational Efficient". Other answers are indicating Cost efficient

upvoted 1 times

cookieMr 2 years, 11 months ago

AWS Backup is a fully managed backup service that simplifies the process of creating and managing backups across various AWS services, including DynamoDB. It allows you to define backup schedules and retention policies to automatically take backups and retain them for the desired duration. By using AWS Backup, you can offload the operational overhead of managing backups to the service itself, ensuring that your data is protected and retained according to the specified retention period.

This solution is more efficient compared to the other options because it provides a centralized and automated backup management approach specifically designed for AWS services. It eliminates the need to manually configure and maintain backup processes, making it easier to ensure data retention compliance without significant operational effort.

upvoted 2 times

Rahul2212 2 years, 12 months ago

A

PITR is used to recover your table to any point in time in a rolling 35 day window, which is used to help customers mitigate accidental deletes or writes to their tables from bad code, malicious access, or user error. (==> A is the answer)

upvoted 1 times

Abrar2022 3 years ago

using AWS Backup cheaper than DynamoDB point-in-time recovery

upvoted 1 times

A company is planning to use an Amazon DynamoDB table for data storage. The company is concerned about cost optimization. The table will not be used on most mornings. In the evenings, the read and write traffic will often be unpredictable. When traffic spikes occur, they will happen very quickly.

What should a solutions architect recommend?

- A. Create a DynamoDB table in on-demand capacity mode. **Most Voted**
- B. Create a DynamoDB table with a global secondary index.
- C. Create a DynamoDB table with provisioned capacity and auto scaling.
- D. Create a DynamoDB table in provisioned capacity mode, and configure it as a global table.

Correct Answer: A

Community vote distribution

A (78%)

C (22%)

Comments

SimonPark **Highly Voted** 3 years, 7 months ago

Selected Answer: A

On-demand mode is a good option if any of the following are true:

- You create new tables with unknown workloads.
- You have unpredictable application traffic.
- You prefer the ease of paying for only what you use.

upvoted 53 times

123jh10 **Highly Voted** 3 years, 7 months ago

Selected Answer: A

****A**** - On demand is the answer -

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.ReadWriteCapacityMode.html#HowItWorks.OnDemand>

B - not related with the unpredictable traffic

C - provisioned capacity is recommended for known patterns. Not the case here.

D - same as C

upvoted 26 times

NasosoAuxtyno 3 years, 3 months ago

Thanks. Your reference link perfectly supports the option "A". 100% correct

upvoted 3 times

patdevops **Most Recent** 1 year, 3 months ago

Selected Answer: A

The correct answer is A, keyword "unpredictable workloads"

upvoted 1 times

AshishDhole 1 year, 4 months ago

Selected Answer: A

Unknown Load/traffic = On demand mode

upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: A

On-demand mode of DynamoDB is the most suitable for unpredictable workloads.

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: C

after browsing through all the responses, I decided to vote for C for the following 2 reasons:

- Unpredictable traffic: Since the evening traffic is often unpredictable with quick spikes, using provisioned capacity with auto scaling allows the table to automatically adjust to meet the fluctuating demand, preventing potential throttling issues during high traffic periods.
 - Cost optimization: By utilizing auto scaling, the company only pays for the capacity it actually uses, which is ideal for a scenario where the table experiences low usage during mornings and high, unpredictable usage in the evenings.
- upvoted 2 times

JA2018 1 year, 6 months ago

For Option A, come to think about it, while it is true that on-demand capacity can handle spikes, it may not be the most cost-effective option as you still have to pay for the capacity used even during low traffic periods.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - handles unpredictable workloads/traffic; pay on demand
C - provisioned capacity for known data I/O - not the case here

upvoted 2 times

SaurabhTiwari1 1 year, 9 months ago

Selected Answer: A

Unpredictable = On-demand

upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

cost is concern so CD are not right as provisioning is costly

B is irrelevant

A on-demand is correct as it will scale according to the usage pattern which is from low to very abrupt high

upvoted 3 times

MiniYang 2 years, 6 months ago

Selected Answer: C

Choosing the On-Demand Capacity model (Option A) may cause performance issues during peak periods because it relies on DynamoDB to automatically adjust throughput based on actual usage, which may not be able to cope with sudden traffic increases in time.

Choosing a DynamoDB table with a global secondary index (option B) is independent of the capacity model and does not directly solve the problem of peak traffic.

Choosing to build DynamoDB tables in provisioned capacity mode and configure them as global tables (option D) may increase costs in some cases without necessarily providing the flexibility to accommodate unpredictable peak traffic.

upvoted 1 times

MiniYang 2 years, 6 months ago

sorry I want to change my answer to A . Because the point is the " cost "

upvoted 7 times

BartoszGolebiowski24 2 years, 7 months ago

Selected Answer: A

DynamoDB autoscaling takes 2 minutes to increase capacity. We need to handle it immediately.

"Application Auto Scaling automatically scales the provisioned capacity only when the consumed capacity is higher than target utilization for two consecutive minutes".

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/TroubleshootingThrottling.html>

upvoted 4 times

xdkonorek2 2 years, 7 months ago

This is important, thank you for the link. A definitely

upvoted 2 times

Wayne23Fang 2 years, 8 months ago

Selected Answer: A

The costly part of (C) is you need to pay for what you order not what you have used for (A) On-Demand: A reserved capacity purchase is an agreement to pay for a minimum amount of provisioned throughput capacity, for the duration of the term of the agreement, in exchange for discounted pricing. If you use less than your reserved capacity, you will still be charged each month for that minimum amount of provisioned throughput capacity.

upvoted 2 times

clark777 2 years, 8 months ago

<https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html>

With on-demand capacity mode, DynamoDB charges you for the data reads and writes your application performs on your tables. You do not need to

specify how much read and write throughput you expect your application to perform because DynamoDB instantly accommodates your workloads as they ramp up or down.

With provisioned capacity mode, you specify the number of reads and writes per second that you expect your application to require, and you are billed based on that. Furthermore if you can forecast your capacity requirements you can also reserve a portion of DynamoDB provisioned capacity and optimize your costs even further.

upvoted 3 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: A

With on-demand capacity mode, DynamoDB instantly accommodates your workloads as they ramp up or down.

upvoted 2 times

ontheyun 2 years, 11 months ago

on-demand capacity : unpredictable application traffic

provisioned capacity : predictable application traffic, run applications whose traffic is consistent, and ramps up or down gradually.

<https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html>

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: C

By choosing provisioned capacity, you can allocate a specific amount of read and write capacity units based on your expected usage during peak times. This helps in cost optimization as you only pay for the provisioned capacity, which can be adjusted according to your anticipated traffic.

Enabling auto scaling allows DynamoDB to automatically adjust the provisioned capacity up or down based on the actual usage. This is beneficial in handling quick traffic spikes without manual intervention and ensuring that the required capacity is available to handle increased load efficiently. Auto scaling helps to optimize costs by dynamically adjusting the capacity to match the demand, avoiding overprovisioning during periods of low usage.

A. Creating a DynamoDB table in on-demand capacity mode, may not be the most cost-effective solution in this scenario. On-demand capacity mode charges you based on the actual usage of read and write requests, which can be beneficial for sporadic or unpredictable workloads. However, it may not be the optimal choice if the table is not used on most mornings.

upvoted 10 times

beginnercloud 3 years ago

Selected Answer: A

Correct answer is A

- You create new tables with unknown workloads. - You have unpredictable application traffic. - You prefer the ease of paying for only what you use.

upvoted 2 times

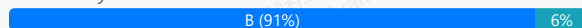
A company recently signed a contract with an AWS Managed Service Provider (MSP) Partner for help with an application migration initiative. A solutions architect needs to share an Amazon Machine Image (AMI) from an existing AWS account with the MSP Partner's AWS account. The AMI is backed by Amazon Elastic Block Store (Amazon EBS) and uses an AWS Key Management Service (AWS KMS) customer managed key to encrypt EBS volume snapshots.

What is the MOST secure way for the solutions architect to share the AMI with the MSP Partner's AWS account?

- A. Make the encrypted AMI and snapshots publicly available. Modify the key policy to allow the MSP Partner's AWS account to use the key.
- B. Modify the launchPermission property of the AMI. Share the AMI with the MSP Partner's AWS account only. Modify the key policy to allow the MSP Partner's AWS account to use the key. **Most Voted**
- C. Modify the launchPermission property of the AMI. Share the AMI with the MSP Partner's AWS account only. Modify the key policy to trust a new KMS key that is owned by the MSP Partner for encryption.
- D. Export the AMI from the source account to an Amazon S3 bucket in the MSP Partner's AWS account, Encrypt the S3 bucket with a new KMS key that is owned by the MSP Partner. Copy and launch the AMI in the MSP Partner's AWS account.

Correct Answer: B

Community vote distribution



Comments

Sauran **Highly Voted** 3 years, 7 months ago

Selected Answer: B

Share the existing KMS key with the MSP external account because it has already been used to encrypt the AMI snapshot.

<https://docs.aws.amazon.com/kms/latest/developerguide/key-policy-modifying-external-accounts.html>

upvoted 20 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: B

CORRECT

B. Modify the launchPermission property of the AMI.

The most secure way for the solutions architect to share the AMI with the MSP Partner's AWS account would be to modify the launchPermission property of the AMI and share it with the MSP Partner's AWS account only. The key policy should also be modified to allow the MSP Partner's AWS account to use the key. This ensures that the AMI is only shared with the MSP Partner and is encrypted with a key that they are authorized to use.

upvoted 15 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, making the AMI and snapshots publicly available, is not a secure option as it would allow anyone with access to the AMI to use it.

Option C, modifying the key policy to trust a new KMS key owned by the MSP Partner, is also not a secure option as it would involve sharing the key with the MSP Partner, which could potentially compromise the security of the data encrypted with the key.

Option D, exporting the AMI to an S3 bucket in the MSP Partner's AWS account and encrypting the S3 bucket with a new KMS key owned by the MSP Partner, is also not the most secure option as it involves sharing the AMI and a new key with the MSP Partner, which could potentially compromise the security of the data.

upvoted 17 times

Gizmo2022 1 year, 6 months ago

Thank u so much for explaining this Buruguduy

upvoted 3 times

jerry1 **Most Recent** 1 year, 2 months ago

Selected Answer: C

Can anyone explain to me? I am not familiar with the terminology
I will go for C, because B mention sharing the key to partner, which might expose other resource which use the same key for encryption, or is this scenario not possible to happen (i.e. key will not be reused across resources)?
Thanks

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: B

I got this wrong! I thought C is right but it turns out that is not secure.
In this question, or any question with encryption, one must consider which of the available options is the most secure way, and it turns out that in this case it is B.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: B

I got this wrong, select D initially, but after read the comments, I agree B is the right solution.
1. <https://docs.aws.amazon.com/toolkit-for-visual-studio/latest/user-guide/tkv-set-ami-launch-perms.html>
2. <https://docs.aws.amazon.com/toolkit-for-visual-studio/latest/user-guide/tkv-set-ami-launch-perms.html>

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: B

Updating the key policy is the most secure way to get access here.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - keep the control simple by only allowing MSP Partner access to the key

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

AD are unsecure.
I was confused between B and C but read the article (link below). You have to allow the other account to use your key somehow otherwise they won't be able to use your AMI. C just allows a trust relationship with MSP's KMS, it won't give them access to your key.
<https://aws.amazon.com/blogs/security/how-to-share-encrypted-amis-across-accounts-to-launch-encrypted-ec2-instances/>

upvoted 4 times

xdkonorek2 2 years, 7 months ago

Selected Answer: B

when you export AMI to s3 bucket it remains encrypted, so partner couldn't launch ec2 instance

upvoted 3 times

Ruffyit 2 years, 7 months ago

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/sharingamis-explicit.html>

upvoted 2 times

TariqKipkemei 2 years, 10 months ago

Selected Answer: B

Share the AMI and Key with the MSP Partner's AWS account only

upvoted 2 times

tamefi5512 2 years, 11 months ago

Selected Answer: B

B - is the Answer
<https://docs.aws.amazon.com/kms/latest/developerguide/key-policy-modifying-external-accounts.html>

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: B

By modifying the launchPermission property of the AMI and sharing it with the MSP Partner's account only, the solutions architect restricts access to the AMI and ensures that it is not publicly available.

Additionally, modifying the key policy to allow the MSP Partner's account to use KMS customer managed key used for encrypting the EBS snapshots ensures that the MSP Partner has the necessary permissions to access and use the key for decryption.

upvoted 3 times

Abrar2022 3 years ago

CORRECTION to my last comment Option B is correct not A.

Explanation why..

Making the AMI and snapshots publicly available, is not a secure option as it would allow anyone with access to the AMI to use it. Best practice would be to share the AMI with the MSP Partner's AWS account then Modify launchPermission property of the AMI. This ensures that the AMI is shared only with the MSP Partner and is encrypted with a key that they are authorised to use.

upvoted 3 times

Abrar2022 3 years ago

Selected Answer: A

Option A, making the AMI and snapshots publicly available, is not a secure option as it would allow anyone with access to the AMI to use it. Best practice would be to share the AMI with the MSP Partner's AWS account then Modify launchPermission property of the AMI. This ensures that the AMI is shared only with the MSP Partner and is encrypted with a key that they are authorised to use.

upvoted 1 times

draum010 3 years, 2 months ago

Selected Answer: D

Option D

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

A solutions architect is designing the cloud architecture for a new application being deployed on AWS. The process should run in parallel while adding and removing application nodes as needed based on the number of jobs to be processed. The processor application is stateless. The solutions architect must ensure that the application is loosely coupled and the job items are durably stored.

Which design should the solutions architect use?

- A. Create an Amazon SNS topic to send the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch configuration that uses the AMI. Create an Auto Scaling group using the launch configuration. Set the scaling policy for the Auto Scaling group to add and remove nodes based on CPU usage.
- B. Create an Amazon SQS queue to hold the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch configuration that uses the AMI. Create an Auto Scaling group using the launch configuration. Set the scaling policy for the Auto Scaling group to add and remove nodes based on network usage.
- C. Create an Amazon SQS queue to hold the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch template that uses the AMI. Create an Auto Scaling group using the launch template. Set the scaling policy for the Auto Scaling group to add and remove nodes based on the number of items in the SQS queue. **Most Voted**
- D. Create an Amazon SNS topic to send the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch template that uses the AMI. Create an Auto Scaling group using the launch template. Set the scaling policy for the Auto Scaling group to add and remove nodes based on the number of messages published to the SNS topic.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Marge_Simpson **Highly Voted** 3 years, 6 months ago

Selected Answer: C

decoupled = SQS
Launch template = AMI
Launch configuration = EC2
upvoted 57 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: C

This design follows the best practices for loosely coupled and scalable architecture. By using SQS, the jobs are durably stored in the queue, ensuring they are not lost. The processor application is stateless, which aligns with the design requirement. The AMI allows for consistent deployment of the application. The launch template and ASG facilitate the dynamic scaling of the application based on the number of items in the SQS, ensuring parallel processing of jobs.

Options A and D suggest using SNS, which is a publish/subscribe messaging service and may not provide the durability required for job storage.

Option B suggests using network usage as a scaling metric, which may not be directly related to the number of jobs to be processed. The number of items in the SQS provides a more accurate metric for scaling based on the workload.

upvoted 9 times

PaulGa **Most Recent** 1 year, 8 months ago

Selected Answer: C

Ans C - decoupled and durable (SQS), auto-scaling based on number of messages.

upvoted 3 times

setout4saa 2 years ago

Selected Answer: C

C for sure

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

SNS is not reliable in case of processing failure which is why none of the SNS options are useful. SQS (not in FIFO mode) allows parallel message processing but reliability/durability of messages is guaranteed. AMI/EC2 scaling is good choice and scaling parameter should be number of messages. Hence "C" is the correct answer

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: C

"Based on the number of jobs to be processed" -> that alone rules out ABD because only C is based on queue length. (D is based on "number of messages published to the queue", not number of messages currently in queue.)

"Job items are durably stored" also speaks for SQS over SNS.

upvoted 3 times

clumsyninja4life 2 years, 5 months ago

so many words...

upvoted 2 times

awsgeek75 2 years, 4 months ago

... yet only one answer is correct

upvoted 3 times

slimen 2 years, 7 months ago

Selected Answer: C

from my perspective, I didn't go for D even though it provides decoupled architecture is because in the question they said "parallel processing" SNS sends the same message to all the subscribers, but in this case we don't want all the nodes to process the same message instead we want them to process as much jobs as possible in a parallel fashion.

SQS in this case is more suitable because each job will get a message and process it and the next message will be taken by another job and so on..

upvoted 4 times

darekw 2 years, 9 months ago

<https://aws.amazon.com/about-aws/whats-new/2021/03/aws-certificate-manager-provides-certificate-expiry-monitoring-through-amazon-cloudwatch/>

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

Loosely coupled = Amazon SQS queue

New application being deployed = deploy on Amazon Machine Image

Adding and removing application nodes as needed based on the number of jobs to be processed = Auto Scaling group with launch template

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

The recommended design is to use an SQS queue to store jobs (option C):

SQS provides a durable and decoupled queue to store job items

An Auto Scaling group with scaling policies based on SQS queue length will add/remove nodes as needed

Launch templates provide flexibility to update AMIs

The key points:

SQS enables loose coupling and stores jobs durably

Auto Scaling provides parallel processing

Scaling based on queue length manages nodes effectively

upvoted 3 times

Bmarodi 3 years ago

Selected Answer: C

C for sure

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: C

CORRECT

The correct design is Option C. Create an Amazon SQS queue to hold the jobs that need to be processed. Create an Amazon Machine Image (AMI) that consists of the processor application. Create a launch template that uses the AMI. Create an Auto Scaling group using the launch template. Set the scaling policy for the Auto Scaling group to add and remove nodes based on the number of items in the SQS queue.

This design satisfies the requirements of the application by using Amazon Simple Queue Service (SQS) as durable storage for the job items and Amazon Elastic Compute Cloud (EC2) Auto Scaling to add and remove nodes based on the number of items in the queue. The processor application can be run in parallel on multiple nodes, and the use of launch templates allows for flexibility in the configuration of the EC2 instances.

upvoted 6 times

Buruguduystunstugudunstuy 3 years, 5 months ago

WRONG

Option A is incorrect because it uses Amazon Simple Notification Service (SNS) instead of SQS, which is not a durable storage solution.

Option B is incorrect because it uses CPU usage as the scaling trigger instead of the number of items in the queue.

Option D is incorrect for the same reasons as option A.

upvoted 7 times

graveend 2 years, 10 months ago

SNS provides durable storage of all messages that it receives.

Ref: <https://aws.amazon.com/sns/faqs/#:~:text=SNS%20provides%20durable%20storage%20of%20all%20messages%20that%20it%20receives.>

Why use SQS instead of SNS? In the question it says parallel execution of processes. SNS has that ability.

upvoted 1 times

cyber_bedouin 2 years, 7 months ago

SQS satisfies the decoupling requirement

upvoted 1 times

awsgeek75 2 years, 4 months ago

SNS is not a durable storage. SQS stores the messages until they are process. SNS just notifies the subscribers and doesn't care if the notification is processed or not. It's kind of "fire and forget"

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: C

SQS with EC2 autoscaling policy based number of messages in the queue.

upvoted 1 times

Uhrien 3 years, 6 months ago

Selected Answer: C

C is correct

upvoted 2 times

kelljons 3 years, 6 months ago

what about the word "coupled"

upvoted 1 times

kewl 3 years, 6 months ago

Selected Answer: C

AWS strongly recommends that you do not use launch configurations hence answer is C

https://docs.amazonaws.cn/en_us/autoscaling/ec2/userguide/launch-configurations.html

upvoted 3 times

A company hosts its web applications in the AWS Cloud. The company configures Elastic Load Balancers to use certificates that are imported into AWS Certificate Manager (ACM). The company's security team must be notified 30 days before the expiration of each certificate.

What should a solutions architect recommend to meet this requirement?

- A. Add a rule in ACM to publish a custom message to an Amazon Simple Notification Service (Amazon SNS) topic every day, beginning 30 days before any certificate will expire.
- B. Create an AWS Config rule that checks for certificates that will expire within 30 days. Configure Amazon EventBridge (Amazon CloudWatch Events) to invoke a custom alert by way of Amazon Simple Notification Service (Amazon SNS) when AWS Config reports a noncompliant resource. **Most Voted**
- C. Use AWS Trusted Advisor to check for certificates that will expire within 30 days. Create an Amazon CloudWatch alarm that is based on Trusted Advisor metrics for check status changes. Configure the alarm to send a custom alert by way of Amazon Simple Notification Service (Amazon SNS).
- D. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to detect any certificates that will expire within 30 days. Configure the rule to invoke an AWS Lambda function. Configure the Lambda function to send a custom alert by way of Amazon Simple Notification Service (Amazon SNS).

Correct Answer: B

Community vote distribution

D (49%)

B (49%)

2

Comments

LeGloupier **Highly Voted** 3 years, 7 months ago

B
AWS Config has a managed rule
named acm-certificate-expiration-check
to check for expiring certificates
(configurable number of days)
upvoted 99 times

LeGloupier 3 years, 7 months ago

<https://aws.amazon.com/premiumsupport/knowledge-center/acm-certificate-expiration/>
upvoted 15 times

ChrisG1454 3 years, 3 months ago

Answer B and answer D are possible according to this article.
So, need to read B & D carefully to determine the most suitable answer.

Reference: <https://aws.amazon.com/premiumsupport/knowledge-center/acm-certificate-expiration/>
upvoted 9 times

TTaws 2 years, 11 months ago

Its B, simply because in option D - event bridge cannot "detect" anything.
upvoted 9 times

RupeC 2 years, 10 months ago

My understanding is that the ACM sends a Cert Expiration event to EventBridge. Thus EB. does not need to detect anything.
upvoted 3 times

pentium75 2 years, 5 months ago

"ACM sends a Cert Expiration event to EventBridge" yes, but 45 (not 30) days before expiration.
upvoted 2 times

mrkmtai 2 years, 1 month ago

This can be configured to whatever you need in the Days to expiry box
upvoted 2 times

darekw 2 years, 9 months ago

AWS Certificate Manager (ACM) now publishes certificate metrics and events through Amazon CloudWatch and Amazon EventBridge.

<https://aws.amazon.com/about-aws/whats-new/2021/03/aws-certificate-manager-provides-certificate-expiry-monitoring-through-amazon-cloudwatch/>

upvoted 4 times

Bayebrymo 2 years, 1 month ago

But from the link shared. It doesn't state that you should invoke lambda function as stated in option D. Option B is explicitly stated in the article as also worded in option B. So I think B should be the answer. My thought though. I stand to be corrected.

upvoted 8 times

824c449 2 years, 1 month ago

It does not have a built-in rule for checking the expiration of ACM certificates directly.

upvoted 2 times

Mia2009687 2 years, 11 months ago

B costs more than D

To get a notification that your certificate is about to expire, use one of the following methods:

Use the ACM API in Amazon EventBridge to configure the ACM Certificate Approaching Expiration event.
Create a custom EventBridge rule to receive email notifications when certificates are nearing the expiration date.
Use AWS Config to check for certificates that are nearing the expiration date.
If you use AWS Config for this resolution, then be aware of the following:

Before you set up the AWS Config rule, create the Amazon Simple Notification Service (Amazon SNS) topic and EventBridge rule. This makes sure that all non-compliant certificates invoke a notification before the expiration date.

Activating AWS Config incurs an additional cost based on usage. For more information, see AWS Config pricing.

<https://repost.aws/knowledge-center/acm-certificate-expiration>

upvoted 5 times

pentium75 2 years, 5 months ago

Nobody asked for cost optimization.

upvoted 15 times

ManoAni **Highly Voted** 3 years, 7 months ago

Selected Answer: B

<https://aws.amazon.com/premiumsupport/knowledge-center/acm-certificate-expiration/>

upvoted 20 times

In2718 **Most Recent** 9 months, 4 weeks ago

Selected Answer: B

Breaking the tie with Chat AI (even rhymes)) - Use the managed AWS Config rule acm-certificate-expiration-check with daysToExpiration = 30. When the rule flags a noncompliant certificate, have EventBridge trigger an SNS notification to the security team.

upvoted 2 times

OrLe 10 months, 1 week ago

Selected Answer: B

aws config provide a managed rule

upvoted 1 times

caca783 11 months, 3 weeks ago

Selected Answer: B

AWS Config provides a managed rule called acm-certificate-expiration-check. This rule checks if ACM certificates in your account are marked for expiration within a specified number of days (default is 30 days)

upvoted 2 times

kaleido 12 months ago

Selected Answer: D

I chose option D. Option B works, but is recommended for beyond 45 days.

upvoted 2 times

Charlesvg 1 year ago

Selected Answer: D

i asked ChatGPT about the best between B & D

It said D because eventBridge is natively connected with ACM whereas AWS Config does not natively support certificates expiration
upvoted 2 times

sagarvh 1 year ago

Selected Answer: D

Answer is D

upvoted 2 times

network_enthusiast 1 year, 1 month ago

Selected Answer: D

Amazon Event Bridge can detect certificate expiry event for certificate managed by ACM.

upvoted 1 times

dattateja8 1 year, 2 months ago

Selected Answer: B

B

AWS Config has a managed rule named acm-certificate-expiration-check to check for expiring certificates (configurable number of days)

upvoted 1 times

dattateja8 1 year, 2 months ago

Selected Answer: D

AWS Config does not have in-built feature to check certificate expiry and it has to use out of the box managed rule to check like AWS Lambda. In Answer B it is not mentioned to use AWS Lambda. So the best option here is D.

upvoted 1 times

gigilla 1 year, 3 months ago

Selected Answer: B

B

AWS Config has a managed rule named acm-certificate-expiration-check to check for expiring certificates (configurable number of days)

upvoted 1 times

kg508 1 year, 3 months ago

Selected Answer: D

B와 헷갈렸는데, 알람에 대한 구체적인 방법이 D에 있어 이것으로 선택했습니다
eventbridge만으로는 보안팀에게 알람을 전달하기에 부족하다고 생각합니다

upvoted 1 times

iamroyalty_k 1 year, 3 months ago

Selected Answer: D

Why Option D is Correct?

- ✓ EventBridge can trigger an alert when an ACM certificate is about to expire.
- ✓ Lambda can process the event and send a notification through Amazon SNS to notify the security team.
- ✓ SNS ensures the notifications are delivered via email, SMS, or other means based on the team's preference.
- ✓ This is an automated, scalable, and AWS-native solution.

Why Not the Other Options?

- A - ✗ ACM does not support custom rules or direct SNS integration for expiring certificates.
- B - ✗ AWS Config does not have a built-in rule to check ACM certificate expiration. Creating a custom rule would add unnecessary complexity.
- C - ✗ Trusted Advisor does not check ACM certificates for expiration. It only monitors IAM certificate expiration.

upvoted 2 times

Dharmarajan 1 year, 4 months ago

Selected Answer: B

B or D both will work. Per the documentation:

To get a notification that your certificate is about to expire, use one of the following methods:

Use the ACM API in Amazon EventBridge to configure the ACM Certificate Approaching Expiration event.
Create a custom EventBridge rule to receive email notifications when certificates are near the expiration date.
Use AWS Config to check for certificates that are near the expiration date.
Create an Amazon CloudWatch alarm based on a static threshold when certificates are near the expiration date.

upvoted 1 times

itsmeDiyan 1 year, 4 months ago

Selected Answer: D

Use EventBridge, Lambda, and SNS
upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: C

Based on this article it is C if the notification is within 45 days, otherwise use B
<https://repost.aws/knowledge-center/acm-certificate-expiration>
upvoted 1 times

A company's dynamic website is hosted using on-premises servers in the United States. The company is launching its product in Europe, and it wants to optimize site loading times for new European users. The site's backend must remain in the United States. The product is being launched in a few days, and an immediate solution is needed. What should the solutions architect recommend?

- A. Launch an Amazon EC2 instance in us-east-1 and migrate the site to it.
- B. Move the website to Amazon S3. Use Cross-Region Replication between Regions.
- C. Use Amazon CloudFront with a custom origin pointing to the on-premises servers. **Most Voted**
- D. Use an Amazon Route 53 geoproximity routing policy pointing to on-premises servers.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: C

CORRECT

C. Use Amazon CloudFront with a custom origin pointing to the on-premises servers.

Amazon CloudFront is a content delivery network (CDN) that speeds up the delivery of static and dynamic web content, such as HTML, CSS, JavaScript, images, and videos. By using CloudFront, the company can distribute the content of their website from edge locations that are closer to the users in Europe, reducing the loading times for these users.

To use CloudFront, the company can set up a custom origin pointing to their on-premises servers in the United States. CloudFront will then cache the content of the website at edge locations around the world and serve the content to users from the location that is closest to them. This will allow the company to optimize the loading times for their European users without having to move the backend of the website to a different region.

upvoted 40 times

Euowelllima 2 years, 9 months ago

excelente explicação

upvoted 3 times

TariqKipkemei 3 years, 3 months ago

good explanation..thanks

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

WRONG

Option A (launch an Amazon EC2 instance in us-east-1 and migrate the site to it) would not address the issue of optimizing loading times for European users.

Option B (move the website to Amazon S3 and use Cross-Region Replication between Regions) would not be an immediate solution as it would require time to set up and migrate the website.

Option D (use an Amazon Route 53 geoproximity routing policy pointing to on-premises servers) would not be suitable because it would not improve the loading times for users in Europe.

upvoted 19 times

SVDK 2 years, 5 months ago

S3 doesn't support dynamic website hosting. Therefore, can be ruled out.

upvoted 5 times

allanm 10 months, 1 week ago

So you agree that option C is correct then?

upvoted 1 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: C

C. This solution leverages the global network of CloudFront edge locations to cache and serve the website's static content from the edge locations closest to the European users.

A. Hosting the website in a single region would still result in increased latency for European users accessing the site.

B. Moving the website to S3 and implementing Cross-Region Replication would distribute the website's content across multiple regions, including Europe. S3 is primarily used for static content hosting, and it does not provide server-side processing capabilities necessary for dynamic website functionality.

D. Using a geoproximity routing policy in Route 53 would allow you to direct traffic to the on-premises servers based on the geographic location of the users. However, this option does not optimize site loading times for European users as it still requires them to access the website from the on-premises servers in the United States. It does not leverage the benefits of content caching and edge locations for improved performance.

upvoted 5 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: C

CloudFront is the most suitable for reducing latency here.

upvoted 2 times

ChymKuBoy 1 year, 7 months ago

Selected Answer: C

C for sure

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: C

Ans C...

I was going for D, then I saw (CookieMr, 1yr 2mth ago) "Route 53 geoproximity routing policy in does allow you to direct traffic to the on-premises servers based on the geographic location of the users - BUT does not optimised site loading times for European users as it still requires them to access the website from the on-prem servers in the US."

upvoted 2 times

Hkayne 2 years, 1 month ago

Selected Answer: C

C is the appropriate answer

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

Immediate solution is C

A: Requires site migration so won't be immediate

B: Dynamic site cannot work from S3

D: Geoproximity routing policy finds a server close to the user so this makes no sense

C: Not ideal but best option given the "immediate" requirement as CloudFront is a CDN so it will cache whatever is possible as close to the user giving best performance in this case (i.e. when compared to other options)

upvoted 3 times

Ruffyit 2 years, 7 months ago

C. Use Amazon CloudFront with a custom origin pointing to the on-premises servers.

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

The key reasons are:

CloudFront can cache static content close to European users using edge locations, improving site performance.

The custom origin feature allows seamlessly integrating the CloudFront CDN with existing on-premises servers.

No changes are needed to the site backend or servers. CloudFront just acts as a globally distributed cache.

This can be set up very quickly, meeting the launch deadline.

Other options like migrating to EC2 or S3 would require more time and changes. CloudFront is an easier lift.

Route 53 geoproximity routing alone would not improve performance much without a CDN.

upvoted 3 times

Bmarodi 3 years ago

Selected Answer: C

C is best solution.

upvoted 1 times

gustavtd 3 years, 5 months ago

Selected Answer: C

Within few days you can not do more than using CloudFront

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

kajal1206 3 years, 6 months ago

Selected Answer: C

C is correct answer

upvoted 1 times

koreanmonkey 3 years, 6 months ago

Selected Answer: C

CloudFront = CDN Service

upvoted 3 times

Liliwood 3 years, 6 months ago

C.

S3 Cross region Replication minimize latency but also copies objects across Amazon S3 buckets in different AWS Regions(data has to remain in origin) so B wrong.

Route 53 geo, does not help reducing the latency.

upvoted 3 times

Wpcorgan 3 years, 6 months ago

C is correct

upvoted 1 times

Hunkie 3 years, 7 months ago

Same question with detailed explanation

<https://www.examttopics.com/discussions/amazon/view/27898-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 4 times

A company wants to reduce the cost of its existing three-tier web architecture. The web, application, and database servers are running on Amazon EC2 instances for the development, test, and production environments. The EC2 instances average 30% CPU utilization during peak hours and 10% CPU utilization during non-peak hours.

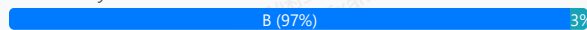
The production EC2 instances run 24 hours a day. The development and test EC2 instances run for at least 8 hours each day. The company plans to implement automation to stop the development and test EC2 instances when they are not in use.

Which EC2 instance purchasing solution will meet the company's requirements MOST cost-effectively?

- A. Use Spot Instances for the production EC2 instances. Use Reserved Instances for the development and test EC2 instances.
- B. Use Reserved Instances for the production EC2 instances. Use On-Demand Instances for the development and test EC2 instances. **Most Voted**
- C. Use Spot blocks for the production EC2 instances. Use Reserved Instances for the development and test EC2 instances.
- D. Use On-Demand Instances for the production EC2 instances. Use Spot blocks for the development and test EC2 instances.

Correct Answer: B

Community vote distribution



Comments

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: B

Spot blocks are not longer available, and you can't use spot instances on Prod machines 24x7, so option B should be valid.
upvoted 20 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: B

Option B, would indeed be the most cost-effective solution. Reserved Instances provide cost savings for instances that run consistently, such as the production environment in this case, while On-Demand Instances offer flexibility and are suitable for instances with variable usage patterns like the development and test environments. This combination ensures cost optimization based on the specific requirements and usage patterns described in the question.

upvoted 9 times

devmon 2 years, 9 months ago

In addition to this, we can set up an automated process to start and stop the EC2 instances in the test and dev environment

upvoted 3 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: B

Using Reserved Instances is suitable for predictable workloads while using On-demand is suitable for dev and test instances as we want to reduce their costs by paying for only resources that we use.

upvoted 1 times

EllenLiu 1 year, 6 months ago

Selected Answer: B

production - reserved
test/dev - on-demand

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B -

A: Spot instances are not ideal for production due to possibility of not running

C,D: Spot blocks discontinued.
That leaves option B

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: B

B is the right answer

upvoted 2 times

KTEgghead 1 year, 11 months ago

Either a trick question, or an old question. The answer is B; spot blocks were discontinued by Amazon, July 2021.

upvoted 3 times

TheTeaBoy 1 year, 7 months ago

Indeed, and anyway Spot blocks only run for a finite duration (1-6 hours), the question says these development and test instances need to run for a least 8, so that makes B the correct answer (even if Spot Blocks were still supported).

<https://aws.amazon.com/blogs/aws/new-ec2-spot-blocks-for-defined-duration-workloads/>

upvoted 2 times

MehulKapadia 2 years, 2 months ago

Selected Answer: B

Option A,C: Production workload must not run on spot instances.

Option D: No savings(compared to reserved Instances) for prod as its on-demand, Spot Blocks are no longer available.

So correct answer is B

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

Isn't this simple or am I thinking wrong?

"The production EC2 instances run 24 hours a day"

AC are not going to give the 24 hour usage as spot is for intermittent pattern.

D is just normal cost without any discounts for production

B use "reserved" instances so there is an option for discount in billing. On-demand for dev/test is ok as their usage patter doesn't really fall in reserved or spot usage discounts

upvoted 3 times

MrPCarrot 2 years, 6 months ago

B = Reserved for Prod and On Demand for Dev

upvoted 5 times

Bmarodi 3 years ago

Selected Answer: B

B meets the requirements, and most cost-effective.

upvoted 2 times

ChanghyeonYoon 3 years, 1 month ago

Selected Answer: B

Spot instances are not suitable for production due to the possibility of not running.

upvoted 3 times

alexiscloud 3 years, 2 months ago

Answer B:

Spot block are no longer available and you can't use spot instace on production

upvoted 2 times

Nandan747 3 years, 5 months ago

Selected Answer: B

Well, AWS has DISCONTINUED the Spot-Block option. so that rules out the two options that use spot-block. Wait, this question must be from SAA-C02 or even 01. STALE QUESTION. I don't think this will feature in SAA-C03. Anyhow, the most cost-effective solution would be Option "b"

upvoted 7 times

Wajif 3 years, 5 months ago

Selected Answer: B

Choosing B as spot blocks (Spot instances with a finite duration) are no longer offered since July 2021

upvoted 2 times

sparky231 3 years ago

https://aws.amazon.com/ec2/spot/?cards.sort-by=item.additionalFields.startDateTime&cards.sort-order=asc&trk=8e336330-37e5-41e0-8438-bc1c75320d09&sc_channel=ps&ef_id=CjwKCAjw67ajBhAVEiwA2g_jECglX_lcbqawbH-wVx2Y_EozBm8xv3g3Ci1eps0V49XcZRyfuy9xPhoCOkcQAvD_BwE:G:s&s_kwcid=AL!4422!3!517520538467!p!!g!!aws%20ec2%20spot!12831094520!1z2300635918

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: A

The most cost-effective solution for the company's requirements would be to use Spot Instances for the development and test EC2 instances and Reserved Instances for the production EC2 instances.

Spot Instances are a cost-effective choice for non-critical, flexible workloads that can be interrupted. Since the development and test EC2 instances are only needed for at least 8 hours per day and can be stopped when not in use, they would be a good fit for Spot Instances.

upvoted 2 times

PassNow1234 3 years, 5 months ago

The production EC2 instances run 24 hours a day.

upvoted 2 times

vincent2023 1 year, 4 months ago

Spot instances can't be used for production environment as they are not long lived in nature.

upvoted 1 times

pentium75 2 years, 5 months ago

A says to use Spot instances for production, which is nonsense as production must run "24 hours a day." Even DEV and TEST can't use Spot because you can't guarantee that they are available for "at least 8 hours each day". That rules out everything but B.

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Reserved Instances are a good fit for production EC2 instances that need to run 24 hours a day, as they offer a significant discount compared to On Demand Instances in exchange for a one-time payment and a commitment to use the instances for a certain period of time.

Option A is the correct answer because it meets the company's requirements for cost-effectively running the development and test EC2 instances and the production EC2 instances.

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option B is not the most cost-effective solution because it suggests using On-Demand Instances for the development and test EC2 instances, which would be more expensive than using Spot Instances. On-Demand Instances are a good choice for workloads that require a guaranteed capacity and can't be interrupted, but they are more expensive than Spot Instances.

Option C is not the correct solution because Spot blocks are a variant of Spot Instances that offer a guaranteed capacity and duration, but they are not available for all instance types and are not necessarily the most cost-effective option in all cases. In this case, it would be more cost-effective to use Spot Instances for the development and test EC2 instances, as they can be interrupted when not in use.

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option D is not the correct solution because it suggests using On-Demand Instances for the production EC2 instances, which would be more expensive than using Reserved Instances. On-Demand Instances are a good choice for workloads that require a guaranteed capacity and can't be interrupted, but they are more expensive than Reserved Instances in the long run. Using Reserved Instances for the production EC2 instances would offer a significant discount compared to On-Demand Instances in exchange for a one-time payment and a commitment to use the instances for a certain period of time.

upvoted 1 times

WherecanIstart 3 years, 3 months ago

Can't use Spot instances for Production environment that needs to run 24/7. That should tell you that Production instances can't have a downtime. Spot instances are used when an application or service can allow disruption and 24/7 production environment won't allow that.

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

A company has a production web application in which users upload documents through a web interface or a mobile app. According to a new regulatory requirement, new documents cannot be modified or deleted after they are stored. What should a solutions architect do to meet this requirement?

- A. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning and S3 Object Lock enabled. **Most Voted**
- B. Store the uploaded documents in an Amazon S3 bucket. Configure an S3 Lifecycle policy to archive the documents periodically.
- C. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning enabled. Configure an ACL to restrict all access to read-only.
- D. Store the uploaded documents on an Amazon Elastic File System (Amazon EFS) volume. Access the data by mounting the volume in read-only mode.

Correct Answer: A

Community vote distribution

A (100%)

Comments

123jh10 **Highly Voted** 3 years, 7 months ago

Selected Answer: A

You can use S3 Object Lock to store objects using a write-once-read-many (WORM) model. Object Lock can help prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. You can use S3 Object Lock to meet regulatory requirements that require WORM storage, or add an extra layer of protection against object changes and deletion.

Versioning is required and automatically activated as Object Lock is enabled.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock-overview.html>

upvoted 31 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: A

CORRECT

A. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning and S3 Object Lock enabled.

S3 Versioning allows multiple versions of an object to be stored in the same bucket. This means that when an object is modified or deleted, the previous version is preserved. S3 Object Lock adds additional protection by allowing objects to be placed under a legal hold or retention period, during which they cannot be deleted or modified. Together, S3 Versioning and S3 Object Lock can be used to meet the requirement of not allowing documents to be modified or deleted after they are stored.

upvoted 12 times

Buruguduystunstugudunstuy 3 years, 5 months ago

WRONG

Option B, storing the documents in an S3 bucket and configuring an S3 Lifecycle policy to archive them periodically, would not prevent the documents from being modified or deleted.

Option C, storing the documents in an S3 bucket with S3 Versioning enabled and configuring an ACL to restrict all access to read-only, would also not prevent the documents from being modified or deleted, since an ACL only controls access to the object and does not prevent it from being modified or deleted.

Option D, storing the documents on an Amazon Elastic File System (Amazon EFS) volume and accessing the data in read-only mode, would prevent the documents from being modified, but would not prevent them from being deleted.

upvoted 8 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: A

S3 Object Lock for preventing deletion as it uses Write-Once-Read-Many (WORM) and Bucket Versioning for preserving the older files and thereby preventing modification.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - 'S3 Object Lock' allows objects to be stored under legal retention regime during which they cannot be changed/deleted. Versioning helps tracking...

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A is the correct answer

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

"S3 Object Lock can help prevent Amazon S3 objects from being deleted or overwritten for a fixed amount of time or indefinitely. "

B is archiving which won't stop deletion

C ACL can be modified

D Sounds like there will be a write volume and a read volume which means write volume will have permissions for deletion

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

S3 Versioning ensures that all versions of an object are retained when overwritten or deleted - this prevents deletion.

S3 Object Lock can be used to apply a retention period and legal hold on objects to prevent them from being overwritten or deleted, even by users with full permissions.

Option B only archives objects on a schedule but does not prevent modification or deletion.

Option C uses ACLs which can still be overridden by users with full permissions.

Option D relies on the application to enforce mounting as read-only, which is not as robust as using S3 Object Lock.

upvoted 3 times

Subhrangsu 2 years, 8 months ago

Liked the explanation for option C.Thanks!

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: A

S3 Versioning allows you to preserve every version of a document as it is uploaded or modified. This prevents accidental or intentional modifications or deletions of the documents.

S3 Object Loc allows you to set a retention period or legal hold on the objects, making them immutable during the specified period. This ensures that the stored documents cannot be modified or deleted, even by privileged users or administrators.

B. Configuring an S3 Lifecycle policy to archive documents periodically does not guarantee the prevention of document modification or deletion after they are stored.

C. Enabling S3 Versioning alone does not prevent modifications or deletions of objects. Configuring an ACL does not guarantee the prevention of modifications or deletions by authorized users.

D. Using EFS does not prevent modifications or deletions of the documents by users or processes with write permissions.

upvoted 3 times

Bmarodi 3 years ago

Selected Answer: A

S3 Versioning and S3 Object Lock enabled meet the requirements, hence A is correct ans.

upvoted 3 times

SilentMilli 3 years, 5 months ago

Selected Answer: A

Option A. Store the uploaded documents in an Amazon S3 bucket with S3 Versioning and S3 Object Lock enabled. This will ensure that the documents cannot be modified or deleted after they are stored, and will meet the regulatory requirement. S3 Versioning allows you to store multiple versions of an object in the same bucket, and S3 Object Lock enables you to apply a retention policy to objects in the bucket to prevent their deletion.

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: A

Option A. Object Lock will prevent modifications to documents

upvoted 2 times

HarryZ 3 years, 6 months ago

Why not C

upvoted 3 times

JayBee65 3 years, 5 months ago

Configure an ACL to restrict all access to read-only would be you could not write the docs to the bucket in the first place.

upvoted 3 times

Wpcorgan 3 years, 6 months ago

A is correct

upvoted 1 times

flbcobra 3 years, 7 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock-overview.html>

upvoted 2 times

Evangelia 3 years, 7 months ago

Selected Answer: A

aaaaaaaaa

upvoted 1 times

Evangelia 3 years, 7 months ago

aaaaaaaaaaa

upvoted 1 times

A company has several web servers that need to frequently access a common Amazon RDS MySQL Multi-AZ DB instance. The company wants a secure method for the web servers to connect to the database while meeting a security requirement to rotate user credentials frequently.

Which solution meets these requirements?

- A. Store the database user credentials in AWS Secrets Manager. Grant the necessary IAM permissions to allow the web servers to access AWS Secrets Manager. **Most Voted**
- B. Store the database user credentials in AWS Systems Manager OpsCenter. Grant the necessary IAM permissions to allow the web servers to access OpsCenter.
- C. Store the database user credentials in a secure Amazon S3 bucket. Grant the necessary IAM permissions to allow the web servers to retrieve credentials and access the database.
- D. Store the database user credentials in files encrypted with AWS Key Management Service (AWS KMS) on the web server file system. The web server should be able to decrypt the files and access the database.

Correct Answer: A

Community vote distribution

A (100%)

Comments

123jh10 **Highly Voted** 3 years, 7 months ago

Selected Answer: A

Secrets Manager enables you to replace hardcoded credentials in your code, including passwords, with an API call to Secrets Manager to retrieve the secret programmatically. This helps ensure the secret can't be compromised by someone examining your code, because the secret no longer exists in the code. Also, you can configure Secrets Manager to automatically rotate the secret for you according to a specified schedule. This enables you to replace long-term secrets with short-term ones, significantly reducing the risk of compromise.

<https://docs.aws.amazon.com/secretsmanager/latest/userguide/intro.html>

upvoted 29 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: A

B. SSM OpsCenter is primarily used for managing and resolving operational issues. It is not designed to securely store and manage credentials like AWS Secrets Manager.

C. Storing credentials in an S3 bucket may provide some level of security, but it lacks the additional features and security controls offered by AWS Secrets Manager.

D. While using KMS for encryption is a good practice, managing credentials directly on the web server file system can introduce complexities and potential security risks. It can be challenging to securely manage and rotate credentials across multiple web servers, especially when considering scalability and automation.

In summary, option A is the recommended solution as it leverages AWS Secrets Manager, which is purpose-built for securely storing and managing secrets, and provides the necessary IAM permissions to allow the web servers to access the credentials securely.

upvoted 7 times

VenkatGovindu **Most Recent** 1 year, 2 months ago

Selected Answer: A

Rotate Key ==> Secrets Manager

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: A

AWS Secret's Manager is the most suitable option here as it provides rotation of credentials using Lambda.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - Secrets Manager, auto rotates...
upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A is the right answer, you should never store any of your credentials in files, even if they are encrypted. Also, secrets manager fulfills the credentials rotation condition.
upvoted 2 times

Duckydoo 1 year, 12 months ago

Selected Answer: A

While I agree that A is the most feasible answer, I don't see how it satisfies the "a secure method for the web servers to connect to the database" requirement. ASM is about securely storing and rotating secrets, but has nothing to do with "secure connection" between the web servers and RDS. That would require something like IAM DB authentication, which is not even mentioned.
upvoted 3 times

CanoSys 1 year, 10 months ago

Yeah, It's a stupid question.
upvoted 1 times

Z0r 2 years, 1 month ago

rotation = SM
upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

AWS Secrets Manager is best for storing credentials and supports auto rotation so A is the best choice
upvoted 4 times

MrPCarrot 2 years, 6 months ago

A = Rotation of user credentials can be automated using Secrets Manager.
upvoted 2 times

Ruffyyit 2 years, 7 months ago

option A is the recommended solution as it leverages AWS Secrets Manager, which is purpose-built for securely storing and managing secrets, and provides the necessary IAM permissions to allow the web servers to access the credentials securely.
upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

AWS Secrets Manager to the rescue....up up and awaaaay
upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

The correct answer is A.

Here is the explanation:

AWS Secrets Manager is a service that helps you store, manage, and rotate secrets. Secrets Manager is a good choice for storing database user credentials because it is secure and scalable.

IAM permissions can be used to grant web servers access to AWS Secrets Manager. This will allow the web servers to retrieve the database user credentials from Secrets Manager and use them to connect to the database.

Rotation of user credentials can be automated using Secrets Manager. This will ensure that the database user credentials are rotated on a regular basis, meeting the security requirement.

upvoted 3 times

Bmarodi 3 years ago

Selected Answer: A

Option A is ans.
upvoted 3 times

vherman 3 years, 3 months ago

Selected Answer: A

A is correct
upvoted 2 times

thensanity 3 years, 5 months ago

literally screams for AWS secrets manager to rotate the credentials

upvoted 5 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: A

CORRECT

Option A. Store the database user credentials in AWS Secrets Manager. Grant the necessary IAM permissions to allow the web servers to access AWS Secrets Manager.

Option A is correct because it meets the requirements specified in the question: a secure method for the web servers to connect to the database while meeting a security requirement to rotate user credentials frequently. AWS Secrets Manager is designed specifically to store and manage secret-like database credentials, and it provides an automated way to rotate secrets every time they are used, ensuring that the secrets are always fresh and secure. This makes it a good choice for storing and managing the database user credentials in a secure way.

upvoted 6 times

Buruguduystunstugudunstuy 3 years, 5 months ago

WRONG

Option B, storing the database user credentials in AWS Systems Manager OpsCenter, is not a good fit for this use case because OpsCenter is a tool for managing and monitoring systems, and it is not designed for storing and managing secrets.

Option C, storing the database user credentials in a secure Amazon S3 bucket, is not a secure option because S3 buckets are not designed to store secrets. While it is possible to store secrets in S3, it is not recommended because S3 is not a secure secrets management service and does not provide the same level of security and automation as AWS Secrets Manager.

upvoted 4 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option D, storing the database user credentials in files encrypted with AWS Key Management Service (AWS KMS) on the web server file system, is not a secure option because it relies on the security of the web server file system, which may not be as secure as a dedicated secrets management service like AWS Secrets Manager. Additionally, this option does not meet the requirement to rotate user credentials frequently because it does not provide an automated way to rotate the credentials.

upvoted 6 times

A company hosts an application on AWS Lambda functions that are invoked by an Amazon API Gateway API. The Lambda functions save customer data to an Amazon Aurora MySQL database. Whenever the company upgrades the database, the Lambda functions fail to establish database connections until the upgrade is complete. The result is that customer data is not recorded for some of the event.

A solutions architect needs to design a solution that stores customer data that is created during database upgrades. Which solution will meet these requirements?

- A. Provision an Amazon RDS proxy to sit between the Lambda functions and the database. Configure the Lambda functions to connect to the RDS proxy.
- B. Increase the run time of the Lambda functions to the maximum. Create a retry mechanism in the code that stores the customer data in the database.
- C. Persist the customer data to Lambda local storage. Configure new Lambda functions to scan the local storage to save the customer data to the database.
- D. Store the customer data in an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Create a new Lambda function that polls the queue and stores the customer data in the database. **Most Voted**

Correct Answer: D

Community vote distribution

D (61%)

A (39%)

Comments

brushek **Highly Voted** 3 years, 7 months ago

Selected Answer: A

<https://aws.amazon.com/rds/proxy/>

RDS Proxy minimizes application disruption from outages affecting the availability of your database by automatically connecting to a new database instance while preserving application connections. When failovers occur, RDS Proxy routes requests directly to the new database instance. This reduces failover times for Aurora and RDS databases by up to 66%.

upvoted 59 times

PassNow1234 3 years, 5 months ago

This is MySQL Database. RDS proxy = no no

upvoted 5 times

Robrobtutu 3 years, 1 month ago

It literally says RDS Proxy is available for Aurora MySQL on the link in the comment you're replying to.

upvoted 6 times

pentium75 2 years, 5 months ago

But still RDS proxy won't help because during upgrades there is no database that it could proxy to.

upvoted 4 times

attila9778 3 years, 6 months ago

Aurora supports RDS proxy!

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 6 times

bgsanata 3 years ago

This is incorrect as nowhere in the question is mentioned the RDS have more than 1 instance. So... when the instance is down for maintenance there is no second instance to which RDS Proxy can redirect the requests.

The correct answer is D.

upvoted 30 times

Abdou1604 2 years, 10 months ago

rDS PROXY Supports RDS (MySQL, PostgreSQL, MariaDB, MS SQL Server) and Aurora (MySQL, PostgreSQL)

upvoted 3 times

aaroncelestin 2 years, 9 months ago

You are going to tell your boss that the customer is going to occasionally lose //only// 33% of their data, as if that's just acceptable?

upvoted 11 times

ekisako 2 years, 7 months ago

lol read and understand carefully, it says REDUCES FAILOVER TIMES BY UP TO 66%.

upvoted 2 times

123jh10 **Highly Voted** 3 years, 7 months ago

Selected Answer: D

The answer is D.

RDS Proxy doesn't support Aurora DBs. See limitations at:

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 36 times

JayBee65 3 years, 5 months ago

It does, according to that link

upvoted 1 times

adeyinkaamole 2 years, 9 months ago

This not RDS supports Aurora mysql database. All the limitations listed in the link you posted above are not related to the question, hence the answer is B

upvoted 2 times

adeyinkaamole 2 years, 9 months ago

I meant the answer answer is A

upvoted 1 times

tinyfoot 3 years, 6 months ago

Actually RDS Proxy supports Aurora DBs running on PostgreSQL and MySQL.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Concepts.Aurora_Fea_Regions_DB-eng.Feature.RDS_Proxy.html

With RDS proxy, you only expose a single endpoint for request to hit and any failure of the primary DB in a Multi-AZ configuration is will be managed automatically by RDS Proxy to point to the new primary DB. Hence RDS proxy is the most efficient way of solving the issue as additional code change is required.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.howitworks.html>

upvoted 11 times

Duke_YU 3 years, 2 months ago

The question doesn't say the RDS is deployed in a Mutli-AZ mode. which means RDS is not accessible during upgrade anyway. RDS proxy couldn't resolve the DB HA issue. The question is looking for a solution to store the data during DB upgrade. I don't know RDS proxy very well, but the RDS proxy introduction doesn't mention it has the capability of storing data. So, answer A couldn't store the data created during the DB upgrade.

I'm assuming this is a bad question design. The expected answer is A, but the question designer missed some important information.

upvoted 7 times

rismail 3 years, 1 month ago

<https://aws.amazon.com/rds/proxy/>, if you go down the page, you will see that RDS is deployed in Multi-AZ (mazon RDS Proxy is highly available and deployed over multiple Availability Zones (AZs) to protect you from infrastructure failure. Each AZ runs on its own physically distinct, independent infrastructure and is engineered to be highly reliable. In the unlikely event of an infrastructure failure, the RDS Proxy endpoint remains online and consistent allowing your application to continue to run database operations.) from the link.

upvoted 1 times

gcmrjbr 3 years, 6 months ago

You can use RDS Proxy with Aurora Serverless v2 clusters but not with Aurora Serverless v1 clusters.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 8 times

jorgemenegaz **Most Recent** 1 year ago

Selected Answer: A

Lambda needs to be in the same network communication to aurora, you can do this using RDS Proxy.

upvoted 1 times

dattateja8 1 year, 2 months ago

Selected Answer: D

Ans D: This is the best solution. By using an SQS FIFO queue, you can store customer data reliably even when the database is temporarily unavailable. The Lambda function can immediately write the data to the queue when the database is not available. A second Lambda function can then poll the queue and process the customer data when the database becomes available again. This ensures that no data is lost, and processing happens in an orderly manner.

upvoted 1 times

Vandaman 1 year, 3 months ago

Selected Answer: D

Scenario describes 1 DB and when it's unavailable due to an upgrade. Best solution is to queue incoming data until DB is available.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: D

Initially I voted A. But that's wrong for one reason. RDS proxy won't help if there is no more than 1 database, but again - if there is just one database, the application is down during the downtime anyways, there is nothing that can be done.

However, a SQS queue is a better option for the reason that the messages persist until marked processed. That guarantee alone is good enough to make D a better choice.

But jeez! talk about trickery!! deceptively misleading!

upvoted 2 times

zdi561 1 year, 4 months ago

Selected Answer: A

D does not work. because during db upgrade the connection is disrupted for a while during which the request in SQS will be consumed and lost

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: A

Google does AWS Aurora DB proxy help DB upgrade, proxy will cache transaction,

upvoted 1 times

AshishDhole 1 year, 4 months ago

Selected Answer: D

decouple the application so that we can save the failed transactions and then pull after the upgrades complete, so D should be the correct answer (SQS Queue)

upvoted 1 times

jayessh 1 year, 7 months ago

Selected Answer: D

this is not failover situation so no need to use RDS proxy. I don't remember RDS proxy to hold data in case for DB unavailability

upvoted 1 times

ChymKuBoy 1 year, 7 months ago

Selected Answer: A

A for sure

upvoted 2 times

lixep 1 year, 8 months ago

Selected Answer: D

It's not a failover situation, it's just temporarily unavailable.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - as excellently explained by SaurabhTiwari1 (9 mth ago) "The original question was about handling a situation where the database is unavailable due to an upgrade, not a failover situation. During a database upgrade, the database instance is not available, and RDS Proxy would not be able to connect to a new database instance because there isn't one."

In this specific scenario, using Amazon SQS as described in option D provides a buffer for the incoming data during the period when the database is unavailable. This ensures that no data is lost, and it can be written to the database once the upgrade is complete

upvoted 4 times

rpmaws 1 year, 9 months ago

Selected Answer: D

A is incorrect

upvoted 1 times

d7d08dc 1 year, 9 months ago

the answer is D:

This solution ensures that customer data is not lost during database upgrades. The data is stored in the FIFO queue until the database is available again. The new Lambda function can then process the data from the queue and store it in the database. This design provides a buffer for the customer data and decouples the data ingestion from the data processing, increasing the resilience of the system during database upgrades.

upvoted 2 times

ChinthaGurumurthi 1 year, 10 months ago

Selected Answer: D

By using Amazon RDS Proxy, you can allow your applications to pool and share database connections to improve their ability to scale. RDS Proxy makes applications more resilient to database failures by automatically connecting to a standby DB instance while preserving application connections. Using RDS Proxy, you can handle unpredictable surges in database traffic. Otherwise, these surges might cause issues due to oversubscribing connections or new connections being created at a fast rate.

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: D

D is the correct answer because it offers storage for the data while the database is updating, using SQS is better than using RDS proxy, because SQS is an independent service and it decouples the system, while RDS proxy might also face some trouble connecting to the database while it's updating.

upvoted 5 times

A survey company has gathered data for several years from areas in the United States. The company hosts the data in an Amazon S3 bucket that is 3 TB in size and growing. The company has started to share the data with a European marketing firm that has S3 buckets. The company wants to ensure that its data transfer costs remain as low as possible.

Which solution will meet these requirements?

- A. Configure the Requester Pays feature on the company's S3 bucket. **Most Voted**
- B. Configure S3 Cross-Region Replication from the company's S3 bucket to one of the marketing firm's S3 buckets.
- C. Configure cross-account access for the marketing firm so that the marketing firm has access to the company's S3 bucket.
- D. Configure the company's S3 bucket to use S3 Intelligent-Tiering. Sync the S3 bucket to one of the marketing firm's S3 buckets.

Correct Answer: A

Community vote distribution

A (46%) B (45%) 8%

Comments

Six_Fingered_Jose **Highly Voted** 3 years, 7 months ago

Selected Answer: B

this question is too vague imho

if the question is looking for a way to incur charges to the European company instead of the US company, then requester pay makes sense.

if they are looking to reduce overall data transfer cost, then B makes sense because the data does not leave the AWS network, thus data transfer cost should be lower technically?

A. makes sense because the US company saves money, but the European company is paying for the charges so there is no overall saving in cost when you look at the big picture

I will go for B because they are not explicitly stating that they want the other company to pay for the charges
upvoted 77 times

rushi0611 3 years, 1 month ago

Agree, B) Cross Region Replication: \$0.02/GB

A) over the internet it is \$0.09/GB

Answer is B

upvoted 9 times

thwvthunder 2 years, 9 months ago

is S3 Cross-Region Replication works between 2 separate aws accounts? shouldn't the answer is C?

upvoted 2 times

awsgeek75 2 years, 5 months ago

"C" can't be the answer as it means the S3 data access survey company will incur the cost when data is accessed by European company.

upvoted 1 times

MutiverseAgent 2 years, 11 months ago

I agree, also the question says that the target firm "has S3 buckets." so I think that is a clue to say they can accept replication data on any of those buckets.

upvoted 2 times

SaurabhTiwari1 2 years, 5 months ago

The company wants to ensure that (its) data transfer costs remain as low as possible.

means Company in US wants to ensure...

So its Option A

upvoted 3 times

123jh10 **Highly Voted** 3 years, 7 months ago

Selected Answer: A

"Typically, you configure buckets to be Requester Pays buckets when you want to share data but not incur charges associated with others accessing the data. For example, you might use Requester Pays buckets when making available large datasets, such as zip code directories, reference data, geospatial information, or web crawling data."

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/RequesterPaysBuckets.html>

upvoted 53 times

In2718 **Most Recent** 9 months, 4 weeks ago

Selected Answer: A

And for answer B/D - Replication/sync would make the survey company pay ongoing transfer and extra storage—more expensive.

upvoted 1 times

In2718 9 months, 4 weeks ago

Selected Answer: A

Chat AI for the tie)) - would answer C be less expensive for the owner than the current status

Current status (no cross-account, just sharing data somehow):

If the European firm downloads the data from the U.S. company's S3 bucket, the bucket owner (U.S. company) pays the data transfer out charges.

Option C (cross-account access, but no Requester Pays):

Cross-account access only controls permissions. It does not change the billing model.

→ The U.S. company would still pay the data transfer out charges when the European firm retrieves the objects.

So: Cost for the U.S. company = same as current status. Not less expensive.

upvoted 1 times

network_enthusiast 1 year, 1 month ago

Selected Answer: A

A is the correct answer

upvoted 1 times

dattateja8 1 year, 2 months ago

Selected Answer: A

The Requester Pays feature in Amazon S3 allows the company to charge the marketing firm for the data transfer costs when they access the company's S3 bucket. By enabling this feature, the marketing firm will be responsible for the data transfer costs when they access the data, rather than the company. This ensures that the company does not incur high data transfer costs when sharing large datasets with an external party, especially across regions.

upvoted 1 times

mc0226 1 year, 3 months ago

Selected Answer: A

- In general, bucket owners pay for all Amazon S3 storage and data transfer costs associated with their bucket
- With Requester Pays buckets, the requester instead of the bucket owner pays the cost of the request and the data download from the bucket

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

The question is clear and answer too. undoubtedly A - one can have "Requester pays" feature on a S3 bucket! My retention isn't great but I remember this from the S3 lesson!

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/RequesterPaysBuckets.html>

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: A

B is not right because we do not know the usage pattern by the marketing company. copying whole data may cost much more

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: A

The survey company wants to share its data transfer costs as low as possible, to replicate the data to the Europe company's account, there will still be some data transfer costs for survey company to pay, so it will be better to request the Europe's market company to pay.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Another reason to select A is the A can scale in the case of more than one company want their survey dataset. B cannot scale.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

For B, When you use S3 Cross-Region Replication (CRR), the source bucket owner (in this case, the company sharing the data) is responsible for paying the data replication costs, including:

Data transfer fees: The cost of transferring data from the source S3 bucket to the destination bucket in a different AWS Region.

Replication requests: Charges for each PUT request made during replication to the destination bucket.

upvoted 1 times

chirag_a_parikh 1 year, 6 months ago

Selected Answer: B

Answer is B, cross region replication.

upvoted 1 times

Balliache520505 1 year, 6 months ago

Selected Answer: A

Option A. When you use S3 Cross-Region Replication (CRR), you will incur data transfer costs. These costs include a fee for transferring data between regions, which is approximately \$0.02 per GB.

upvoted 1 times

Ode7d1b 1 year, 6 months ago

Selected Answer: A

The Requester Pays feature is the most cost-effective solution for this scenario, as it shifts the data transfer costs to the marketing firm while keeping the data accessible.

upvoted 1 times

Carlini2020 1 year, 6 months ago

Selected Answer: A

B has actually more cost. A is the answer.

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: A

If I understand the question correctly, the owner is already paying to maintain the data (size is still growing) in its S3 bucket. The owner wants to ensure that its data transfer costs remain as low as possible (implies that it is best if the transfer cost is \$0).

For option A, the requestor will have to bear the cost of the data transfer request from the bucket, with the data owner incurring a \$0 transfer cost, which is optimal for the data owner.

This question does not state that the original data owner must also consider the cost incurred by the requestor and then find an optimal ways to share the data transfer cost for both parties.

My 2 cents' worth.

upvoted 2 times

Neilossi 1 year, 6 months ago

Selected Answer: A

The company wants to ensure that "its" data transfer costs remain as low as possible, so I choose A

upvoted 1 times

DevanshGupta 1 year, 7 months ago

Per my understanding, the company is already paying to maintain the data(data keeps growing) in the S3 bucket. The company wants to ensure that its data transfer costs remain as low as possible (implies that it is best if the transfer cost is \$0).

As per option A, the requestor would bear the cost of the request and the data downloaded from the bucket, causing the data owner to incur a \$0 transfer cost, which is optimal for the data owner.

The question does not say that the data owner must also consider the cost incurred by the requestor and then find an optimal cost solution.

upvoted 2 times

A company uses Amazon S3 to store its confidential audit documents. The S3 bucket uses bucket policies to restrict access to audit team IAM user credentials according to the principle of least privilege. Company managers are worried about accidental deletion of documents in the S3 bucket and want a more secure solution.

What should a solutions architect do to secure the audit documents?

- A. Enable the versioning and MFA Delete features on the S3 bucket. **Most Voted**
- B. Enable multi-factor authentication (MFA) on the IAM user credentials for each audit team IAM user account.
- C. Add an S3 Lifecycle policy to the audit team's IAM user accounts to deny the s3:DeleteObject action during audit dates.
- D. Use AWS Key Management Service (AWS KMS) to encrypt the S3 bucket and restrict audit team IAM user accounts from accessing the KMS key.

Correct Answer: A

Community vote distribution

A (100%)

Comments

123jhl0 **Highly Voted** 3 years, 7 months ago

Selected Answer: A

Same as Question #44

upvoted 16 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: A

Using MFA Delete is the most suitable option here as it avoids deletion of files.

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A is the correct answer.

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

Accidental deletion is the key. Deletion is allowed but MFA deletion ensures that deletion requires an additional step.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/MultiFactorAuthenticationDelete.html>

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

Enable the versioning to ensure restoration in case of accidental deletion and MFA Delete for double verification before deletion.

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

Versioning will keep multiple variants of an object in case one is accidentally or intentionally deleted - the previous versions can still be restored.

MFA Delete requires additional authentication to permanently delete an object version. This prevents accidental deletion

upvoted 3 times

cookieMr 2 years, 11 months ago

B. Enabling MFA on the IAM user credentials adds an extra layer of security to the user authentication process. However, it does not specifically address the concern of accidental deletion of documents in the S3 bucket.

C. Adding an S3 Lifecycle policy to deny the delete action during audit dates would prevent intentional deletions during specific time periods. However, it does not address accidental deletions that can occur at any time.

D. Using KMS for encryption and restricting access to the KMS key provides additional security for the data stored in the S3 . However, it does not directly prevent accidental deletion of documents in the S3.

Enabling versioning and MFA Delete on the S3 (option A) is the most appropriate solution for securing the audit documents. Versioning ensures that multiple versions of the documents are stored, allowing for easy recovery in case of accidental deletions. Enabling MFA Delete requires the use of multi-factor authentication to authorize deletion actions, adding an extra layer of protection against unintended deletions.

upvoted 3 times

beginnercloud 3 years ago

Selected Answer: A

A is answer.

upvoted 1 times

Bmarodi 3 years ago

Selected Answer: A

A is answer.

upvoted 1 times

Robrobtutu 3 years, 1 month ago

Selected Answer: A

A is correct.

upvoted 1 times

remand 3 years, 4 months ago

Selected Answer: A

only accidental deletion should be avoided. IAM policy will completely remove their access.hence, MFA is the right choice.

upvoted 2 times

karbob 3 years, 5 months ago

what about : IAM policies are used to specify permissions for AWS resources, and they can be used to allow or deny specific actions on those resources.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyDeleteObject",
      "Effect": "Deny",
      "Action": "s3:DeleteObject",
      "Resource": [
        "arn:aws:s3:::my-bucket/my-object",
        "arn:aws:s3:::my-bucket"
      ]
    }
  ]
}
```

upvoted 2 times

remand 3 years, 4 months ago

only accidental deletion should be avoided. IAM policy will completely remove their access.hence, MFA is the right choice.

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: A

The solution architect should do Option A: Enable the versioning and MFA Delete features on the S3 bucket.

This will secure the audit documents by providing an additional layer of protection against accidental deletion. With versioning enabled, any deleted or overwritten objects in the S3 bucket will be preserved as previous versions, allowing the company to recover them if needed. With MFA Delete enabled, any delete request made to the S3 bucket will require the use of an MFA code, which provides an additional layer of security.

upvoted 4 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option B: Enable multi-factor authentication (MFA) on the IAM user credentials for each audit team IAM user account, would not provide protection against accidental deletion.

Option C: Adding an S3 Lifecycle policy to the audit team's IAM user accounts to deny the s3:DeleteObject action during audit dates, which would not provide protection against accidental deletion outside of the specified audit dates.

Option D: Use AWS Key Management Service (AWS KMS) to encrypt the S3 bucket and restrict audit team IAM user accounts from accessing the KMS key, would not provide protection against accidental deletion.

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: A

A is the right answer

upvoted 1 times

Wpcorgan 3 years, 6 months ago

A is correct

upvoted 1 times

Jtic 3 years, 7 months ago

Selected Answer: A

Enable the versioning and MFA Delete features on the S3 bucket.

upvoted 1 times

A company is using a SQL database to store movie data that is publicly accessible. The database runs on an Amazon RDS Single-AZ DB instance. A script runs queries at random intervals each day to record the number of new movies that have been added to the database. The script must report a final total during business hours.

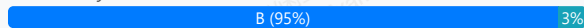
The company's development team notices that the database performance is inadequate for development tasks when the script is running. A solutions architect must recommend a solution to resolve this issue.

Which solution will meet this requirement with the LEAST operational overhead?

- A. Modify the DB instance to be a Multi-AZ deployment.
- B. Create a read replica of the database. Configure the script to query only the read replica. **Most Voted**
- C. Instruct the development team to manually export the entries in the database at the end of each day.
- D. Use Amazon ElastiCache to cache the common queries that the script runs against the database.

Correct Answer: B

Community vote distribution



Comments

alvarez100 **Highly Voted** 3 years, 7 months ago

Selected Answer: B

Elasti Cache if for reading common results. The script is looking for new movies added. Read replica would be the best choice.
upvoted 45 times

Gil80 **Highly Voted** 3 years, 7 months ago

Selected Answer: B

- You have a production DB that is taking on a normal load
- You want to run a reporting application to run some analytics
- You create a read replica to run the new workload there
- The prod application is unaffected
- Read replicas are used for SELECT (=read) only kind of statements

Therefore I believe B to be the better answer.

As for "D" - ElastiCache use cases are:

1. Your data is slow or expensive to get when compared to cache retrieval.
2. Users access your data often.
3. Your data stays relatively the same, or if it changes quickly staleness is not a large issue.

- 1 - Somewhat true.
- 2 - Not true for our case.
- 3 - Also not true. The data changes throughout the day.

For my understanding, caching has to do with millisecond results, high-performance reads. These are not the issues mentioned in the questions, therefore B.

upvoted 18 times

NitiATOS 3 years, 4 months ago

I will support this by point to the question : " with the LEAST operational overhead?"

Configuring the read replica is much easier than configuring and integrating new service.

upvoted 6 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: B

Read replicas are replications of original databases and are the most suitable option for running scripts here.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - as per Gil80 (1 year, 10mth ago)

"Selected Answer: B

- You have a production DB that is taking on a normal load
 - You want to run a reporting application to run some analytics
 - You create a read replica to run the new workload there
 - The prod application is unaffected
 - Read replicas are used for SELECT (=read) only kind of statements
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For my understanding, caching has to do with millisecond results, high-performance reads. These are not the issues mentioned in the questions, therefore B."

upvoted 3 times

jaradat02 1 year, 10 months ago

Selected Answer: B

option B for sure, read replica is designed for this very use case, improving performance, on the other hand, enabling multi-AZ improves availability and not performance.

upvoted 3 times

MehulKapadia 2 years, 2 months ago

Selected Answer: B

- A. Having multi-AZ database would increase availability, bu not performance.
C. Not practical. Huge Operational Overhead. (Solution should be LEAST operational overhead)
D. Good for fixed queries with fixed results. not a good fit in this case as script is looking for new results in DB. It has to scan the database.

Correct Answer B. Read replica ensure you have dedicated read instance with its own resources.

upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

Just read from read replica.

- A: This will make it HA but won't solve any problems
C: We want an AWS solution not change the development team's ways of working
D: Elasticache is cache of read queries when data doesn't change. It's useless for finding new data.

upvoted 3 times

bujuman 2 years, 5 months ago

Selected Answer: A

Answer C is inconceivable according to LEAST operational overhead?

We will exclude answer D because question is about RDS databases and ElastiCache is not.

Between answers A and B , A is the most appropriate answer due to 2 foolowing points:

- Possible to transfrom a Single-AZ RDS to Multi-AZ
- LEAST operational overhead

upvoted 2 times

smdrouiss 2 years, 6 months ago

Selected Answer: A

It is A , because the scenario mention "single db instance" which is not possible to enable read replica

upvoted 1 times

pentium75 2 years, 5 months ago

Doesn't it become a multi-instance DB when you add a read replica? A can't be because you can't read from the passive replica of a multi-AZ DB.

upvoted 3 times

slimen 2 years, 7 months ago

Selected Answer: B

lol seriously the person who wrote the answer wants us to fail

upvoted 7 times

tom_cruise 2 years, 8 months ago

Selected Answer: B

This is what we do in the real world.

upvoted 1 times

joshik 2 years, 8 months ago

Selected Answer: B

- Cached data might not always be up-to-date, so you need to manage cache expiry and invalidation carefully.
- It may require some code changes to implement caching logic in your script.
- ElastiCache comes with additional costs, so you should assess the cost implications based on your usage.

upvoted 2 times

underdogpex 2 years, 9 months ago

Selected Answer: B

Why not D:

While ElastiCache can be relatively easy to set up, it still requires ongoing management, monitoring, and potentially scaling as the dataset and query load grow. This introduces operational overhead that may not align with the goal of minimizing operational work.

upvoted 2 times

Router 2 years, 9 months ago

the correct answer should be A, you can't create a read replica on a single-AZ DB instance

upvoted 1 times

effiecancode 1 year, 11 months ago

Actually you can

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

a read replica is always fit for these type of scenarios.

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: B

The key requirements are:

The script must report a final total during business hours

Resolve the issue of inadequate database performance for development tasks when the script is running

With the least operational overhead

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: B

A. Modifying the DB to be a Multi-AZ deployment improves high availability and fault tolerance but does not directly address the performance issue during the script execution.

C. Instructing the development team to manually export the entries in the database introduces manual effort and is not a scalable or efficient solution.

D. While using ElastiCache for caching can improve read performance for common queries, it may not be the most suitable solution for the scenario described. Caching is effective for reducing the load on the database for frequently accessed data, but it may not directly address the performance issue during the script execution.

Creating a read replica of the database (option B) provides a scalable solution that offloads read traffic from the primary database. The script can be configured to query the read replica, reducing the impact on the primary database during the script execution.

upvoted 5 times

A company has applications that run on Amazon EC2 instances in a VPC. One of the applications needs to call the Amazon S3 API to store and read objects. According to the company's security regulations, no traffic from the applications is allowed to travel across the internet.

Which solution will meet these requirements?

- A. Configure an S3 gateway endpoint. **Most Voted**
- B. Create an S3 bucket in a private subnet.
- C. Create an S3 bucket in the same AWS Region as the EC2 instances.
- D. Configure a NAT gateway in the same subnet as the EC2 instances.

Correct Answer: A

Community vote distribution

A (100%)

Comments

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: A

Gateway endpoints provide reliable connectivity to Amazon S3 and DynamoDB without requiring an internet gateway or a NAT device for your VPC. It should be option A.

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html>

upvoted 33 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: A

CORRECT

The correct solution is Option A (Configure an S3 gateway endpoint.)

A gateway endpoint is a VPC endpoint that you can use to connect to Amazon S3 from within your VPC. Traffic between your VPC and Amazon S3 never leaves the Amazon network, so it doesn't traverse the internet. This means you can access Amazon S3 without the need to use a NAT gateway or a VPN connection.

WRONG

Option B (creating an S3 bucket in a private subnet) is not a valid solution because S3 buckets do not have subnets.

Option C (creating an S3 bucket in the same AWS Region as the EC2 instances) is not a requirement for meeting the given security regulations.

Option D (configuring a NAT gateway in the same subnet as the EC2 instances) is not a valid solution because it would allow traffic to leave the VPC and travel across the Internet.

upvoted 19 times

PaulGa **Most Recent** 1 year, 8 months ago

Selected Answer: A

Ans A - S3 gateway endpoint: dedicated end-end and private

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A is the correct answer

upvoted 1 times

effiecancode 1 year, 11 months ago

it's definitely A

upvoted 1 times

JohnZh 2 years, 2 months ago

A. Configure an S3 gateway endpoint.

Correct: <https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html>

With a gateway endpoint, you can access Amazon S3 from your VPC, without requiring an internet gateway or NAT device for your VPC.

Additional you need to configure the route table for the subnet that EC2 stays, but we have the key word here.

B. Create an S3 bucket in a private subnet.

I am not aware that we can create S3 bucket in certain subnet.

C. Create an S3 bucket in the same AWS Region as the EC2 instances.

Not enough. Without VPC gateway endpoint, access will through go out to the internet.

D. Configure a NAT gateway in the same subnet as the EC2 instances.

NAT gateway outbound traffic should also go out to the internet.

upvoted 2 times

Charumathi 2 years, 4 months ago

Selected Answer: A

You can access Amazon S3 from your VPC using gateway VPC endpoints. After you create the gateway endpoint, you can add it as a target in your route table for traffic destined from your VPC to Amazon S3.

There is no additional charge for using gateway endpoints.

Amazon S3 supports both gateway endpoints and interface endpoints. With a gateway endpoint, you can access Amazon S3 from your VPC, without requiring an internet gateway or NAT device for your VPC, and with no additional cost. However, gateway endpoints do not allow access from on-premises networks, from peered VPCs in other AWS Regions, or through a transit gateway. For those scenarios, you must use an interface endpoint, which is available for an additional cost.

<https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html>

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

EC2 to S3 without public interne = S3 gatwat

B: Cannot be implemented

C: Even if you create EC2 and S3 in same region, without a S3 gateway it will use the public internet

D: Makes no sense, NAT gateway in the subnet as EC2 instance to do what?

upvoted 2 times

Ruffyt 2 years, 7 months ago

A gateway endpoint is a VPC endpoint that you can use to connect to Amazon S3 from within your VPC. Traffic between your VPC and Amazon S3 never leaves the Amazon network, so it doesn't traverse the internet. This means you can access Amazon S3 without the need to use a NAT gateway or a VPN connection

upvoted 2 times

David_Ang 2 years, 8 months ago

Selected Answer: A

Answer "A" is correct because an endpoint create a way for the data to travel in the VPC

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

Prevent traffic from traversing the internet = Gateway VPC endpoint for S3.

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

Configure an S3 gateway endpoint

upvoted 1 times

tamefi5512 2 years, 11 months ago

Selected Answer: A

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html>

upvoted 2 times

cookieMr 2 years, 11 months ago

B. Creating an S3 in a private subnet restricts direct internet access to the bucket but does not provide a direct and secure connection between the EC2 and the S3. The application would still need to traverse the internet to access the S3 API.

C. Creating an S3 in the same Region as the EC2 does not inherently prevent traffic from traversing the internet.

D. Configuring a NAT gateway allows outbound internet connectivity for resources in private subnets, but it does not provide a direct and secure connection to the S3 service. The traffic from the EC2 to the S3 API would still traverse the internet.

The most suitable solution is to configure an S3 gateway endpoint (option A). It provides a secure and private connection between the VPC and the S3 service without requiring the traffic to traverse the internet. With an S3 gateway endpoint, the EC2 can access the S3 API directly within the VPC, meeting the security requirement of preventing traffic from traveling across the internet.

upvoted 3 times

Bmarodi 3 years ago

Selected Answer: A

Configure an S3 gateway endpoint is answer.

upvoted 2 times

gustavtd 3 years, 5 months ago

Selected Answer: A

S3 Gateway Endpoint is a VPC endpoint,

upvoted 2 times

langiac 3 years, 6 months ago

Selected Answer: A

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html>

upvoted 2 times

A company is storing sensitive user information in an Amazon S3 bucket. The company wants to provide secure access to this bucket from the application tier running on Amazon EC2 instances inside a VPC.

Which combination of steps should a solutions architect take to accomplish this? (Choose two.)

- A. Configure a VPC gateway endpoint for Amazon S3 within the VPC. **Most Voted**
- B. Create a bucket policy to make the objects in the S3 bucket public.
- C. Create a bucket policy that limits access to only the application tier running in the VPC. **Most Voted**
- D. Create an IAM user with an S3 access policy and copy the IAM credentials to the EC2 instance.
- E. Create a NAT instance and have the EC2 instances use the NAT instance to access the S3 bucket.

Correct Answer: AC

Community vote distribution

AC (90%)

9%

Comments

awsgeek75 **Highly Voted** 2 years, 4 months ago

Selected Answer: AC

A: VPC S3 gateway for direct connection (no public internet) to access S3

C: Bucket policy to secure access and only allow the VPC application tier to access it

B: Opens up to public

D: Not secure to copy credentials

E: NAT instance (obsolete now) is not useful for limiting resource access, it's for subnet connections

upvoted 6 times

ArielSchivo **Highly Voted** 3 years, 7 months ago

Selected Answer: AC

Options A and C.

<https://aws.amazon.com/premiumsupport/knowledge-center/s3-private-connection-no-authentication/>

upvoted 5 times

satyaamm **Most Recent** 1 year, 5 months ago

Selected Answer: AC

VPC Gateway endpoint provides secure access to S3 and DynamoDB while the bucket policy allows access only to the application tier making it a secure connection.

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: AD

A,D - altho' I stand corrected: D means copying credentials which introduces a security risk... so that means A,C

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: AC

A removes the need for a NAT gateway and keeps the connection private, C restricts access to the bucket.

upvoted 4 times

rityoui 2 years, 5 months ago

no one mentioned the translation issue, "limit access to sth" sounds like limit this but allow others, confusing for non-English speaker.

upvoted 3 times

Ruffyt 2 years, 7 months ago

) Configure a VPC gateway endpoint for Amazon S3 within the VPC.

C) Create a bucket policy that limits access to only the application tier running in the VPC.

The key requirements are secure access to the S3 bucket from EC2 instances in the VPC.

A VPC endpoint for S3 allows connectivity from the VPC to S3 without needing internet access. The bucket policy should limit access only to the VPC by whitelisting the VPC endpoint.

upvoted 2 times

David_Ang 2 years, 8 months ago

Selected Answer: AC

These are correct because "A" and "C" ensure secure access and secure connectivity between the S3 and the EC2 instances

upvoted 4 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: AC

The key requirements are to provide secure access to the S3 bucket only from the application tier EC2 instances inside the VPC.

A VPC gateway endpoint allows private access to S3 from within the VPC without needing internet access. This keeps the traffic secure within the AWS network.

The bucket policy should limit access to only the application tier, not make the objects public. This restricts access to the sensitive data to only the authorized application tier.

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: AC

The correct options are:

A) Configure a VPC gateway endpoint for Amazon S3 within the VPC.

C) Create a bucket policy that limits access to only the application tier running in the VPC.

The key requirements are secure access to the S3 bucket from EC2 instances in the VPC.

A VPC endpoint for S3 allows connectivity from the VPC to S3 without needing internet access. The bucket policy should limit access only to the VPC by whitelisting the VPC endpoint.

upvoted 3 times

sohailn 2 years, 10 months ago

ac is the correct answer, as per my knowledge people are confused with IAM user we can use IAM role for secure access.

upvoted 2 times

tamefi5512 2 years, 11 months ago

Selected Answer: AC

AC is the right answer

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: AC

A. This eliminates the need for the traffic to go over the internet, providing an added layer of security.

B. It is important to restrict access to the bucket and its objects only to authorized entities.

C. This helps maintain the confidentiality of the sensitive user information by limiting access to authorized resources.

D. In this case, since the EC2 instances are accessing the S3 bucket from within the VPC, using IAM user credentials is unnecessary and can introduce additional security risks.

E. a NAT instance to access the S3 bucket adds unnecessary complexity and overhead.

In summary, the recommended steps to provide secure access to the S3 from the application tier running on EC2 inside a VPC are to configure a VPC gateway endpoint for S3 within the VPC (option A) and create a bucket policy that limits access to only the application tier running in the VPC (option C).

upvoted 3 times

Bmarodi 3 years ago

Selected Answer: AC

A & C the correct solutions.

upvoted 2 times

TillieEhaung 3 years ago

Selected Answer: AC

A and C

upvoted 1 times

annabellehiro 3 years, 2 months ago

Selected Answer: AC

A and C

upvoted 1 times

Help2023 3 years, 3 months ago

Selected Answer: AC

The key part that many miss out on is 'Combination'

The other answers are not wrong but

A works with C and not with the rest as they need an internet connection.

upvoted 3 times

A company runs an on-premises application that is powered by a MySQL database. The company is migrating the application to AWS to increase the application's elasticity and availability.

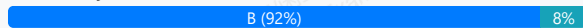
The current architecture shows heavy read activity on the database during times of normal operation. Every 4 hours, the company's development team pulls a full export of the production database to populate a database in the staging environment. During this period, users experience unacceptable application latency. The development team is unable to use the staging environment until the procedure completes.

A solutions architect must recommend replacement architecture that alleviates the application latency issue. The replacement architecture also must give the development team the ability to continue using the staging environment without delay. Which solution meets these requirements?

- A. Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Populate the staging database by implementing a backup and restore process that uses the mysqldump utility.
- B. Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Use database cloning to create the staging database on-demand. **Most Voted**
- C. Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Use the standby instance for the staging database.
- D. Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Populate the staging database by implementing a backup and restore process that uses the mysqldump utility.

Correct Answer: B

Community vote distribution



Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 11 months ago

Selected Answer: B

The recommended solution is Option B: Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Use database cloning to create the staging database on-demand.

To alleviate the application latency issue, the recommended solution is to use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production, and use database cloning to create the staging database on-demand. This allows the development team to continue using the staging environment without delay, while also providing elasticity and availability for the production application.

Therefore, Options A, C, and D are not recommended

upvoted 25 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Option A: Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Populating the staging database by implementing a backup and restore process that uses the mysqldump utility is not the recommended solution because it involves taking a full export of the production database, which can cause unacceptable application latency.

Option C: Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Using the standby instance for the staging database is not the recommended solution because it does not give the development team the ability to continue using the staging environment without delay. The standby instance is used for failover in case of a production instance failure, and it is not intended for use as a staging environment.

upvoted 20 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Option D: Use Amazon RDS for MySQL with a Multi-AZ deployment and read replicas for production. Populating the staging database by implementing a backup and restore process that uses the mysqldump utility is not the recommended solution because it involves taking a full export of the production database, which can cause unacceptable application latency.

upvoted 9 times

MutiverseAgent 2 years, 5 months ago

Agree, solution it seems to be the B)

1) Because the company wants "elasticity and availability" as the question mentioned, so I think this leaves us in the two questions related to Aurora: discarding the RDS Mysql solution.

2) According AWS documentation (<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Managing.Clone.html>)

"Aurora cloning is especially useful for quickly setting up test environments using your production data, without risking data corruption"

upvoted 9 times

arashjs1993 **Highly Voted** 2 years, 1 month ago

Selected Answer: B

Aurora MySQL is very fast in comparison to RDS for creating a clone of DB, you can create an even clone of a clone while you still work on your own clone, this will allow the dev team to continue working during cloning step.

<https://aws.amazon.com/blogs/aws/amazon-aurora-fast-database-cloning/>

upvoted 12 times

Rcosmos **Most Recent** 1 year, 5 months ago

Selected Answer: B

Conclusão:

A configuração B com o Amazon Aurora MySQL oferece uma solução eficiente, escalável e de alto desempenho, eliminando o problema de latência do aplicativo e permitindo que a equipe de desenvolvimento use o ambiente de preparo sem atrasos. A clonagem de banco de dados Aurora é a chave para atender a esses requisitos com o mínimo impacto operacional.

upvoted 1 times

MehulKapadia 1 year, 8 months ago

Selected Answer: B

Answer B:

Few Points about Aurora Database Cloning:

- Create a new Aurora DB Cluster from an existing one
- Faster than snapshot & restore
- Uses copy-on-write protocol
- Very fast & cost-effective
- Useful to create a "staging" database from a "production" database without impacting the production database

upvoted 3 times

ronntsai127 1 year, 8 months ago

When the question did not say about "cost efficient", always choose Aurora MySQL > RDS MySQL, because AWS can earn more money in Aurora

upvoted 3 times

awsgeek75 1 year, 11 months ago

Selected Answer: B

I'll go for B

AD: looks time consuming as mysqldump is like a table dump

C: You cannot use a standby for anything apart from read-only database. This would be an option if dev team was specifically using it for read-only mode.

<https://aws.amazon.com/blogs/database/readable-standby-instances-in-amazon-rds-multi-az-deployments-a-new-high-availability-option/>

upvoted 2 times

Ruffyit 2 years, 1 month ago

B. With Aurora, you can create a clone of the production database quickly and efficiently, without the need for time-consuming backup and restore processes. The development team can spin up the staging database on-demand, eliminating delays and allowing them to continue using the staging environment without interruption.

upvoted 2 times

Modulopi 2 years, 2 months ago

Selected Answer: B

B is the correct

upvoted 1 times

TariqKipkemei 2 years, 3 months ago

Selected Answer: C

No mention of cost, so technically both options B & C would work.

C. <https://aws.amazon.com/blogs/database/readable-standby-instances-in-amazon-rds-multi-az-deployments-a-new-high-availability-option/#:~:text=read%20replicas,-,Amazon%20RDS,-now%20offers%20Multi>

B. <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Managing.Clone.html#:~:text=cloning%20works,-,Aurora%20cloning,-is%20especially%20useful>

upvoted 2 times

Guru4Cloud 2 years, 4 months ago

Selected Answer: B

Option B is the best solution that meets all the requirements:

Use Amazon Aurora MySQL with Multi-AZ Aurora Replicas for production. Use database cloning to create the staging database on-demand.

The key requirements are to:

Alleviate application latency caused by database exports
Give development immediate access to a staging environment
Aurora Multi-AZ replicas improves availability and provides fast failover.

Database cloning creates an instantly available copy of the production database that can be used for staging. This avoids any export or restoration delay.

upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: B

A. Populating the staging database through a backup and restore process using the mysqldump utility would still result in delays and impact application latency.

B. With Aurora, you can create a clone of the production database quickly and efficiently, without the need for time-consuming backup and restore processes. The development team can spin up the staging database on-demand, eliminating delays and allowing them to continue using the staging environment without interruption.

C. Using the standby instance for the staging database would not provide the development team with the ability to use the staging environment without delay. The standby instance is designed for failover purposes and may not be readily available for immediate use.

D. Relying on a backup and restore process using the mysqldump utility would still introduce delays and impact application latency during the data population phase.

upvoted 3 times

linux_admin 2 years, 8 months ago

Selected Answer: B

With Amazon Aurora MySQL, creating a staging database using database cloning is an easy process. Using database cloning will eliminate the performance issues that occur when a full export is done, and the new database is created. In addition, Amazon Aurora's high availability is provided through Multi-AZ deployment, and read replicas can be used to serve the heavy read traffic without affecting the production database. This solution provides better scalability, elasticity, and availability than the current architecture.

upvoted 5 times

alexiscloud 2 years, 8 months ago

Answer B:

upvoted 1 times

bdp123 2 years, 9 months ago

Selected Answer: B

<https://aws.amazon.com/blogs/aws/amazon-aurora-fast-database-cloning/>

upvoted 3 times

john2323 2 years, 10 months ago

Selected Answer: B

Database cloning is the best answer

upvoted 1 times

techhb 2 years, 11 months ago

Selected Answer: B

Database cloning is right answer here.

upvoted 1 times

career360guru 2 years, 12 months ago

Option B is right.

You can not access Standby instance for Read in RDS Multi-AZ Deployments.

upvoted 4 times

aadi7 2 years, 11 months ago

This is correct, standby instances cannot be used for read/write and is for failover targets. Read Replicas can be used for that so B is correct.

upvoted 3 times

aadi7 2 years, 11 months ago

In a RDS Multi-AZ deployment, you can use the standby instance for read-only purposes, such as running queries and reporting. This is known as a "read replica." You can create one or more read replicas of a DB instance and use them to offload read traffic from the primary instance.

<https://aws.amazon.com/about-aws/whats-new/2018/01/amazon-rds-read-replicas-now-support-multi-az-deployments/>
upvoted 4 times

A company is designing an application where users upload small files into Amazon S3. After a user uploads a file, the file requires one-time simple processing to transform the data and save the data in JSON format for later analysis.

Each file must be processed as quickly as possible after it is uploaded. Demand will vary. On some days, users will upload a high number of files. On other days, users will upload a few files or no files.

Which solution meets these requirements with the LEAST operational overhead?

- A. Configure Amazon EMR to read text files from Amazon S3. Run processing scripts to transform the data. Store the resulting JSON file in an Amazon Aurora DB cluster.
- B. Configure Amazon S3 to send an event notification to an Amazon Simple Queue Service (Amazon SQS) queue. Use Amazon EC2 instances to read from the queue and process the data. Store the resulting JSON file in Amazon DynamoDB.
- C. Configure Amazon S3 to send an event notification to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function to read from the queue and process the data. Store the resulting JSON file in Amazon DynamoDB. **Most Voted**
- D. Configure Amazon EventBridge (Amazon CloudWatch Events) to send an event to Amazon Kinesis Data Streams when a new file is uploaded. Use an AWS Lambda function to consume the event from the stream and process the data. Store the resulting JSON file in an Amazon Aurora DB cluster.

Correct Answer: C

Community vote distribution

C (100%)

Comments

rjam **Highly Voted** 3 years, 7 months ago

Option C

Dynamo DB is a NoSQL-JSON supported

upvoted 17 times

rjam 3 years, 7 months ago

also Use an AWS Lambda - serverless - less operational overhead

upvoted 12 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: C

A. Configuring EMR and an Aurora DB cluster for this use case would introduce unnecessary complexity and operational overhead. EMR is typically used for processing large datasets and running big data frameworks like Apache Spark or Hadoop.

B. While using S3 event notifications and SQS for decoupling is a good approach, using EC2 to process the data would introduce operational overhead in terms of managing and scaling the EC2.

D. Using EventBridge and Kinesis Data Streams for this use case would introduce additional complexity and operational overhead compared to the other options. EventBridge and Kinesis are typically used for real-time streaming and processing of large volumes of data.

In summary, option C is the recommended solution as it provides a serverless and scalable approach for processing uploaded files using S3 event notifications, SQS, and Lambda. It offers low operational overhead, automatic scaling, and efficient handling of varying demand. Storing the resulting JSON file in DynamoDB aligns with the requirement of saving the data for later analysis.

upvoted 10 times

Dharmarajan **Most Recent** 1 year, 4 months ago

Selected Answer: C

C is the simplest and most widely used architecture, for example, processing photos to generate thumbnails.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: C

C is most suitable here as we need DynamoDB for low latency, AWS Lambda for scaling as per the demand.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: C

Ans C - as per cookieMr (1 yr, 2 mth ago)

"...In summary, option C is the recommended solution as it provides a serverless and scalable approach for processing uploaded files using S3 event notifications, SQS, and Lambda. It offers low operational overhead, automatic scaling, and efficient handling of varying demand. Storing the resulting JSON file in DynamoDB aligns with the requirement of saving the data for later analysis."

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: C

Option C, fulfills the least operational overhead condition.

upvoted 2 times

TilTil 2 years, 2 months ago

Selected Answer: C

B where we use EC2 instances for processing would be ideal in situations where runtime is > 15 minutes. However the question mentions 'simple processing', hence we go for Lambda.

upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

LEAST operational overhead

A: EMR is massive programming effort for this

B: EC2 is considerable overhead

D: Nice solution but why would you use Kinesis as there is no streaming scenario here

C: Simplest and all managed services so least operational overhead compared to other options

upvoted 2 times

Ruffyit 2 years, 7 months ago

Option C is the best solution that meets the requirements with the least operational overhead:

Configure Amazon S3 to send event notification to SQS queue

Use Lambda function triggered by SQS to process each file

Store output JSON in DynamoDB

This leverages serverless components like S3, SQS, Lambda, and DynamoDB to provide automated file processing without needing to provision and manage servers.

SQS queues the notifications and Lambda scales automatically to handle spikes and drops in file uploads. No EMR cluster or EC2 Fleet is needed to manage.

upvoted 2 times

Modulopi 2 years, 8 months ago

Selected Answer: C

C: Lambdas are made for that

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

C is best

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: C

Option C is the best solution that meets the requirements with the least operational overhead:

Configure Amazon S3 to send event notification to SQS queue

Use Lambda function triggered by SQS to process each file

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This leverages serverless components like S3, SQS, Lambda, and DynamoDB to provide automated file processing without needing to provision and manage servers.

SQS queues the notifications and Lambda scales automatically to handle spikes and drops in file uploads. No EMR cluster or EC2 Fleet is needed to manage.

upvoted 2 times

beginnercloud 3 years ago

Selected Answer: C

Option C is correct - Dynamo DB is a NoSQL-JSON supported
upvoted 1 times

Abrar2022 3 years ago

Selected Answer: C

SQS + LAMDA + JSON >>>>> Dynamo DB
upvoted 2 times

Bmarodi 3 years ago

Selected Answer: C

The option C is right answer.
upvoted 1 times

jy190 3 years, 1 month ago

can someone explain why SQS? it's a poll-based messaging, does it guarantee reacting the event asap?
upvoted 1 times

Zerotn3 3 years, 5 months ago

Selected Answer: C

Dynamo DB is a NoSQL-JSON supported
upvoted 1 times

An application allows users at a company's headquarters to access product data. The product data is stored in an Amazon RDS MySQL DB instance. The operations team has isolated an application performance slowdown and wants to separate read traffic from write traffic. A solutions architect needs to optimize the application's performance quickly.

What should the solutions architect recommend?

- A. Change the existing database to a Multi-AZ deployment. Serve the read requests from the primary Availability Zone.
- B. Change the existing database to a Multi-AZ deployment. Serve the read requests from the secondary Availability Zone.
- C. Create read replicas for the database. Configure the read replicas with half of the compute and storage resources as the source database.
- D. Create read replicas for the database. Configure the read replicas with the same compute and storage resources as the source database. **Most Voted**

Correct Answer: D

Community vote distribution

D (99%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: D

The solutions architect should recommend option D: Create read replicas for the database. Configure the read replicas with the same compute and storage resources as the source database.

Creating read replicas allows the application to offload read traffic from the source database, improving its performance. The read replicas should be configured with the same compute and storage resources as the source database to ensure that they can handle the read workload effectively.

upvoted 29 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: D

A. In a Multi-AZ deployment, a standby replica of the database is created in a different AZ for high availability and automatic failover purposes. However, serving read requests from the primary AZ alone would not effectively separate read and write traffic. Both read and write traffic would still be directed to the primary database instance, which might not fully optimize performance.

B. The secondary instance in a Multi-AZ deployment is intended for failover and backup purposes, not for actively serving read traffic. It operates in a standby mode and is not optimized for handling read queries efficiently.

C. Configuring the read replicas with half of the compute and storage resources as the source database might not be optimal. It's generally recommended to configure the read replicas with the same compute and storage resources as the source database to ensure they can handle the read workload effectively.

D. Configuring the read replicas with the same compute and storage resources as the source database ensures that they can handle the read workload efficiently and provide the required performance boost.

upvoted 8 times

Dharmarajan **Most Recent** 1 year, 4 months ago

Selected Answer: D

The read replica should be created with same capacity so it serves as a failover site as well, improving the availability of the database.

upvoted 1 times

Rcosmos 1 year, 5 months ago

Selected Answer: D

A criação de réplicas de leitura com os mesmos recursos de computação e armazenamento que o banco de dados de origem resolve o problema de forma eficiente, pois:

Tráfego de leitura separado:

Direcionar o tráfego de leitura para as réplicas reduz a carga na instância primária, otimizando o desempenho geral.
Manutenção da performance:

Garantir que as réplicas tenham os mesmos recursos evita gargalos, mesmo sob tráfego intenso.
Implementação rápida:

A configuração de réplicas de leitura no Amazon RDS é rápida e gerenciada, permitindo uma solução quase imediata.
Essa abordagem atende à necessidade da empresa de otimizar o desempenho do aplicativo de forma rápida e escalável.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: D

Multi-AZ's offer availability but Read Replicas offers read performance.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - can't be C because we don't know how much CPU write/read respectively consumes; we'll have to monitor to find out...

upvoted 3 times

jaradat02 1 year, 10 months ago

Selected Answer: D

D makes the most sense.

upvoted 1 times

MehulKapadia 2 years, 2 months ago

Selected Answer: D

Keyword: "separate read traffic from write traffic" = Read Replica = Option A and B are not the correct answer.

Option C: Why would you try to have half the resource for read replicas ?. It must be equal resources to ensure read load can be served consistently.

Correct Answer is D: Read replica with same compute power as source database instance.

upvoted 5 times

drich22 2 years ago

By default, a read replica is created with the same storage type as the source DB instance.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReadRepl.html

upvoted 4 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

A: This will not have any change as you are still reading from same instance as you are writing to

B: Not possible (<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Concepts.MultiAZSingleStandby.html>)

C: Why would you do that even if that was possible? No one asked to save on cost

D: Read replicas are normally for handling read-only traffic

upvoted 3 times

ignajtpolandstrong 2 years, 5 months ago

Selected Answer: D

In a Multi-AZ deployment, the standby instance is kept in sync with the primary instance and is used for failover purposes only. You cannot read data from the standby instance in a Multi-AZ deployment. If you need to offload read traffic from the primary instance, you can create one or more Read Replicas. Read Replicas are read-only copies of your database that can be used to offload read traffic from the primary instance, which can help improve performance

upvoted 3 times

Ruffiyit 2 years, 7 months ago

D. Configuring the read replicas with the same compute and storage resources as the source database ensures that they can handle the read workload efficiently and provide the required performance boost.

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

Both B and D would work.

Amazon RDS now offers Multi-AZ deployments with readable standby instances (also called Multi-AZ DB cluster deployments) . You should consider using Multi-AZ DB cluster deployments with two readable DB instances if you need additional read capacity in your Amazon RDS Multi-AZ deployment and if your application workload has strict transaction latency requirements such as single-digit milliseconds transactions.

<https://aws.amazon.com/blogs/database/readable-standby-instances-in-amazon-rds-multi-az-deployments-a-new-high-availability-option/#:~:text=read%20replicas-,Amazon%20RDS,-now%20offers%20Multi>

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

The best solution is to create read replicas for the database and configure them with the same compute and storage resources as the source database.

The key requirements are to quickly optimize performance by isolating reads from writes.

Read replicas allow read-only workloads to be directed to one or more replicas of the source RDS instance. This separates reporting or analytics queries from transactional workloads.

The read replicas should have the same compute and storage as the source to provide equivalent performance for reads. Scaling down the replicas would limit read performance.

Using Multi-AZ alone does not achieve read/write separation. The secondary AZ instance is for disaster recovery, not performance.

upvoted 5 times

MNotABot 2 years, 11 months ago

Read replica + Same resources as we may need to turn replica to primary in few cases

upvoted 1 times

Bmarodi 3 years ago

Selected Answer: D

D meets the requirements.

upvoted 1 times

Adeshina 3 years, 1 month ago

Option C suggests creating read replicas for the database and configuring them with half of the compute and storage resources as the source database. This is a better option as it allows read traffic to be offloaded from the primary database, separating read traffic from write traffic. Configuring the read replicas with half the resources will also save on costs.

upvoted 1 times

Charlesleeee 3 years ago

Err, just curious, what if the production database is 51% full? Your half storage read replica would explode...?

upvoted 5 times

NSA_Poker 2 years ago

If the source database is already 40% full, the read replica's performance will degrade as it is @ 80% capacity. This will not optimize the apps performance.

upvoted 1 times

Oldman2023 3 years, 2 months ago

Can anyone explain why B is not an option?

upvoted 5 times

draum010 3 years, 2 months ago

CHATGPT says:

To optimize the application's performance and separate read traffic from write traffic, the solutions architect should recommend creating read replicas for the database and configuring them to serve read requests. Option C and D both suggest creating read replicas, but option D is a better choice because it configures the read replicas with the same compute and storage resources as the source database.

Option A and B suggest changing the existing database to a Multi-AZ deployment, which would provide high availability by replicating the database across multiple Availability Zones. However, it would not separate read and write traffic, so it is not the best solution for optimizing application performance in this scenario.

upvoted 4 times

caffee 3 years, 2 months ago

Multi-AZ: Synchronous replication occurs, meaning that synchronizing data between DB instances immediately can slow down application's performance. But this method increases High Availability.

Read Replicas: Asynchronous replication occurs, meaning that replicating data in other moments rather than in the writing will maintain application's performance. Although the data won't be HA as Multi-AZ kind of deployment, this method increases Scalability. Good for read heavy workloads.

upvoted 3 times

An Amazon EC2 administrator created the following policy associated with an IAM group containing several users:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:TerminateInstances",
      "Resource": "*",
      "Condition": {
        "IpAddress": {
          "aws:SourceIp": "10.100.100.0/24"
        }
      }
    },
    {
      "Effect": "Deny",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringNotEquals": {
          "ec2:Region": "us-east-1"
        }
      }
    }
  ]
}
```

What is the effect of this policy?

- A. Users can terminate an EC2 instance in any AWS Region except us-east-1.
- B. Users can terminate an EC2 instance with the IP address 10.100.100.1 in the us-east-1 Region.
- C. Users can terminate an EC2 instance in the us-east-1 Region when the user's source IP is 10.100.100.254. **Most Voted**
- D. Users cannot terminate an EC2 instance in the us-east-1 Region when the user's source IP is 10.100.100.254.

Correct Answer: C

Community vote distribution

C (72%)

D (28%)

Comments

Joxtat **Highly Voted** 3 years, 5 months ago

What the policy means:

1. Allow termination of any instance if user's source IP address is 100.100.254.

2. Deny termination of instances that are not in the us-east-1 Combining this two, you get:

"Allow instance termination in the us-east-1 region if the user's source IP address is 10.100.100.254. Deny termination operation on other regions."

upvoted 73 times

KMohsoe 3 years, 1 month ago

Nice explanation. Thanks

upvoted 4 times

Subh_fidelity **Highly Voted** 3 years, 6 months ago

C is correct.
0.0/24 , the following five IP addresses are reserved:
0.0: Network address.
0.1: Reserved by AWS for the VPC router.
0.2: Reserved by AWS. The IP address of the DNS server is the base of the VPC network range plus two. ...
0.3: Reserved by AWS for future use.
0.255: Network broadcast address.
upvoted 35 times

Bmarodi 3 years ago

A good explanation!
upvoted 3 times

c12ab95 1 year ago

Also, it is the USER's IP that matters, the policy provided does not check or restrict the IP address of the EC2 instance itself
upvoted 1 times

Dharmarajan Most Recent 1 year, 4 months ago

Selected Answer: C

It is C. Because the keyword "SourceIp". That's what distinguishes C and D.
upvoted 1 times

itsmeDiyan 1 year, 4 months ago

Selected Answer: C

Uncle Trump says so
upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: C

Ans C - must be in us-east-1 region and CIDR address is in allowable range (/24)
upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: C

The first rule allows users with the specified IP CIDR to terminate instances, and the second rule specifies that the region must be us-east-1 for the termination process to be allowed, hence C is the correct answer.
upvoted 2 times

jatric 1 year, 11 months ago

Selected Answer: C

policy allow us-east-1 and with the specific IP address in the range
upvoted 3 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: C

C for sure
upvoted 1 times

jhoiti 2 years, 2 months ago

Selected Answer: D

D. Users cannot terminate an EC2 instance in the us-east-1 region when the user's source IP is 10.100.100.254.

This option corresponds to the second statement in the policy, where all EC2 actions in the "us-east-1" region are denied permission when the user's source IP is "10.100.100.254".
upvoted 2 times

lofzee 2 years ago

but it says "StringNotEquals" meaning everything is denied apart from us-east-1
upvoted 4 times

vip2 2 years, 3 months ago

Selected Answer: C

Clearly the answer is C.
D is 'Deny' 'String NOT equal' == only allow us-east-1
upvoted 5 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

Here is how I interpreted this
first part: terminate instance is allowed for the given CIDR block
second part: deny all ec2 actions when region is not us-east-1

so second part is like double negative which means allow for us-east-1 region

You combine both (remember deny always take priority which is why this is written in double negative) and you get:
[allow us-east-region1 to do any action on ec2] when [action is terminate instance and CIDR block is match]
so C is the answer

D is there to confuse you with the double negative

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: C

Deny takes precedence over Allow. Thus the flow is as follows:

IF region of the EC2 instance is not "us-east-1" -> Deny
ELSE if request is coming from 10.100.100.0/24 -> Allow
ELSE: implicit deny (what is not allowed is denied)

upvoted 4 times

Cyberkayu 2 years, 6 months ago

if

IP = 10.100.100.0/24

allow terminate EC2

Else

Deny EC2 termination permission

- with the condition "String NOT equal" to us-east-1

Answer C

upvoted 5 times

Bjfikky 2 years, 6 months ago

Selected Answer: D

The first statement allows users to terminate EC2 instances (ec2:TerminateInstances) from any IP address within the range 10.100.100.0/24.
The second statement denies users the ability to perform any EC2 actions (ec2:*) in any region other than us-east-1.
So, the correct interpretation is:

D. Users cannot terminate an EC2 instance in the us-east-1 Region when the user's source IP is 10.100.100.254

upvoted 1 times

pentium75 2 years, 5 months ago

D denies "the ability to perform any actions in any region OTHER than us-east-1". Thus the user CAN terminate instances IN us-east-1. Thus C.

upvoted 2 times

sweetheatmn 2 years, 7 months ago

Selected Answer: C

C because the explicit deny blocks other regions than us-east-1

upvoted 1 times

tom_cruise 2 years, 8 months ago

Selected Answer: C

The first statement is a subset of the second statement.

upvoted 1 times

prabhjot 2 years, 8 months ago

ans D - This policy denies EC2 instance termination for users with the source IP address 10.100.100.254 in the us-east-1 Region.

upvoted 1 times

A company has a large Microsoft SharePoint deployment running on-premises that requires Microsoft Windows shared file storage. The company wants to migrate this workload to the AWS Cloud and is considering various storage options. The storage solution must be highly available and integrated with Active Directory for access control.

Which solution will satisfy these requirements?

- A. Configure Amazon EFS storage and set the Active Directory domain for authentication.
- B. Create an SMB file share on an AWS Storage Gateway file gateway in two Availability Zones.
- C. Create an Amazon S3 bucket and configure Microsoft Windows Server to mount it as a volume.
- D. Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication.

Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: D

D. Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication.

Amazon FSx for Windows File Server is a fully managed file storage service that is designed to be used with Microsoft Windows workloads. It is integrated with Active Directory for access control and is highly available, as it stores data across multiple availability zones. Additionally, FSx can be used to migrate data from on-premises Microsoft Windows file servers to the AWS Cloud. This makes it a good fit for the requirements described in the question.

upvoted 27 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: D

A. EFS does not provide native integration with AD for access control. While you can configure EFS to work with AD, it requires additional setup and is not as straightforward as using a dedicated Windows file system like FSx for Windows File Server.

B. It may introduce additional complexity for this use case. Creating an SMB file share using AWS Storage Gateway would require maintaining the gateway and managing the synchronization between on-premises and AWS storage.

C. S3 does not natively provide the SMB file protocol required for MS SharePoint and Windows shared file storage. While it is possible to mount an S3 as a volume using 3rd-party tools or configurations, it is not the recommended.

D. FSx for Windows File Server is a fully managed, highly available file storage service that is compatible with MSWindows shared file storage requirements. It provides native integration with AD, allowing for seamless access control and authentication using existing AD user accounts.

upvoted 9 times

Dharmarajan **Most Recent** 1 year, 4 months ago

Selected Answer: D

This Q is simple if we remember - FSx is the only readily windows NTFS/SMB/AD compatible FS in Amazon offering. Please, correct me.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: D

FSx is most suitable for Windows File servers whereas Active Directory is also important for access control.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - Amazon FSx for Windows; set Active Directory

I was initially going for A, but EFS doesn't provide native integration with AD for access control; in any event, its Windows so the obvious answer is FSx

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: D

D for sure

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/aws-ad-integration-fsxW.html>

"When you create a file system with Amazon FSx, you join it to your Active Directory domain to provide user authentication and file- and folder-level access control."

upvoted 2 times

Ruffyyit 2 years, 7 months ago

D. Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication.

Amazon FSx for Windows File Server is a fully managed file storage service that is designed to be used with Microsoft Windows workloads. It is integrated with Active Directory for access control and is highly available, as it stores data across multiple availability zones. Additionally, FSx can be used to migrate data from on-premises Microsoft Windows file servers to the AWS Cloud. This makes it a good fit for the requirements described in the question.

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: D

Microsoft Windows shared file storage = Amazon FSx for Windows File Server

upvoted 4 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

The best solution that satisfies the requirements is D) Create an Amazon FSx for Windows File Server file system on AWS and set the Active Directory domain for authentication.

The key requirements are:

Shared Windows file storage for SharePoint

High availability

Integrated Active Directory authentication

upvoted 2 times

cheese929 3 years, 1 month ago

Selected Answer: D

D is correct. FSx is for windows and supports AD authentication

upvoted 2 times

kakka22 3 years, 1 month ago

Why not B? Migrating the workload? Maybe is needed a hybrid cloud solution

upvoted 1 times

gx2222 3 years, 2 months ago

Selected Answer: D

One solution that can satisfy the mentioned requirements is to use Amazon FSx for Windows File Server. Amazon FSx is a fully managed service that provides highly available and scalable file storage for Windows-based applications. It is designed to be fully integrated with Active Directory, which allows you to use your existing domain users and groups to control access to your file shares.

Amazon FSx provides the ability to migrate data from on-premises file servers to the cloud, using tools like AWS DataSync, Robocopy or PowerShell. Once the data is migrated, you can continue to use the same tools and processes to manage and access the file shares as you would on-premises.

Amazon FSx also provides features such as automatic backups, data encryption, and native multi-Availability Zone (AZ) deployments for high availability. It can be easily integrated with other AWS services, such as Amazon S3, Amazon EFS, and AWS Backup, for additional functionality and backup options.

upvoted 3 times

psr83 3 years, 5 months ago

Selected Answer: D

FSx is for Windows

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

xeun88 3 years, 6 months ago

Im going for D as the answer because FXs is compatible with windows

upvoted 2 times

kajal1206 3 years, 6 months ago

Selected Answer: D

Answer is D

upvoted 1 times

An image-processing company has a web application that users use to upload images. The application uploads the images into an Amazon S3 bucket. The company has set up S3 event notifications to publish the object creation events to an Amazon Simple Queue Service (Amazon SQS) standard queue. The SQS queue serves as the event source for an AWS Lambda function that processes the images and sends the results to users through email.

Users report that they are receiving multiple email messages for every uploaded image. A solutions architect determines that SQS messages are invoking the Lambda function more than once, resulting in multiple email messages.

What should the solutions architect do to resolve this issue with the LEAST operational overhead?

- A. Set up long polling in the SQS queue by increasing the ReceiveMessage wait time to 30 seconds.
- B. Change the SQS standard queue to an SQS FIFO queue. Use the message deduplication ID to discard duplicate messages.
- C. Increase the visibility timeout in the SQS queue to a value that is greater than the total of the function timeout and the batch window timeout. **Most Voted**
- D. Modify the Lambda function to delete each message from the SQS queue immediately after the message is read before processing.

Correct Answer: C

Community vote distribution

C (81%)

Other

Comments

Six_Fingered_Jose **Highly Voted** 3 years, 7 months ago

Selected Answer: C

answer should be C,

users get duplicated messages because -> lambda polls the message, and starts processing the message.

However, before the first lambda can finish processing the message, the visibility timeout runs out on SQS, and SQS returns the message to the poll, causing another Lambda node to process that same message.

By increasing the visibility timeout, it should prevent SQS from returning a message back to the poll before Lambda can finish processing the message

upvoted 68 times

MutiverseAgent 2 years, 11 months ago

I agree it seems solution is C, as thought the SQS FIFO makes sense deduplication id would make NO sense as the system who put messages in the queue is S3 events; and as far as I know S3 do not send duplicated events. Also, the question mention that users are complaining about receiving multiple emails for each email, which is different to say they are receiving occasionally a repeated email; so my guess is SQS FIFO is not needed.

upvoted 2 times

JoeGuan 2 years, 10 months ago

The FIFO SQS is for solving a different problem, where items in the queue require order. You cannot simply switch from a standard queue to fifo queue. Duplicate emails are a common issue with a standard queue. The documentation consistently reminds us that duplicate emails can occur, and the solution is not to create a FIFO queue, but rather adjust the configuration parameters accordingly.

upvoted 5 times

PLN6302 2 years, 9 months ago

amazon s3 doesn't support fifo queues

upvoted 3 times

Ello2023 3 years, 4 months ago

I am confused. If the email has been sent many times already why would they need more time?

I believe SQS Queue Fifo will keep in order and any duplicates with same ID will be deleted. Can you tell me where i am going wrong? Thanks

upvoted 9 times

MrAWS 3 years, 4 months ago

I tend to agree with you. See my comments above.

upvoted 2 times

Robrobtutu 3 years, 1 month ago

Increasing the visibility timeout would give time to the lambda function to finish processing the message, which would make it disappear from the queue, and therefore only one email would be sent to the user.

If the visibility timeout ends while the lambda function is still processing the message, the message will be returned to the queue and there another lambda function would pick it up and process it again, which would result in the user receiving two or more emails about the same thing.

upvoted 12 times

aadityaravi8 2 years, 11 months ago

I agree with your answer explanation

upvoted 1 times

Abdou1604 2 years, 10 months ago

I agree because the issue is multiple received email for an image uploaded

upvoted 1 times

brushek Highly Voted 3 years, 8 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-visibility-timeout.html>

this is important part:

Immediately after a message is received, it remains in the queue. To prevent other consumers from processing the message again, Amazon SQS sets a visibility timeout, a period of time during which Amazon SQS prevents other consumers from receiving and processing the message. The default visibility timeout for a message is 30 seconds. The minimum is 0 seconds. The maximum is 12 hours.

upvoted 16 times

In2718 Most Recent 9 months, 2 weeks ago

Selected Answer: C

Switching S3 event notifications to FIFO SQS queues is not supported (S3 can only publish to Standard queues)

upvoted 1 times

BennyBrrr 1 year ago

Selected Answer: C

I guessed B because I know FIFO queues have deduplication functionality, but when I looked it up it wouldn't solve this issue.

FIFO deduplication protects against a producer sending a message with the same deduplication ID to the queue multiple times within the deduplication interval (5 minutes). The question doesn't mention duplicates are being added into the queue; it just mentions that messages from the queue are being processed more than once.

As a result, the answer must be C because messages that take longer to process than the configured visibility timeout are made visible on the queue again, and are therefore picked up by another processor.

upvoted 1 times

AwsAbhiKumar 1 year, 3 months ago

Selected Answer: B

I believe answer is B cause FIFO queues provide exactly-once processing and guarantee that messages are delivered in order. They also support message deduplication, which prevents processing duplicate messages within a specified deduplication interval.

Whereas a longer visibility timeout might prevent messages from being processed more than once if processing takes longer than expected. However, it does not eliminate the inherent at-least-once delivery behavior of standard queues, which can still cause duplicates.

upvoted 2 times

jerry1 1 year, 2 months ago

I think there is another question that mention the visibility timeout concept as well

My understanding is for SQS, you have producer (inject data to queue) and consumer (get data from queue), the SQS itself does not know who will be the producer or consumer, this is implemented at the producer/consumer side like a Lambda function

So ensuring the read sequence doesn't mean no duplication at all

Instead you need to ensure a data (element in the queue) not to be read multiple times by consumer

And this is done by adjusting the visible timeout, making it long enough will make sure each consumer (Lambda in this case) can either consume the data completely (or fail) without other consumer's interruption

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: C

SQS feature of "Visibility timeout" sets the time for getting acknowledgement from processor that the message has been successfully processed. I think it should have been called "Invisibility " rather than "Visibility" timeout. Anyways, I think AWS engineers have thought this Queue service through well. So if the processing time > visibility timeout, the message comes back and is picked up by another execution of lambda function. It is such a small thing but it can easily take up lambda usage if not considered!

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: B

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-visibility-timeout.html> , standard queue is delivered at lease once, the duplication can not be avoided

upvoted 1 times

Rcosmos 1 year, 5 months ago

Selected Answer: C

C. Aumentar o tempo limite de visibilidade:

Um tempo limite de visibilidade maior reduz a chance de reprocessamento prematuro, mas não resolve completamente o problema de mensagens duplicadas.

É mais complexo ajustar adequadamente o tempo de visibilidade para sincronizar com o processamento da função Lambda e o tempo limite.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: C

Using Visibility Timeout is the most suitable option here with the least operational overhead.

upvoted 1 times

salman7540 1 year, 5 months ago

Selected Answer: C

FIFO queues are designed to never introduce duplicate messages. However, a message producer might introduce duplicates if another producer (lambda in this case) picks up same fifo message after visibility timeout over but previous lambda was still processing it. Hence correct answer is C.

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: B

it should be B

- Using a FIFO (First In, First Out) queue ensures that messages are processed in the exact order they are received, preventing duplicates from being processed multiple times by the Lambda function. By setting up message deduplication IDs, you can further guarantee that even if the same message is sent multiple times, only one instance will be processed.

- When dealing with potential duplicate messages in an event-driven architecture, switching to an SQS FIFO queue with message deduplication is generally the most efficient and reliable solution to prevent redundant processing.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - is my preferred choice because it removes the record and therefore eliminates the possibility of duplication; the only snag here is ensuring the SQS record is locked.

Ans C, which has the most votes will work but is less robust - but perhaps it is the cheapest option.

Ans A, the author's preference, won't work unless the Lambda function is guaranteed to complete within 30secs - and nowhere is that stated

upvoted 2 times

jaradat02 1 year, 10 months ago

the correct answer is C, although B and D might also work, C is the one that achieves the least operational overhead condition.

upvoted 1 times

Solomon2001 2 years, 1 month ago

Selected Answer: B

B. Change the SQS standard queue to an SQS FIFO queue. Use the message deduplication ID to discard duplicate messages.

Here's why this option is suitable:

SQS FIFO Queue:

Amazon SQS FIFO (First-In-First-Out) queues ensure that the order of messages is preserved.

Each message in a FIFO queue has a unique message deduplication ID.

By using a FIFO queue, you can prevent duplicate messages from being processed.

Message Deduplication ID:

When sending messages to a FIFO queue, set the message deduplication ID to ensure that identical messages are treated as duplicates.

If a message with the same deduplication ID is sent within a 5-minute window, it is considered a duplicate and discarded.

<https://docs.aws.amazon.com/lambda/latest/dg/with-sqs.html>

upvoted 2 times

Anthony_Rodrigues 2 years ago

This wouldn't work as expected because the issue is not the order or the queue receiving multiple times the same message. The issue is that the Lambda executes the same message multiple times.

The FIFO deduplication ID works when the producer sends the message, not when the consumer is receiving the message.

(<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/using-messagededuplicationid-property.html>)
For this to work, the Lambda function would need to know/keep the last N deduplication ID to guarantee that it isn't processing the same message which goes against Lambda's statelessness.

Therefore, for this case, B is the best answer.

upvoted 1 times

824c449 2 years, 1 month ago

Selected Answer: B

B: Directly addresses the issue of duplicates by ensuring exact-once processing and message ordering, which mitigates the risk without requiring adjustments based on anticipated processing times or additional application logic to manage potential duplicates.

So, while increasing the visibility timeout could help in managing when messages are available for processing again, it doesn't provide a structural solution to the problem of duplicate processing in the way that using an SQS FIFO queue does, nor does it ensure the operational simplicity and reliability that comes with eliminating duplicates at the source.

upvoted 2 times

ManikRoy 2 years, 1 month ago

Selected Answer: C

to all those who are considering option B, Its INCORRECT.

Amazon Simple Queue Service FIFO (First-In-First-Out) queues aren't supported as an Amazon S3 event notification destination.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/EventNotifications.html>

upvoted 3 times

Uzbekistan 2 years, 2 months ago

Selected Answer: B

Given the requirement to resolve the issue of multiple email messages being sent to users with the least operational overhead, the most appropriate solution is:

B. Change the SQS standard queue to an SQS FIFO queue. Use the message deduplication ID to discard duplicate messages.

Explanation:

SQS FIFO Queue: FIFO (First-In-First-Out) queues in SQS ensure that the order in which messages are sent and received is strictly preserved and that each message is processed only once. By switching to an SQS FIFO queue, you can prevent the Lambda function from processing duplicate messages.

upvoted 1 times

ManikRoy 2 years, 1 month ago

Its INCORRECT

Amazon Simple Queue Service FIFO (First-In-First-Out) queues aren't supported as an Amazon S3 event notification destination.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/EventNotifications.html>

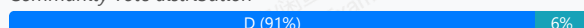
upvoted 1 times

A company is implementing a shared storage solution for a gaming application that is hosted in an on-premises data center. The company needs the ability to use Lustre clients to access data. The solution must be fully managed. Which solution meets these requirements?

- A. Create an AWS Storage Gateway file gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.
- B. Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.
- C. Create an Amazon Elastic File System (Amazon EFS) file system, and configure it to support Lustre. Attach the file system to the origin server. Connect the application server to the file system.
- D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

123jh10 **Highly Voted** 3 years, 7 months ago

Selected Answer: D

Answer is D.

Lustre in the question is only available as FSx

<https://aws.amazon.com/fsx/lustre/>

upvoted 30 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: D

Option D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system.

Amazon FSx for Lustre is a fully managed file system that is designed for high-performance workloads, such as gaming applications. It provides a high-performance, scalable, and fully managed file system that is optimized for Lustre clients, and it is fully integrated with Amazon EC2. It is the only option that meets the requirements of being fully managed and able to support Lustre clients.

upvoted 16 times

Dharmarajan **Most Recent** 1 year, 4 months ago

Selected Answer: D

This one had me stumped! Thanks, community, for the link.

upvoted 1 times

Rcosmos 1 year, 5 months ago

Selected Answer: C

Não é uma prática recomendada:

O padrão em sistemas assíncronos com SQS e Lambda é remover a mensagem apenas após o processamento bem-sucedido. Isso garante que, caso algo falhe, a mensagem possa ser processada novamente.

Não resolve duplicidade real:

Mesmo que você exclua a mensagem imediatamente, o problema de mensagens duplicadas pode ocorrer devido à lógica de envio duplicado no S3 ou por notificações redundantes do evento de criação de objetos. Assim, o problema original pode persistir.

upvoted 1 times

Rcosmos 1 year, 5 months ago

Selected Answer: C

A opção D pode parecer uma solução direta, mas não é recomendada porque introduz riscos significativos para a confiabilidade do sistema. Vamos analisar mais a fundo:

O que a opção D sugere:

Modificar a função Lambda para excluir cada mensagem da fila do SQS imediatamente após a mensagem ser lida, antes do processamento.

Problemas com a opção D:

Perda de mensagens em caso de falhas:

Se a mensagem for excluída do SQS antes de ser processada e algo der errado (como falha na função Lambda ou erro na lógica de processamento), a mensagem será perdida permanentemente.

O SQS é projetado para garantir que a mensagem seja processada com segurança antes de ser removida. Excluir a mensagem antes do processamento viola essa garantia.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: D

Amazon FSx Lustre is the most suitable for managing Lustre Clients.

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: D

D, self explanatory.

upvoted 2 times

lofzee 2 years ago

the question is missing some text i think.... as none of the answers really solve the equation.

if the answer is D, how is the on prem infrastructure accessing Lustre in AWS?

either way the only possible option it could be is D, so D it is.

upvoted 3 times

NullIM 2 years, 1 month ago

Sorry but what is origin server in this context, does it mean AWS server or premise??

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

D: Lustre is key requirement

AB: No support for Lustre

C: Cannot just configure EFS to support Lustre file system

upvoted 2 times

Ruffyit 2 years, 7 months ago

Option D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system.

Amazon FSx for Lustre is a fully managed file system that is designed for high-performance workloads, such as gaming applications. It provides a high-performance, scalable, and fully managed file system that is optimized for Lustre clients, and it is fully integrated with Amazon EC2. It is the only option that meets the requirements of being fully managed and able to support Lustre clients.

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: D

Lustre clients = Amazon FSx for Lustre file system

upvoted 4 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

The correct solution is D) Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system.

The key requirements are:

Shared storage solution

Support Lustre clients

Fully managed service

Amazon FSx for Lustre provides a fully managed file system that is optimized for Lustre workloads. It allows Lustre clients to seamlessly connect to the file system.

upvoted 4 times

RupeC 2 years, 10 months ago

Selected Answer: A

Sorry, but I disagree with everyone. The question states "a gaming application that is hosted in an on-premises data center". Option D does not address this and cannot to my knowledge address it. Thus:

A. Create an AWS Storage Gateway file gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.

By using AWS Storage Gateway in file gateway mode, you can extend your on-premises data center storage into the AWS cloud. The file share created on AWS Storage Gateway can use the necessary client protocol (such as Lustre), which would allow the Lustre clients in your on-premises data center to access the data stored on AWS Storage Gateway.

This solution enables you to use Lustre clients to access data, while still keeping the gaming application hosted in your on-premises data center. AWS Storage Gateway provides a fully managed solution for this hybrid scenario, allowing seamless integration between on-premises and AWS cloud storage.

upvoted 5 times

David_Ang 2 years, 8 months ago

mate if you have an aws service that is meant to be used for this task, there is simply not discussion, is more simple, is more cheap and better option

upvoted 1 times

JoeGuan 2 years, 10 months ago

So, I think that the FSx File Gateway is currently only available for Windows? I don't think Lustre is part of this offering yet as of 8/8/2023

upvoted 1 times

pentium75 2 years, 5 months ago

"The file share created on AWS Storage Gateway can use the necessary client protocol (such as Lustre)", no it can use only NFS or SMB. But NOT Lustre.

upvoted 3 times

james2033 2 years, 11 months ago

Selected Answer: D

Content of "Amazon FSx for Lustre" at this link <https://aws.amazon.com/fsx/lustre/> . Focus at image, section: "On-premises clients".

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: D

A. Lustre client access is not supported by AWS Storage Gateway file gateway.

B. Creating a Windows file share on an EC2 Windows instance is suitable for Windows-based file sharing, but it does not provide the required Lustre client access. Lustre is a high-performance parallel file system primarily used in high-performance computing (HPC) environments.

C. EFS does not natively support Lustre client access. Although EFS is a managed file storage service, it is designed for general-purpose file storage and is not optimized for Lustre workloads.

D. Amazon FSx for Lustre is a fully managed file system optimized for high-performance computing workloads, including Lustre clients. It provides the ability to use Lustre clients to access data in a managed and scalable manner. By choosing this option, the company can benefit from the performance and manageability of Amazon FSx for Lustre while meeting the requirement of Lustre client access.

upvoted 4 times

Musti35 3 years, 1 month ago

Selected Answer: D

https://aws.amazon.com/fsx/lustre/?nc1=h_ls#:~:text=Amazon%20FSx%20for%20Lustre%20provides%20fully%20managed%20shared%20storage%20with%20the%20scalability%20and%20performance%20of%20the%20popular%20Lustre%20file%20system.

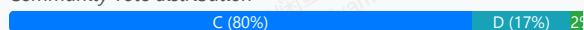
upvoted 2 times

A company's containerized application runs on an Amazon EC2 instance. The application needs to download security certificates before it can communicate with other business applications. The company wants a highly secure solution to encrypt and decrypt the certificates in near real time. The solution also needs to store data in highly available storage after the data is encrypted. Which solution will meet these requirements with the LEAST operational overhead?

- A. Create AWS Secrets Manager secrets for encrypted certificates. Manually update the certificates as needed. Control access to the data by using fine-grained IAM access.
- B. Create an AWS Lambda function that uses the Python cryptography library to receive and perform encryption operations. Store the function in an Amazon S3 bucket.
- C. Create an AWS Key Management Service (AWS KMS) customer managed key. Allow the EC2 role to use the KMS key for encryption operations. Store the encrypted data on Amazon S3. **Most Voted**
- D. Create an AWS Key Management Service (AWS KMS) customer managed key. Allow the EC2 role to use the KMS key for encryption operations. Store the encrypted data on Amazon Elastic Block Store (Amazon EBS) volumes.

Correct Answer: C

Community vote distribution



Comments

Chunsl **Highly Voted** 3 years, 7 months ago

C makes a better sense. Between C (S3) and D (EBS), S3 is highly available with LEAST operational overhead.

upvoted 50 times

MutiverseAgent 2 years, 11 months ago

Agree, also the data in EBS will be accessible only to the EC2 instance and that is not as available as S3 would be.

upvoted 8 times

jerry1 1 year, 2 months ago

agree on this, although the question did not mention where the EBS is attached to, but i assume should be the EC2, then your statement make sense

upvoted 1 times

MXB05 **Highly Voted** 3 years, 8 months ago

Selected Answer: C

Correct Answer is C: EBS is not highly available

upvoted 23 times

FNJ1111 3 years, 5 months ago

Per AWS: "Amazon EBS volumes are designed to be highly available, reliable, and durable"

<https://aws.amazon.com/ebs/features/>

upvoted 2 times

Ello2023 3 years, 4 months ago

EBS is Highly Available as it stores in multi AZ and S3 is regional.

upvoted 2 times

pentium75 2 years, 5 months ago

S3 is also highly available. Within the region, but still. Multi-AZ = HA.

upvoted 3 times

Bayebrymo 2 years, 1 month ago

This is false... S3 is Multi AZ and EBS is only replicated WITHIN an AZ by default.

upvoted 3 times

oguz11 3 years, 4 months ago

EBS also has Multi-AZ capability, but it does not replicate the data across multiple availability zones by default. When Multi-AZ is enabled, it creates a replica of the EBS volume in a different availability zone and automatically failover to the replica in case of a failure. However, this requires additional configuration and management. In comparison, Amazon S3 automatically replicates data across multiple availability zones without any additional configuration. Therefore, storing the data on Amazon S3 provides a simpler and more efficient solution for high availability.

upvoted 11 times

dkw2342 2 years, 3 months ago

This is false. There is no AWS-provided functionality that will replicate EBS volumes across AZs. There are 3rd-party solutions to this, but that's not what's being asked here.

EBS is only replicated WITHIN an AZ by default.

upvoted 3 times

JayBee65 3 years, 5 months ago

Yes it is!

upvoted 1 times

TS1991 1 year, 8 months ago

EBS volumes are not Multi-AZ. EBS io2 types are multi-attach within the same AZ. EFS is multi-AZ

upvoted 2 times

pabloveintimilla **Most Recent** 1 year, 3 months ago

Selected Answer: C

C. Correct, AWS KMS handles encryption with minimal effort and Amazon S3 offers durable and available storage vs D Incorrect. EBS is storage attached to an EC2 instance, which reduces availability compared to S3.

upvoted 1 times

sk1974 1 year, 3 months ago

Selected Answer: C

The answer is in LEAST operational overhead

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: C

Of the given options, C makes most sense. Reason being the rest of the options do not make as much sense due to A. Being not specific enough, B. being insufficient to achieve the objective, D, being on EBS, which needs to be attached to a EC2 instance.

One thing with the questions is that there are many times, some data that is unclear or there is some ambiguity. I feel these scenarios make one to assume things and perhaps even train the mind to evaluate ambiguous situations. This is valuable in my opinion.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: C

S3 is highly available compared to EBS and using AWS KMS is more suitable for managing certificates here

upvoted 1 times

thiahthura 1 year, 6 months ago

Selected Answer: C

Both C and D are correct. C is suitable for this requirement. We've to use S3 because they want to save the data with H.A

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: C

C is the most efficient.

upvoted 1 times

Shub80 1 year, 11 months ago

AWS KMS: Provides a managed service for secure key storage and encryption/decryption operations. This eliminates the need to manage encryption/decryption logic within the application itself.

Customer Managed Key: The company maintains control over the key, ensuring security.

EC2 Role Permissions: Granting permissions to the EC2 role allows the application to use KMS for encryption/decryption without managing individual credentials.

Amazon S3: Offers highly available and scalable storage for the encrypted certificates. S3 is generally cheaper than EBS for data that is not frequently accessed.

upvoted 4 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: C

C for sure

upvoted 1 times

huangyou2003 2 years, 1 month ago

Selected Answer: C

S3: highly available

EBS: lower latency

upvoted 1 times

f761d0e 2 years, 1 month ago

"Amazon S3 is an object storage service that can store large volumes of unstructured data, whereas Amazon EBS is a block storage service that is ideally suited for durable, low-latency data storage associated with EC2 instances."

<https://www.tutorialspoint.com/difference-between-amazon-s3-and-amazon-ebs#:~:text=In%20conclusion%2C%20Amazon%20S3%20is,storage%20associated%20with%20EC2%20instances.>

Seems like D to me. S3 is for large data, EBS is ec2 specific.

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

The language is confusing over here so I'm going by process of elimination

A: Wrong because manual operation and fine grained IAM is overhead

B: What?

D: Between C and D S3 is more HA than EFS so C wins

upvoted 2 times

awsgeek75 2 years, 4 months ago

Sorry meant EBS, not EFS for D

D: Between C and D, S3 is more HA than EBS. So C wins

upvoted 2 times

ignajtpolandstrong 2 years, 5 months ago

Selected Answer: D

I would select D.

you can mount a single Amazon Elastic Block Store (EBS) volume to multiple Docker containers running on the same Amazon Elastic Compute Cloud (EC2) instance.

you can store data from a container running on Amazon Elastic Compute Cloud (EC2) to an Amazon Simple Storage Service (S3) bucket. One way to do this is to use the aws s3 cp command in the command line of the EC2 instance.

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: C

A - does not mention storing the encrypted data at all (though that is a requirement), also involves manual action which is surely NOT "least operational effort"

B - Doesn't make any sense

C - Yes, S3 meets the requirements and is easy to access from containerized app

D - EBS volumes are mounted to the container host, but data is created on containers

upvoted 3 times

xdkonorek2 2 years, 7 months ago

Selected Answer: A

A is OK

secrets manager:

- is highly available

- you can store custom secrets in it like certificate

- automatically encrypts secrets at rest, and can be configured for encryption in transit

- downloading certificate from it is less operational overhead than decrypting it manually with KMS key

arguments againts it that this is more manual than C and D? this manual step is necessary measure and can't be omitted in other options

C and D have this "store the encrypted data in..." to store encrypted certificate you have to: log in to instance, get kms key, get certificate, encrypt it, and load that data this is more operational overhead

upvoted 2 times

pentium75 2 years, 5 months ago

"Least operational overhead" and "manually" (as in A) usually don't go together. Also, A does not say anything about storing the data (which is a requirement).

"C and D have this 'store the encrypted data in'" yes, exactly, the encrypted data, NOT the certificate. You encrypt data with the certificate, and you want to store THAT encrypted data.

upvoted 3 times

David_Ang 2 years, 8 months ago

Selected Answer: C

"C" is more correct because S3 is more efficient and cheaper to store data like certificates, like this case. Also Option D involves using Amazon Elastic Block Store (Amazon EBS) volumes, which is not typically used for storing certificates and may introduce unnecessary complexity and operational overhead.

upvoted 2 times

A solutions architect is designing a VPC with public and private subnets. The VPC and subnets use IPv4 CIDR blocks. There is one public subnet and one private subnet in each of three Availability Zones (AZs) for high availability. An internet gateway is used to provide internet access for the public subnets. The private subnets require access to the internet to allow Amazon EC2 instances to download software updates.

What should the solutions architect do to enable Internet access for the private subnets?

A. Create three NAT gateways, one for each public subnet in each AZ. Create a private route table for each AZ that forwards non-VPC traffic to the NAT gateway in its AZ. **Most Voted**

B. Create three NAT instances, one for each private subnet in each AZ. Create a private route table for each AZ that forwards non-VPC traffic to the NAT instance in its AZ.

C. Create a second internet gateway on one of the private subnets. Update the route table for the private subnets that forward non-VPC traffic to the private internet gateway.

D. Create an egress-only internet gateway on one of the public subnets. Update the route table for the private subnets that forward non-VPC traffic to the egress-only Internet gateway.

Correct Answer: A

Community vote distribution

A (98%)

2

Comments

Gil80 **Highly Voted** 3 years, 7 months ago

Selected Answer: A

NAT Instances - OUTDATED BUT CAN STILL APPEAR IN THE EXAM!

However, given that A provides the newer option of NAT Gateway, then A is the correct answer.

B would be correct if NAT Gateway wasn't an option.

upvoted 19 times

Shrestwt 3 years, 1 month ago

NAT instance or NAT Gateway always created in public subnet to provide internet access to private subnet. In option B. they are creating NAT Instance in private subnet which is not correct.

upvoted 30 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: A

The correct answer is option A.

To enable Internet access for the private subnets, the solutions architect should create three NAT gateways, one for each public subnet in each Availability Zone (AZ). NAT gateways allow private instances to initiate outbound traffic to the Internet but do not allow inbound traffic from the Internet to reach the private instances.

The solutions architect should then create a private route table for each AZ that forwards non-VPC traffic to the NAT gateway in its AZ. This will allow instances in the private subnets to access the Internet through the NAT gateways in the public subnets.

upvoted 13 times

SirDNS **Most Recent** 1 year, 2 months ago

Selected Answer: A

NAT Gateways always have to part of public subnet. We can then create a destination to these NAT Gateways in the private route table.

upvoted 1 times

satyaamm 1 year, 5 months ago

Selected Answer: A

NAT Gateways are much better here as they can easily scale and provide high availability. NAT instances can't provide scalability and HA. Internet Gateway is 2-way and not suitable here. Egress-only is used for IPv6.

upvoted 1 times

kernel1973 1 year, 6 months ago

You have to create 3 NAT gateways but in the private subnet.

Public and Private Subnets are the name of the concept. A public subnet is a subnet with a route to the internet gateway, private subnet doesn't have a route to the internet gateway.

In this case the private subnets must have the NAT Gateway/NAT Instances, not the public subnet because if we are defined a subnet as public it means that this subnet has a route to the IG.

upvoted 1 times

87217f5 9 months, 4 weeks ago

NAT GW is deployed in the public subnet I believe..

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - it can only be A or B and NAT Gateways are preferred over NAT Instances.

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: A

the correct answer is A, to connect a private subnet to the internet using internet gateways is irrelevant, you have to use either NAT gateway or NAT instance, and NAT gateway is the better choice.

upvoted 1 times

soufiyane 2 years, 2 months ago

Selected Answer: A

Nat instances can do the same except it's not cost effective also it needs a lot of management, going with nat gateways makes more sense

upvoted 1 times

ronin201 2 years, 7 months ago

in Azure there is 1 NAT GW multi AZ, 1 per network, I think this is an example for AWS to change

upvoted 1 times

pentium75 2 years, 5 months ago

But in AWS a NAT GW is attached to a subnet, and a subnet resides in a single AZ. Can't create multi-AZ NAT GW without changing whole architecture. You CAN use one NAT GW from multiple subnets in multiple AZs I think, but then it would not be HA.

upvoted 1 times

Ruffyit 2 years, 7 months ago

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-example-private-subnets-nat.html>

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

The best solution is to create a NAT gateway in each public subnet (one per availability zone), and update the route tables for the private subnets to send internet traffic to the NAT gateway.

NAT gateways allow private subnets to access the internet for things like software updates, without exposing those instances directly to the internet. An egress-only internet gateway would allow outbound access, but also allow inbound internet traffic, which is not desired for the private subnets.

upvoted 5 times

james2033 2 years, 10 months ago

Selected Answer: A

"Egress" means outbound connection, remove D. "Second gateway", remove C.

Now has only A and B. The difference between A versus B is "1 NAT gateway, 1 for public subnet in each AZ" (A) and "1 NAT gateway, 1 for private subnet in each AZ" (B).

Choose A.

upvoted 4 times

cookieMr 2 years, 11 months ago

By creating a NAT gateway in each public subnet, the private subnets can route their Internet-bound traffic through the NAT gateways. This allows EC2 in the private subnets to download software updates and access other resources on the Internet.

Additionally, a separate private route table should be created for each AZ. The private route tables should have a default route that forwards non-VPC traffic (0.0.0.0/0) to the corresponding NAT gateway in the same AZ. This ensures that the private subnets use the appropriate NAT gateway for

Internet access.

B is incorrect because NAT instances require manual management and configuration compared to NAT gateways, which are a fully managed service. NAT instances are also being deprecated in favor of NAT gateways.

C is incorrect because creating a second internet gateway on a private subnet is not a valid solution. Internet gateways are associated with public subnets and cannot be directly associated with private subnets.

D is incorrect because egress-only internet gateways are used for IPv6 traffic.

upvoted 5 times

Jeeva28 3 years ago

NAT Gateway will be created Public Subnet and Provide access to Private Subnet

upvoted 1 times

cheese929 3 years, 1 month ago

Selected Answer: A

A is correct.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-example-private-subnets-nat.html>

upvoted 2 times

Heric 3 years, 1 month ago

Selected Answer: A

Now NAT Instances is avoided by AWS. Then choose the NAT Gateway

upvoted 3 times

alexiscloud 3 years, 2 months ago

A: NAT Gateway

upvoted 1 times

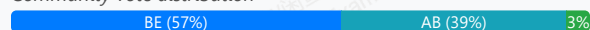
A company wants to migrate an on-premises data center to AWS. The data center hosts an SFTP server that stores its data on an NFS-based file system. The server holds 200 GB of data that needs to be transferred. The server must be hosted on an Amazon EC2 instance that uses an Amazon Elastic File System (Amazon EFS) file system.

Which combination of steps should a solutions architect take to automate this task? (Choose two.)

- A. Launch the EC2 instance into the same Availability Zone as the EFS file system.
- B. Install an AWS DataSync agent in the on-premises data center. **Most Voted**
- C. Create a secondary Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instance for the data.
- D. Manually use an operating system copy command to push the data to the EC2 instance.
- E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server. **Most Voted**

Correct Answer: BE

Community vote distribution



Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: BE

Answer and HOW-TO

B. Install an AWS DataSync agent in the on-premises data center.

E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server.

To automate the process of transferring the data from the on-premises SFTP server to an EC2 instance with an EFS file system, you can use AWS DataSync. AWS DataSync is a fully managed data transfer service that simplifies, automates, and accelerates transferring data between on-premises storage systems and Amazon S3, Amazon EFS, or Amazon FSx for Windows File Server.

To use AWS DataSync for this task, you should first install an AWS DataSync agent in the on-premises data center. This agent is a lightweight software application that you install on your on-premises data source. The agent communicates with the AWS DataSync service to transfer data between the data source and target locations.

upvoted 74 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Next, you should use AWS DataSync to create a suitable location configuration for the on-premises SFTP server. A location represents a data source or a data destination in an AWS DataSync task. You can create a location for the on-premises SFTP server by specifying the IP address, the path to the data, and the necessary credentials to access the data.

Once you have created the location configuration for the on-premises SFTP server, you can use AWS DataSync to transfer the data to the EC2 instance with the EFS file system. AWS DataSync handles the data transfer process automatically and efficiently, transferring the data at high speeds and minimizing downtime.

upvoted 24 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Explanation of other options

A. Launch the EC2 instance into the same Availability Zone as the EFS file system.

This option is not wrong, but it is not directly related to automating the process of transferring the data from the on-premises SFTP server to the EC2 instance with the EFS file system. Launching the EC2 instance into the same Availability Zone as the EFS file system can improve the performance and reliability of the file system, as it reduces the latency between the EC2 instance and the file system. However, it is not necessary for automating the data transfer process.

upvoted 18 times

Buruguduystunstugudunstuy 3 years, 5 months ago